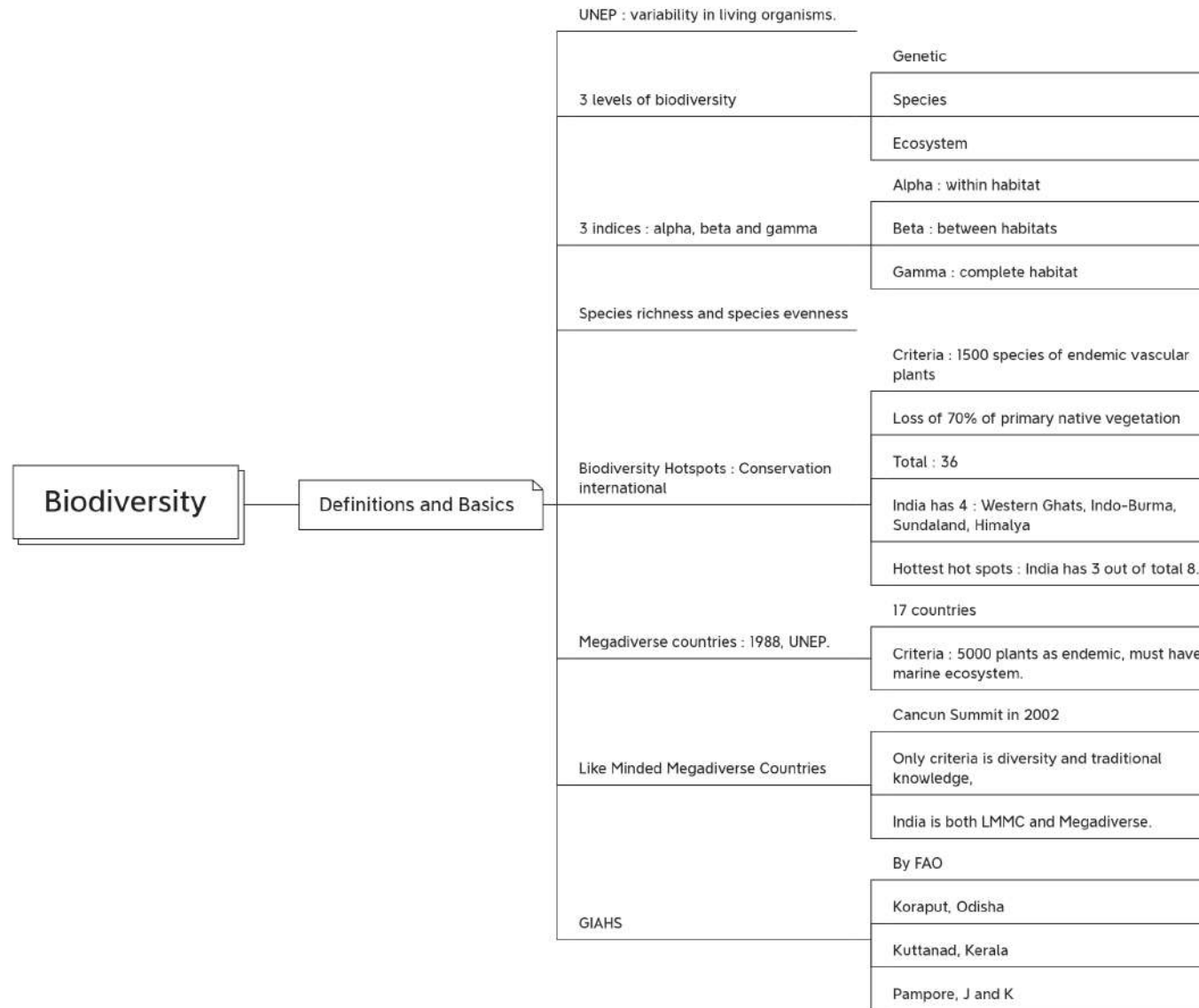
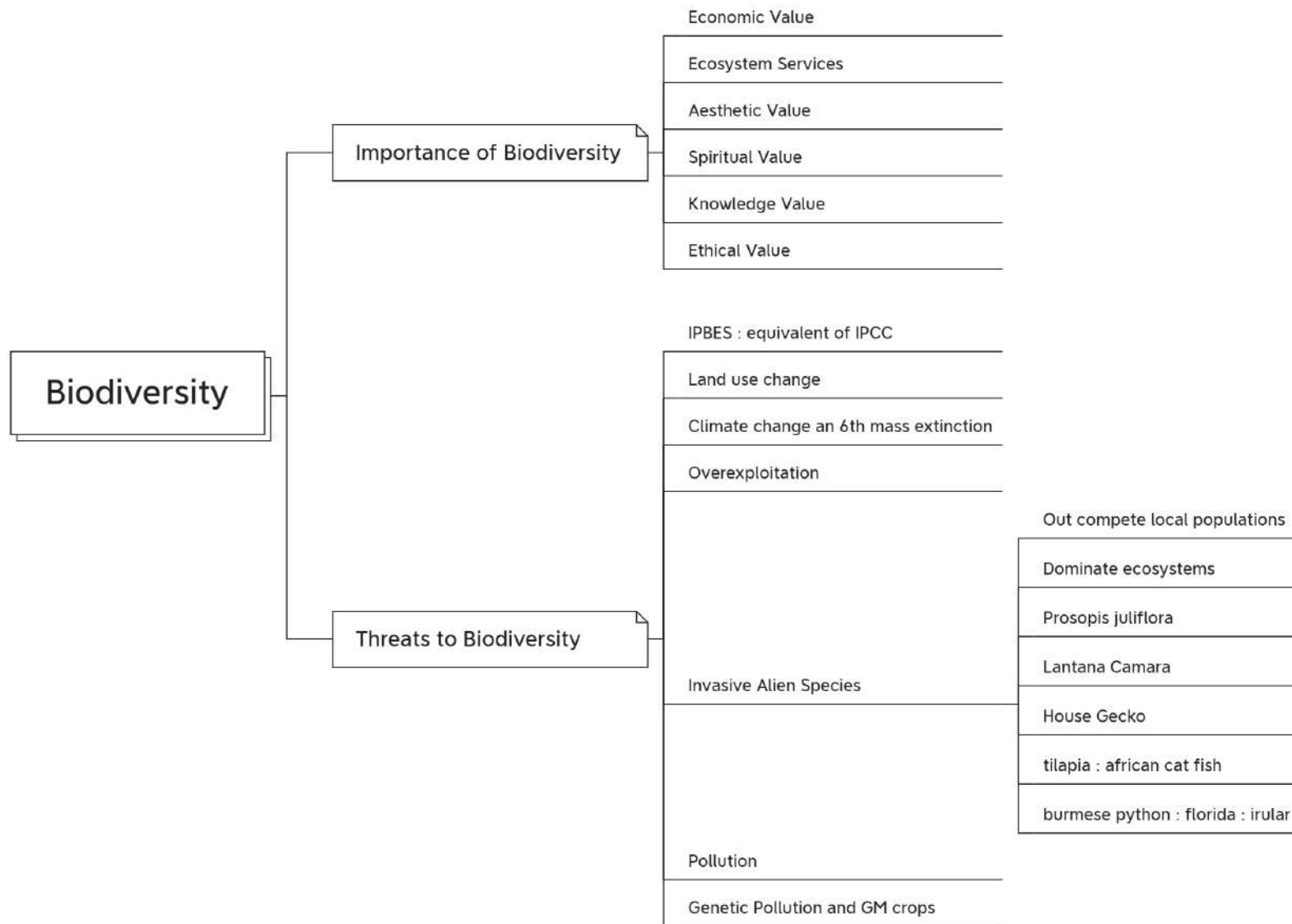
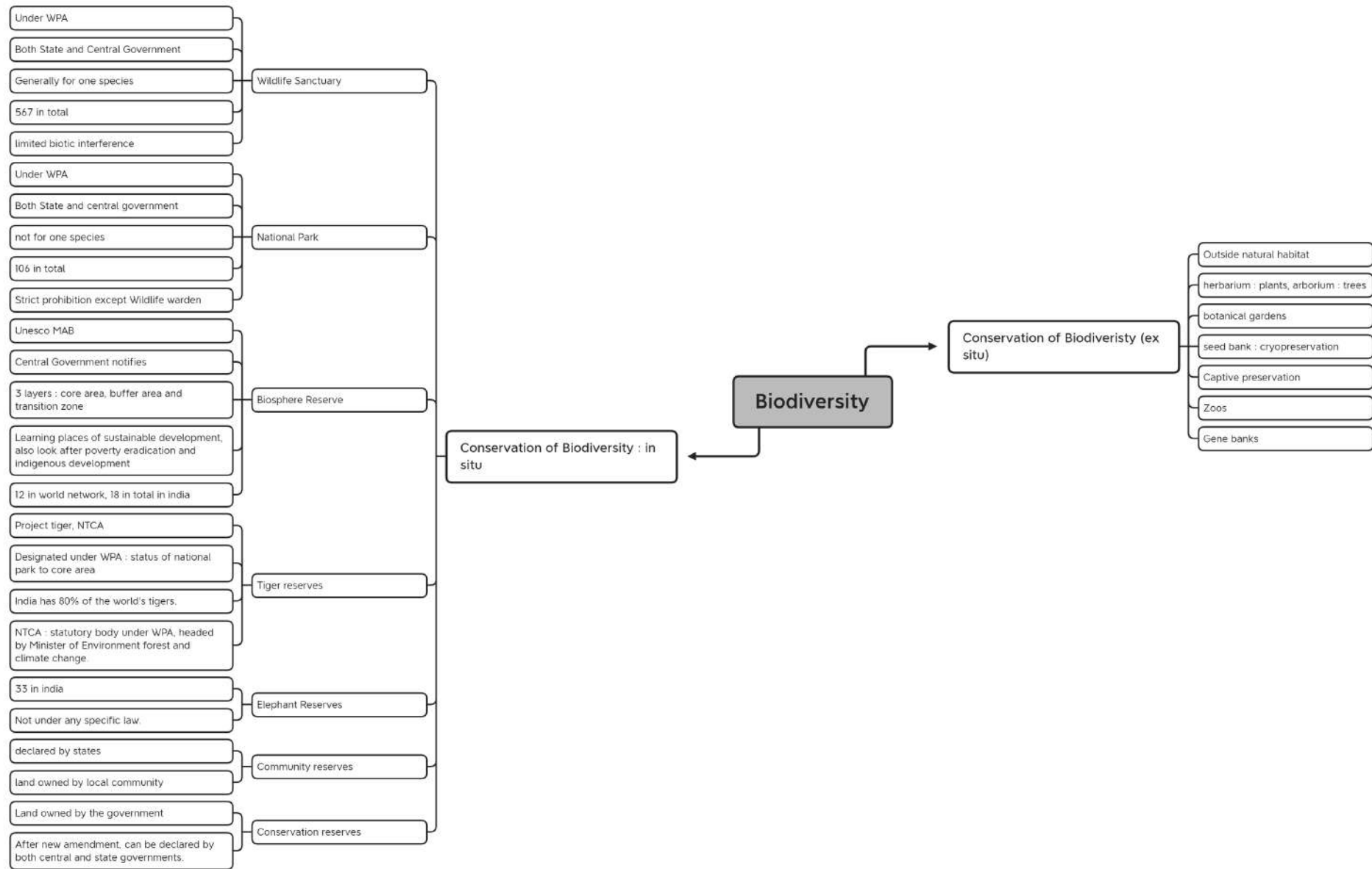


# Biodiversity

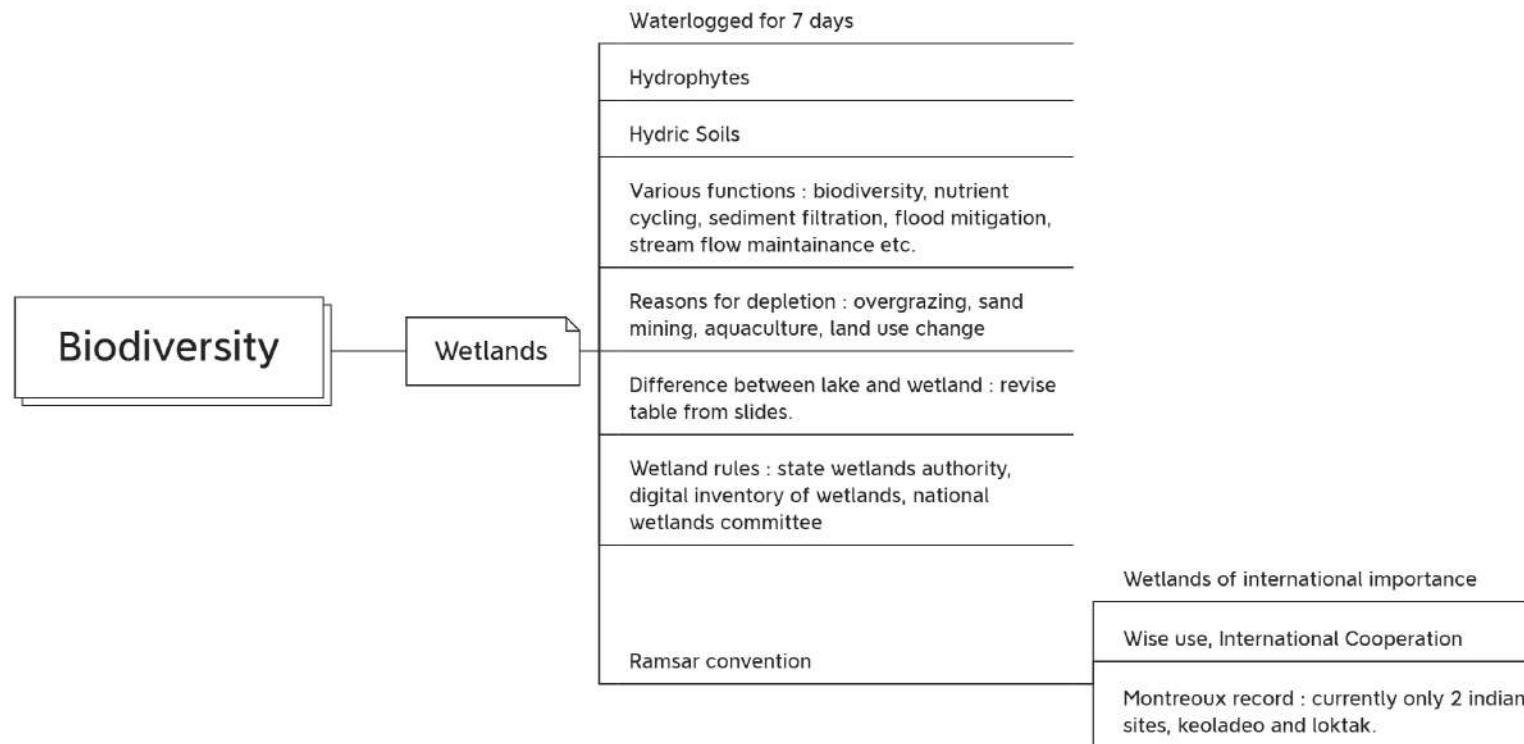


# Biodiversity





# Biodiversity





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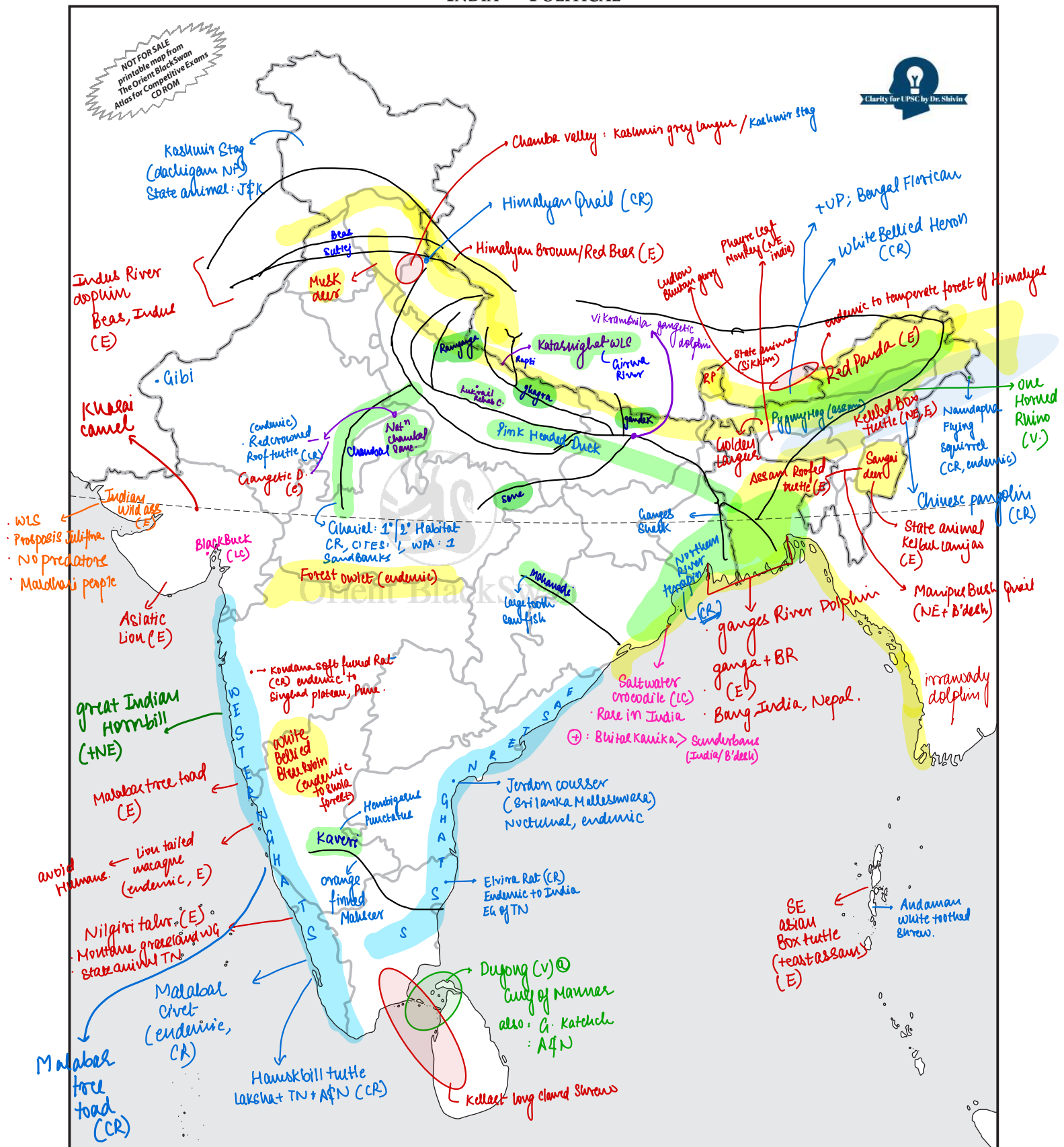
## **Environment Course**

*One-Stop Solution for  
Environment **in UPSC IAS***

**BY DR. SHIVIN CHAUDHARY**



## INDIA — POLITICAL



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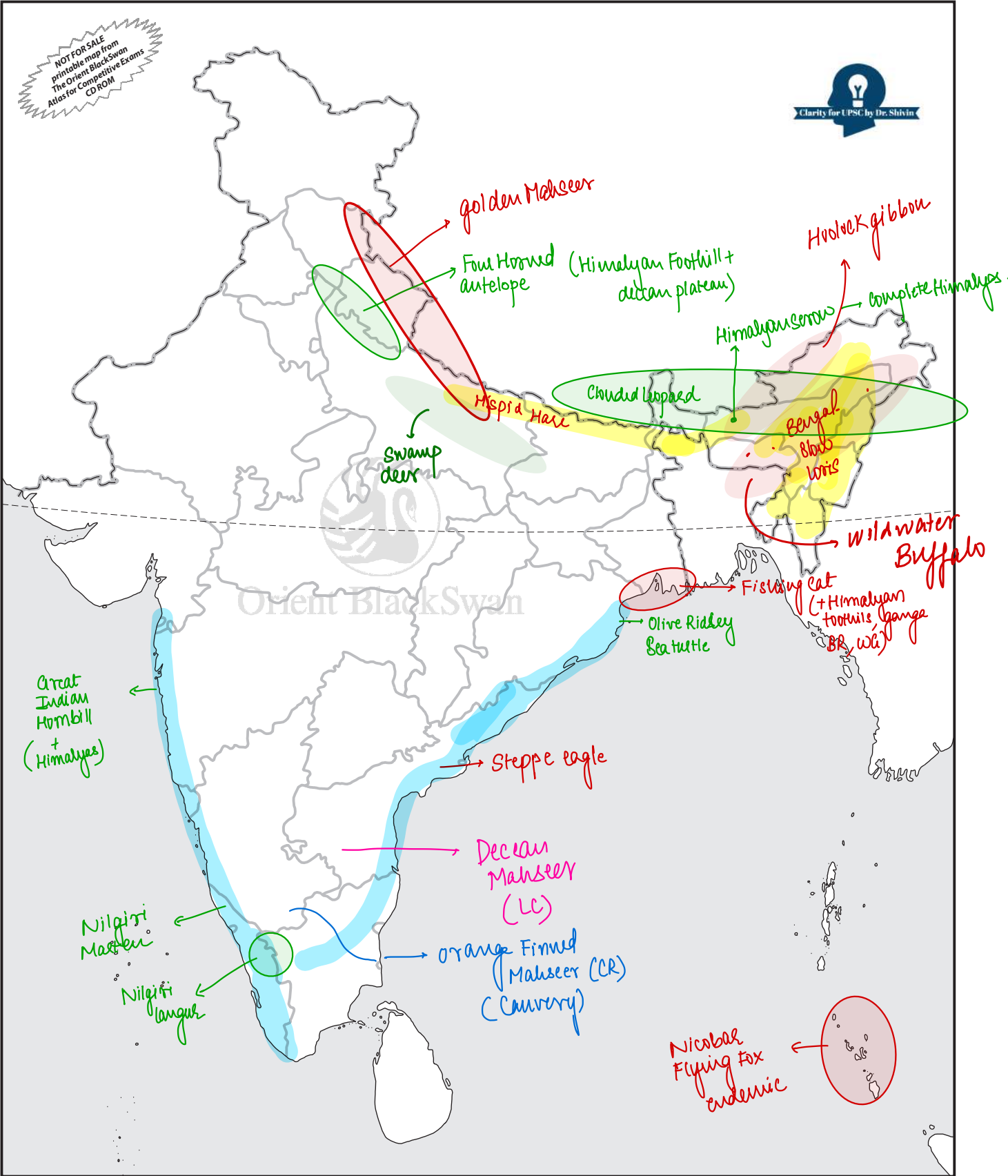
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Indian pangolin:  $\times$   $\begin{cases} NE \\ \text{aid} \\ \text{Himalayas} \end{cases}$   $\odot$   $\begin{cases} \text{Bang} \\ \text{PK} \\ \text{Nepal, SL} \end{cases}$   $-(E)$

Muggers: (V), FW species, various lakes, simlialp.

INDIA — POLITICAL



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$$A^2 = B^2 + C^2$$



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Ishita Rathi

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**Rank 75**



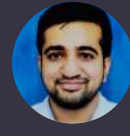
Surbhi Singla

**Rank 81**



Ishu Agrawal

**Rank 84**



Sujawal  
Jagga

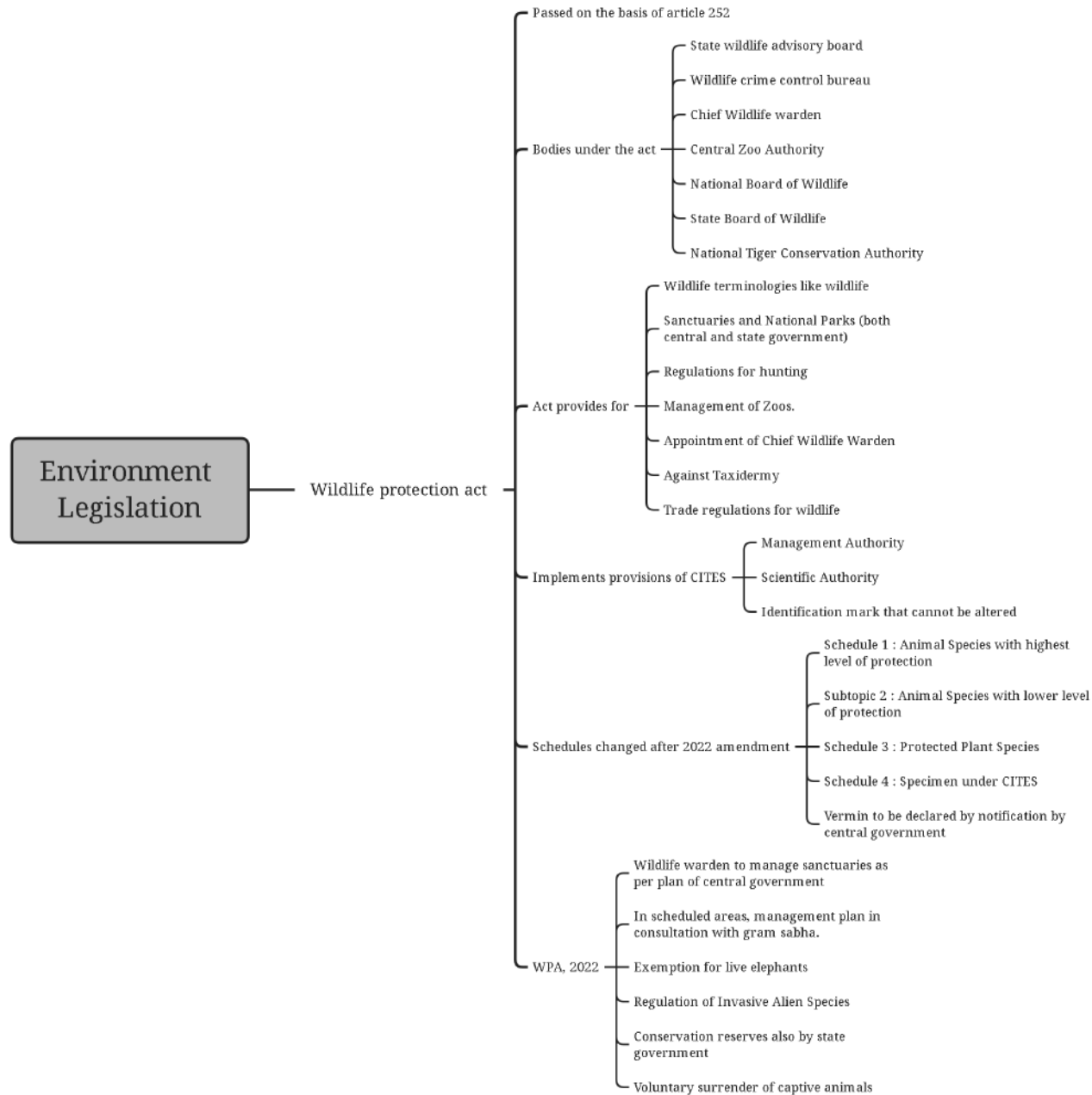


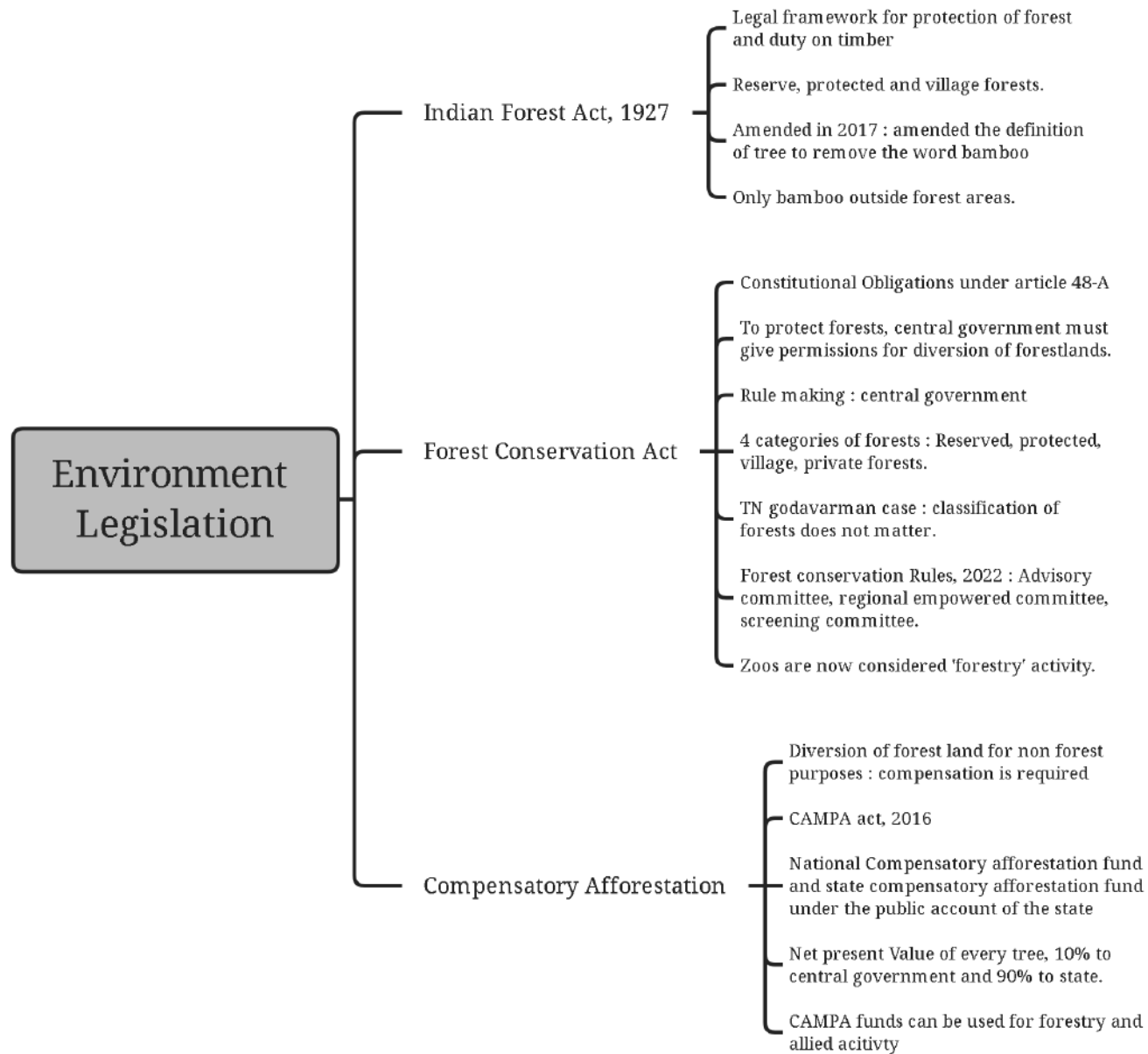
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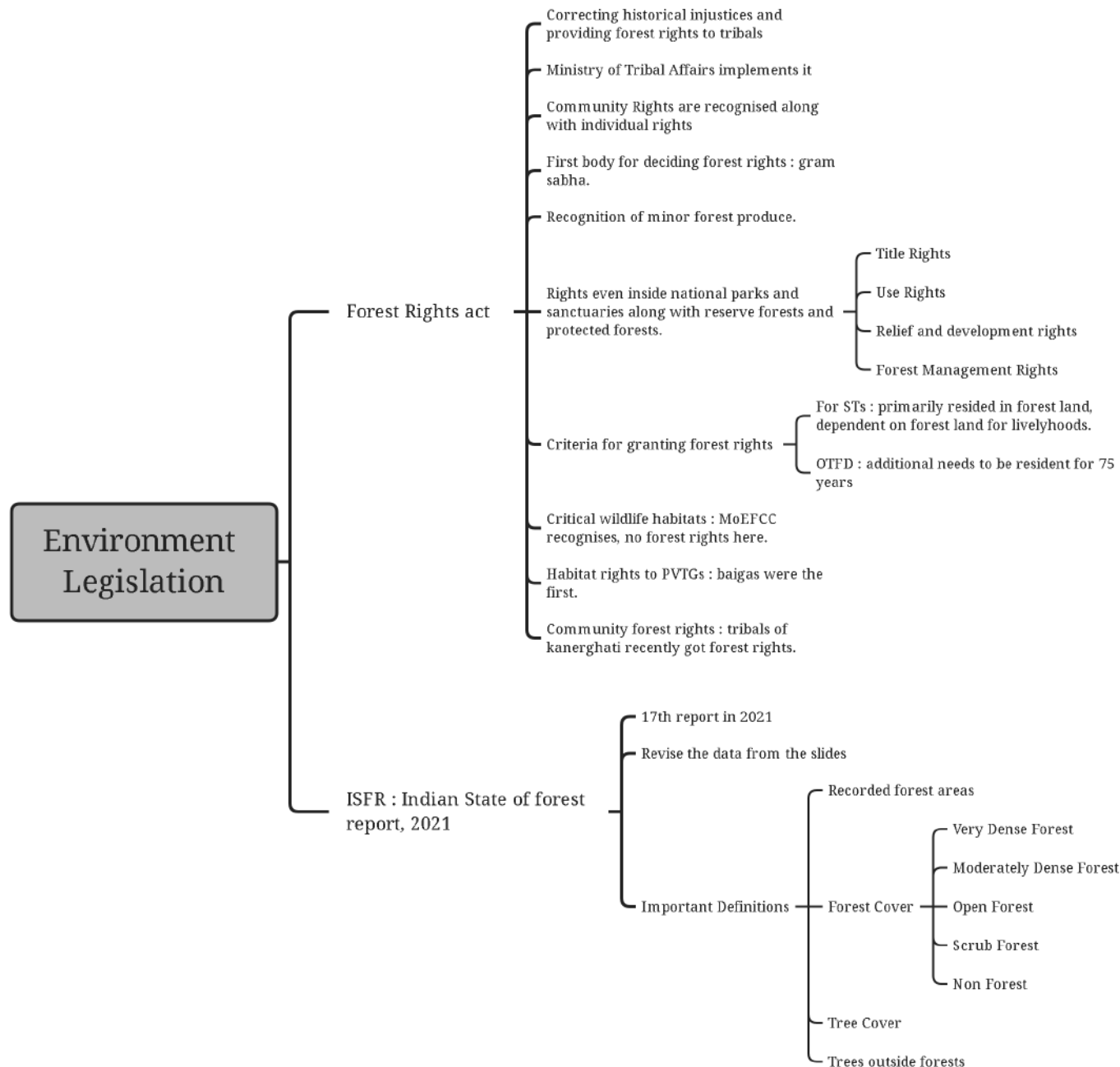
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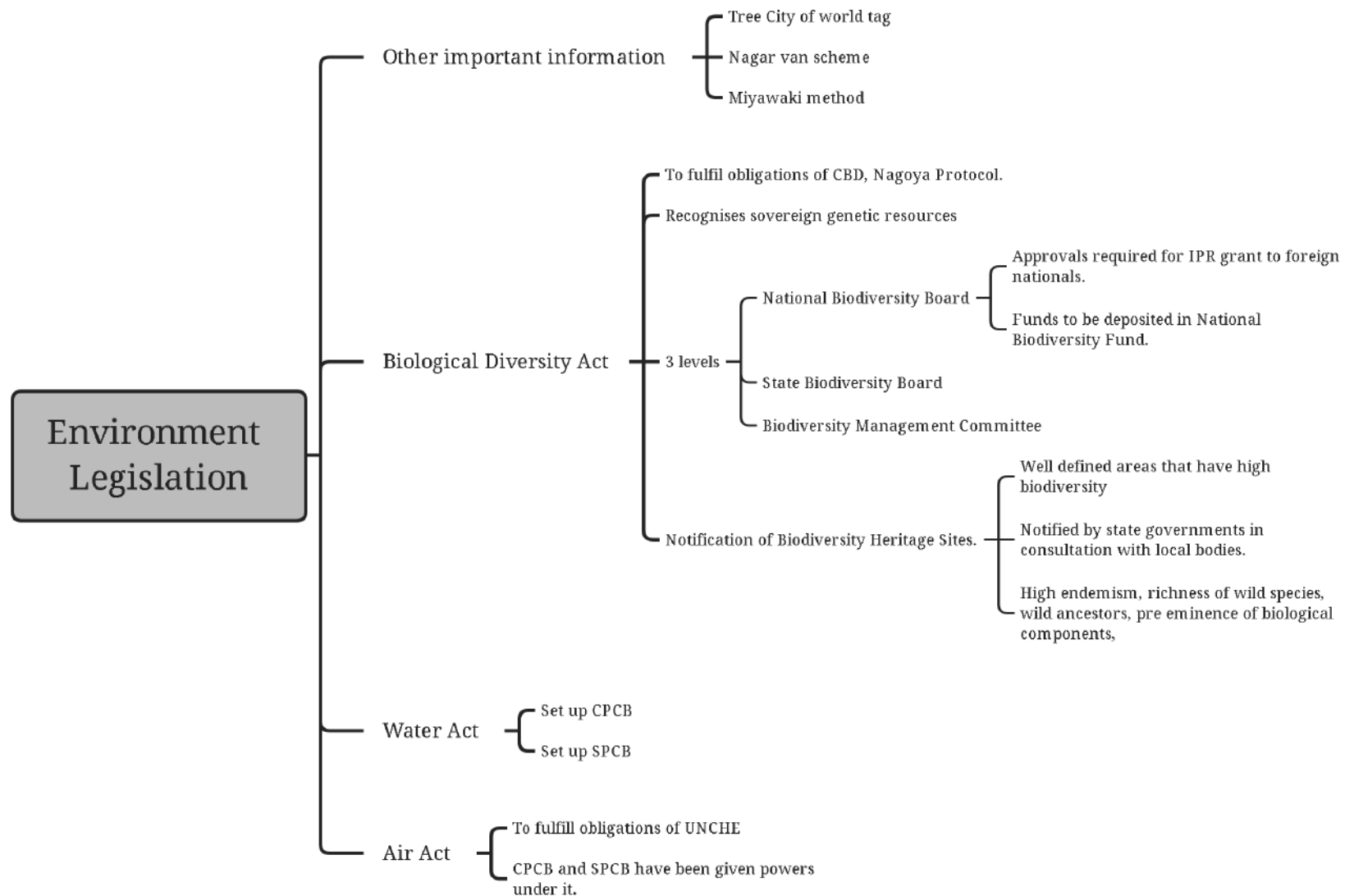
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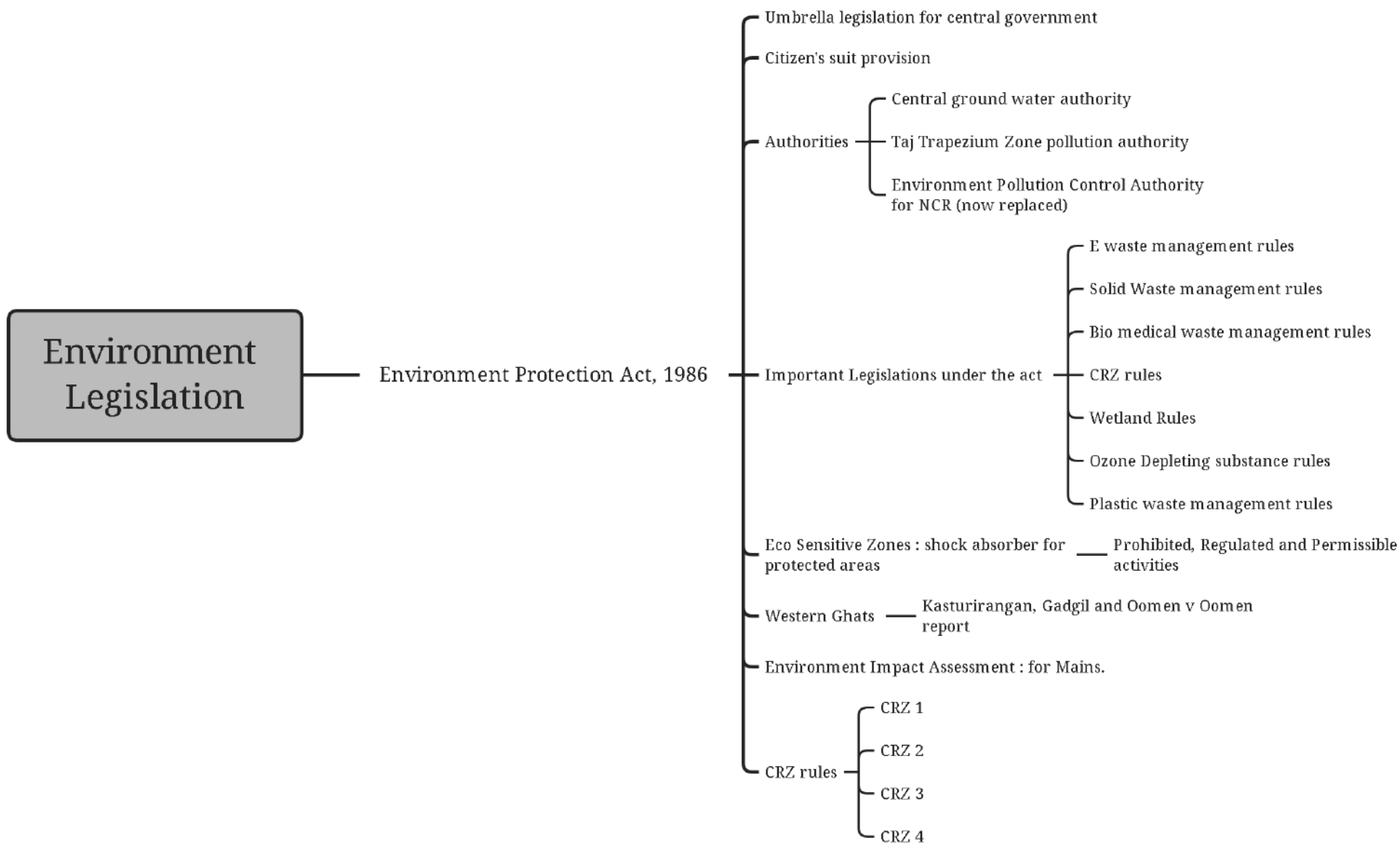


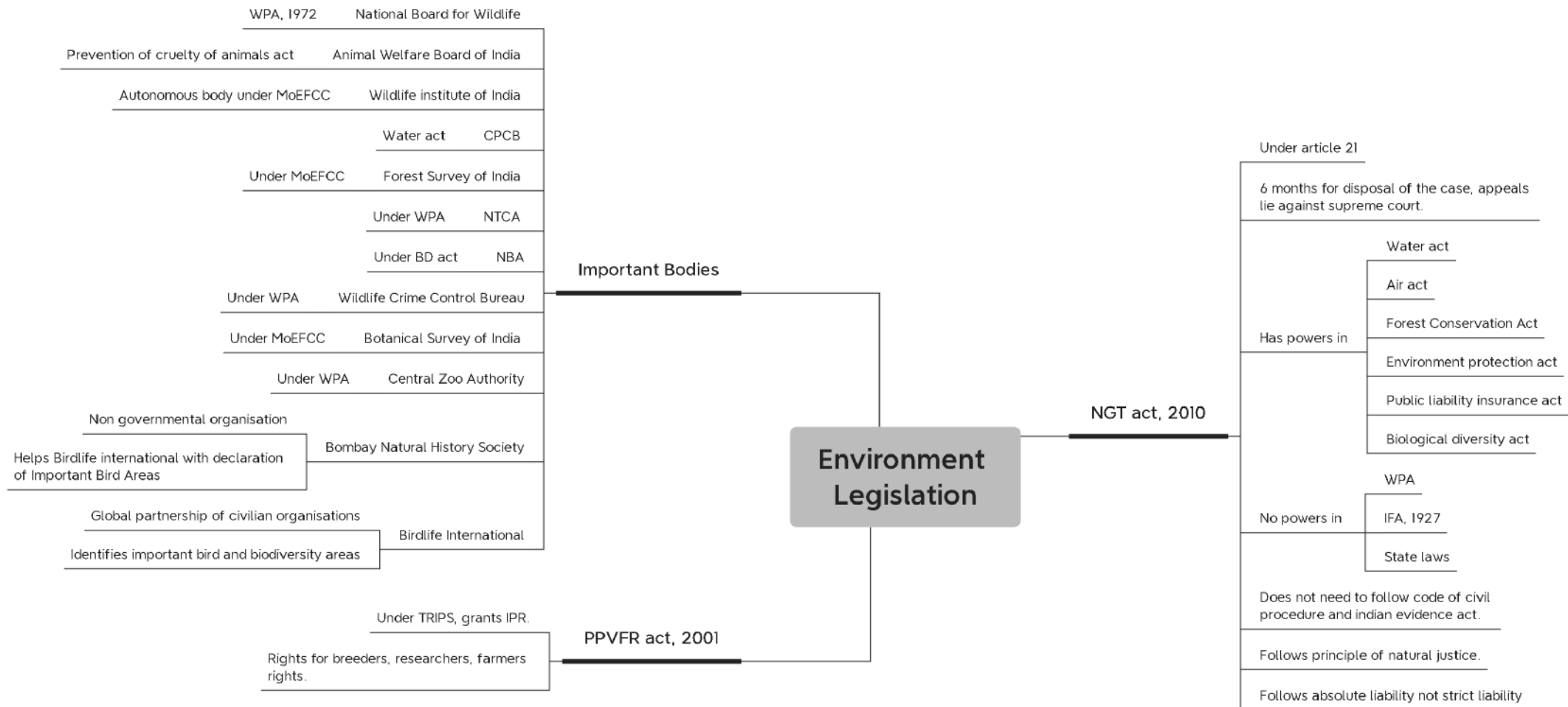


# Environment Legislation









# VALUE ADDITION NOTES

## NATIONAL PARK - 1



## **VALUE ADDITION NOTES**

### **NATIONAL PARK**

#### **PART - 1**

|                                   |    |
|-----------------------------------|----|
| 1. Sri Venkateswara National Park | 2  |
| 2. Namdapha National Park         | 3  |
| 3. Kaziranga National Park        | 4  |
| 4. Dibru-Saikhowa National Park   | 7  |
| 5. Manas National Park            | 8  |
| 6. Orang National Park            | 10 |
| 7. Dehing-Patkai National Park    | 11 |
| 8. Valmiki National Park          | 12 |
| 9. Indravati National Park        | 15 |
| 10. Gir Forest National Park      | 16 |

## 1. Sri Venkateswara National Park

- **Location** - Andhra Pradesh, India.
- **Geography** - The National Park is located in Eastern Ghats spread over the Seshachalam hills of Cuddapah district and Tirumala hills of Chittoor district. The vegetation here is a mix of dry deciduous and moist deciduous types.
- **Important Facts** - As the government of India declared the Seshachalam Hills as one of the biosphere reserves of India in 2010, this national park becomes a part of it.
- **Flora** - The vegetation in the national park consists of dry deciduous mixed forest with patches of moist deciduous forests in the valleys.
- **Fauna** - Endangered yellow-throated and Critically Endangered Oriental white-backed vulture.



## 2. Namdapha National Park

- **Location** - Arunachal Pradesh, North East India
- **Geography** - The national park is located in Changlang district of the northeastern state of Arunachal Pradesh, near the international border with Myanmar. It is located between the Dapha Bum range of the Mishmi Hills and the Patkai range.
  - It is crossed from east to west by the Noa Dihing River that originates at the Chaukan Pass, located on the Indo-Myanmar border. The land cover changes with increasing elevation from tropical evergreen forest to temperate broadleaf and mixed forest
- **Important Facts** - With more than 1,000 floral and about 1,400 faunal species, it is a biodiversity hotspot in the Eastern Himalayas. The national park harbours the northernmost lowland evergreen rainforests in the world at 27°N latitude
- **Fauna** - Namdapha flying squirrel (It is endemic to the park and critically endangered). leopard, snow leopard, tiger and clouded leopard.





### 3. Kaziranga National Park

- **Location** - Assam, India
- **Geography** - Kaziranga is one of the largest tracts of protected land in the sub-Himalayan belt, and due to the presence of highly diverse and visible species, has been described as a "biodiversity hotspot".
  - The park is located in the Indomalayan realm, and the dominant ecoregions of the region are Brahmaputra Valley semi-evergreen forests of the tropical and subtropical moist broadleaf forests biome, and the frequently-flooded Terai-Duar savanna and grasslands of the tropical and subtropical grasslands, savannas, and shrublands biome.
  - Kaziranga is also surrounded by lush green tea plantations, most of them contributing heavily to Assam's economy.

- **Important Facts** - The sanctuary, which hosts two-thirds of the world's great one-horned rhinoceroses, is a World Heritage Site. The park is home to large breeding populations of elephants, wild water buffalo, and swamp deer.
- Kaziranga is recognized as an **Important Bird Area by BirdLife International for conservation of avifaunal species**. When compared with other protected areas in India, Kaziranga has achieved notable success in wildlife conservation. Located on the edge of the Eastern Himalaya biodiversity hotspot, the park combines high species diversity and visibility.
- **Flora** - Kaziranga is a vast expanse of tall elephant grass, marshland, and dense tropical moist broadleaf forests, criss-crossed by four major rivers, including the Brahmaputra, and the park includes numerous small bodies of water. Four main types of vegetation exist in this park.
  - These are alluvial inundated grasslands, alluvial savanna woodlands, tropical moist mixed deciduous forests, and tropical semi-evergreen forests.
- **Fauna** - The park has the distinction of being home to the world's largest population of the Greater One-Horned Rhinoceros, wild Asiatic water buffalo and eastern swamp deer.
  - Significant populations of large herbivores include Indian elephants, gaur and sambar. Small herbivores include the Indian muntjac, wild boar, and hog deer. Kaziranga has the largest population of the Wild water buffalo anywhere accounting for about 57% of the world population.

- The One-Horned rhinoceros, Royal Bengal Tiger, Asian elephant, wild water buffalo and swamp deer are collectively known as the '**Big Five**' of Kaziranga.

**Kaziranga was declared a Tiger Reserve in 2006.** Small mammals include the rare hispid hare, Indian gray mongoose, small Indian mongooses, large Indian civet, small Indian civets, Bengal fox, golden jackal, sloth bear, Chinese pangolin, Indian pangolins, hog badger, Chinese ferret badgers, and particoloured flying squirrel. Kaziranga's rivers are also home to the endangered Ganges dolphin.



#### 4. Dibru-Saikhowa National Park

- **Location** - Assam, India
- **Geography** - The park is bounded by the Brahmaputra and Lohit Rivers in the north and Dibru river in the south. It mainly consists of moist mixed semi-evergreen forests, moist mixed deciduous forests, canebrakes and grasslands.
- **Important Facts** - It was designated a Biosphere Reserve in July 1997.
- **Flora** - The forest of Dibru-Saikhowa consists of semi-evergreen forests, deciduous, littoral and swamp forests and patches of wet evergreen forests
- **Fauna** - Species include Bengal tiger, Indian leopard, clouded leopard, jungle cat, sloth bear, dhole, small Indian civet, Malayan giant squirrel, Chinese pangolin, Gangetic dolphin, slow loris, pig tailed macaque,



## 5. Manas National Park

- **Location** - Assam, India
- **Geography** - Manas is located in the foothills of the Eastern Himalaya and is densely forested. The Manas river flows through the west of the park and is the main river within it. It is a major tributary of Brahmaputra river and splits into two separate rivers, the Bwrsi and Bholkaduba as it reaches the plains.
  - The Manas river also serves as an international border dividing India and Bhutan. The bedrock of the savanna area in the north of the park is made up of limestone and sandstone, whereas the grasslands in the south of the park stand on deep deposits of fine alluvium.
- **Important Facts** - Manas National Park or Manas Wildlife Sanctuary is a UNESCO Natural World Heritage site, a Project Tiger reserve, an elephant reserve and a biosphere reserve in Assam, India. Located in the Himalayan foothills, it is contiguous with the Royal Manas National Park in Bhutan.
- **Flora** - The monsoon forests of Manas lie in the Brahmaputra Valley semi-evergreen forests ecoregion. The combination of Sub-Himalayan Bhabar Terai formation with riverine succession leading up to the Himalayan subtropical broadleaf forests makes it one of the richest biodiversity areas in the world.
  - The main vegetation types are:

- Sub-Himalayan Light Alluvial Semi-Evergreen forests in the northern parts.
  - East Himalayan mixed Moist and Dry Deciduous forests (the most common type).
  - Low Alluvial Savanna Woodland, and
  - Assam Valley Semi-Evergreen Alluvial Grasslands which cover almost 50% of the park.
- **Fauna** - The park is known for its rare and endangered endemic wildlife such as the Assam roofed turtle, hispid hare, golden langur and pygmy hog. Manas is famous for its population of the wild water buffalo.
    - The park is well known for species of rare and endangered wildlife that are not found anywhere else in the world like the Assam roofed turtle, hispid hare, golden langur and pygmy hog.



## 6. Orang National Park

- **Location** - Assam, India
- **Geography** - The park is formed of alluvial flood plains of the many rivers and is an integral part of the Indo-Burma biodiversity hotspot.
  - The total area of the park has been categorized into:
    - Eastern Himalayan Moist Deciduous Forest; Eastern Seasonal Swamp Forest, Eastern Wet Alluvial Grassland, Savannah Grassland, Degraded Grassland, Water body, Moist Sandy area and Dry Sandy area.
- **Important Facts** - It is located on the northern bank of the Brahmaputra River in the Darrang and Sonitpur districts of Assam.
- **Fauna** - It has a rich flora and fauna, including great Indian rhinoceros, pygmy hog, Asian elephant, wild water buffalo and Bengal tiger.



## 7. Dehing-Patkai National Park

- **Location** - Assam, India
- **Geography** - It is located in the Dehing patkai landscape which is a dipterocarp-dominated lowland rainforest.
- **Important Facts** - The Dehing Patkai forms the largest stretch of lowland rainforests in India. The Dehing Patkai Wildlife Sanctuary was declared as Dehing-Patkai Elephant Reserve under Project Elephant.
- **Flora** - Being a completely virgin rainforest, this sanctuary is very rich in biodiversity. It is an ideal habitat for non-human primates.
- **Fauna** - The most common mammal species of this sanctuary are hoolock gibbon, slow loris, Assamese macaque, stump-tailed macaque, capped langur, Asian elephant, Bengal tiger, Indian leopard, gaur, Chinese pangolin, Himalayan black bear, Red giant flying squirrel, leopard cat, clouded leopard, porcupine, crab eating mongoose, sambar, sun bear, binturong, barking deer, Asian golden cat and marbled cat



## 8. Valmiki National Park

- **Location** - Bihar, India
- **Geography** - The landscape of VTR encompasses foothills ranges of Himalayan Sivaliks with mosaic of the cliffs, ridges, gorges, hills, streams and valleys; dense forests, open woodlands, grasslands, swamps and riverine fringe.
  - Situated in Gangetic plains bio-geographic zone of the Country, the forest has combination of bhabar and terai tracts. Boulder and pebble deposits by the Himalayan rivers in foothills characterized the Bhabar tract while the finer sediment deposits feature Terai lands.
- **Important Facts** - located at the India-Nepal border in the West Champaran district of Bihar, India on the bank of river Gandak. It is the only National park in Bihar.
- **Fauna** - The wildlife found in the forest of VTR are the Bengal tiger, Indian rhinoceros, Asiatic black bear, Indian sloth bear, otter, Indian leopard, wild dog, wild water buffalo, and wild boar.
  - There are several species of deer, including barking deer, spotted deer, hog deer, Sambar, blue bull. Apart from that the striped hyena, leopard cat, fishing cat, langur, monkey, flying squirrel, clouded leopard, Indian gaur, mongoose can also be sighted.



## 9. Indravati National Park

- **Location** - Chhattisgarh, India
- **Important Facts** - It derives its name from the nearby Indravati River. It is home to one of the last populations of rare wild buffalo.
- **Flora** - The vegetation of the Indravati National Park is mainly of the tropical moist and dry deciduous type with predominance of bamboo, sal and teak.
  - There are also rich patches of luxuriant grassland providing plenty of food to large herbivores such as wild water buffalos, chital, barking deer, nilgai, and gaurs. The most common trees in the park are teak, lendia, salai, mahua, tendu, semal, haldu, ber and jamun.
- **Fauna** - Indravati National Park has one of the last populations of the endangered wild Asian buffalo.

- Reported from the area are gaur (Indian bison), nilgai, blackbuck, chausingha (four-horned antelope), sambar, chital, Indian muntjac, Indian spotted chevrotain and wild boar.
- Large predators are represented by tigers, leopards, sloth bears, dholes (wild dogs) and striped hyenas.



## 10. Gir Forest National Park

- **Location** - Gujarat, India
- **Geography** - The seven major perennial rivers of the Gir region are Hiran, Shetrunji, Datardi, Shingoda, Machhundri, Godavari and Raval.

- The four reservoirs of the area are at **four dams, one each on Hiran, Machhundri, Raval and Shingoda rivers, including the biggest reservoir in the area, the Kamleshwar Dam, dubbed 'the lifeline of Gir'.**
- **Important Facts** - Asiatic Lions are only Found here
- **Flora** - The Gir forest area is a very dry teak forest. Teak occurs mixed with dry deciduous species. It is part of the Khathiar-Gir dry deciduous forests ecoregion
- **Fauna** - Mainly comprises the Asiatic lion, Indian leopard, jungle cat, striped hyena, golden jackal, Bengal fox, Indian gray mongoose and ruddy mongoose, and honey badger. Asiatic wildcat and rusty-spotted cats occur, but are rarely seen.



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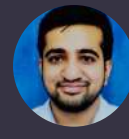
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Jagga



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# VALUE ADDITION NOTES

**NATIONAL PARK - 2**



## **VALUE ADDITION NOTES**

### **NATIONAL PARK**

#### **PART - 2**

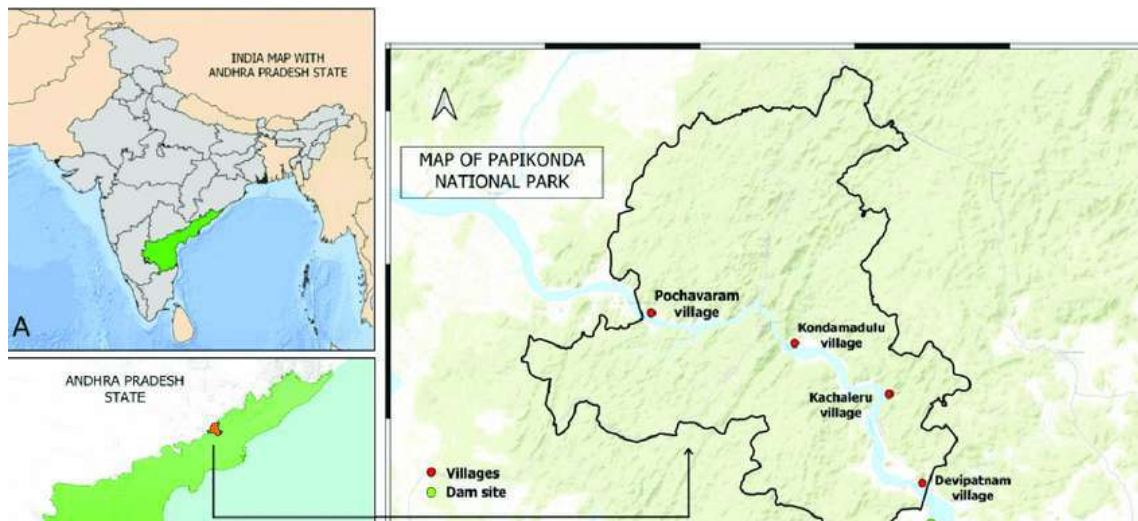
|                                           |    |
|-------------------------------------------|----|
| 1. Papikonda Wildlife Sanctuary           | 2  |
| 2. Sri Lankamalleswara Wildlife Sanctuary | 4  |
| 3. Pakhui Tiger Reserve                   | 4  |
| 4. Dibang Wildlife Sanctuary              | 6  |
| 5. Satkosia Tiger Reserve                 | 7  |
| 6. Parambikulam Tiger Reserve             | 8  |
| 7. Sathyamangalam Wildlife Sanctuary      | 9  |
| 8. Marine National Park                   | 11 |
| 9. Pin Valley National Park               | 12 |
| 10. Great Himalayan National Park         | 13 |
| 11. Dachigam National Park                | 14 |
| 12. Hemis National Park                   | 15 |
| 13. Salim Ali National Park               | 17 |
| 14. Nagarhole National Park               | 17 |
| 15. Bandipur National Park                | 18 |
| 16. Bannerghatta National Park            | 19 |
| 17. Silent Valley National Park           | 20 |
| 18. Periyar National Park                 | 22 |

|                               |    |
|-------------------------------|----|
| 19. Bandhavgarh National Park | 23 |
| 20. Panna National Park       | 24 |
| 21. Pench National Park       | 25 |



## 1. Papikonda Wildlife Sanctuary

- **Location** - Andhra Pradesh, India
- **Geography** - No part of Papikonda remains outside East and West Godavari districts after 2014 and after the construction of Polavaram Dam.
- **Important Facts** - It is an Important Bird and Biodiversity Area and home to some endangered species of flora and fauna. Godavari River flows through the park
- **Flora** - The vegetation of Papikonda national park contains species of moist deciduous and dry deciduous forests
- **Fauna** - Mammals - Bengal tiger, Indian leopard, rusty-spotted cat, jungle cat, leopard cat, sloth bear, small Indian civet, Asian palm civet, wild boar and honey badger.
- **Birds** - Black-bellied tern, Pale-capped pigeon, Yellow-throated bulbul, Oriental darter, Pallid harrier, Great Thick-knee, River lapwing, River tern, Malabar pied hornbill, Alexandrine parakeet.



## 2. Sri Lankamalleswara Wildlife Sanctuary

- **Location** - Andhra Pradesh, India
- **Important Facts** - It is the only habitat in the world which provides home for the Jerdon's courser, a highly endangered bird species

- **Flora** - It has dry deciduous mixed thorn forests with deep gorges and steep slopes. Red Sanders, an endemic species, can be found here

- **Fauna** - Its fauna includes the panther, sloth bear, cheetal, sambar, chowsingha, chinkara, nilgai, wild boar, fox and the Jerdon's courser.



### 3. Pakhui Tiger Reserve

- **Location** - Arunachal Pradesh, India
- **Geography** - The terrain is rugged with mountainous ranges in the north and narrow plains and sloping hill valleys in the south. The sanctuary slopes southwards towards the river valley of the Brahmaputra River.
  - Pakke Tiger Reserve has a subtropical climate with cold weather from November to March. It receives rainfall predominantly from the south-west monsoon in May to September and north-east monsoon from November to April

- **Important Facts** - This Tiger Reserve has won India Biodiversity Award 2016 in the category of 'Conservation of threatened species' for its Hornbill Nest Adoption Programme.
- **Flora** - The habitat types are lowland semi-evergreen, evergreen forest and Eastern Himalayan broadleaf forests. The general vegetation type of the entire tract is classified as Assam Valley tropical semi-evergreen forest. The forests are multi-storeyed and rich in epiphytic flora and woody lianas. The vegetation is dense, with a high diversity and density of woody lianas and climbers. The forest types include tropical semi-evergreen forests
- **Fauna** - The Assam roofed turtle, a highly endangered species, is commonly sighted. Tiger, leopard, clouded leopard, jungle cat, wild dog, jackal, Himalayan black bear, binturong, elephant, gaur, sambar, hog deer, barking deer, wild boar, yellow-throated marten, Malayan giant squirrel, flying squirrel, squirrel, capped langur, rhesus macaque, Assamese macaque, gaur are also found here.



#### 4. Dibang Wildlife Sanctuary

- **Location** - Arunachal Pradesh, India
- **Geography** - Dibang Wildlife Sanctuary is located fully or partly within Dihang-Dibang Biosphere Reserve
- **Flora** - Dibang wildlife sanctuary nestles amidst the dramatic mountains, snow covered peaks, deep gorges, dense forests and shimmering rivers. It is a popular biodiversity hotspot.
- **Fauna** - Rare mammals such as Mishmi takin, red goral, musk deer (at least two species), red panda, Asiatic black bear, occasional tiger and Gongshan muntjac occur while among birds there are the rare Sclater's monal and Blyth's tragopan



## 5. Satkosia Tiger Reserve

- **Location** - Odisha, India
- **Geography** - It is located where the Mahanadi River passes through a 22 km long gorge in the Eastern Ghats mountains.
- **Flora** - The tiger reserve is located in the Eastern Highlands moist deciduous forests ecoregion. The major plant communities are mixed deciduous forests including Sal (*Shorea robusta*), and riverine forest.
- **Fauna** - Mammals found include the leopard, indian wild dog or the (dhole), wild boars , striped hyena, sloth bear, leopard cat and the jungle cat. indian python, indian giant squirrel



## 6. Parambikulam Tiger Reserve

- **Location** - Kerala, India
- **Geography** - It is adjacent to the Anamalai Wildlife Sanctuary to the east in Tamil Nadu.
- **Important Facts** - Parambikulam is one of the premier Tiger Reserves of India and is endowed by nature in terms of species, habitat and ecosystem diversity, characterized by functional human-ecological affinities.
  - It is in the **Sungam range of hills between the Anaimalai Hills and Nelliampathy Hills**. The sanctuary is the home of four different tribes of indigenous peoples including the Kadar, Malasar, Muduvar and Mala Malasar settled in six colonies.
- **Flora** - The sanctuary has a variety of trees mainly teak, neem, sandalwood and rosewood. Even the oldest ever teak tree, Kannimara Teak exists here.
- **Fauna** - The sanctuary has a rich diversity of animal life including: Mammals, Amphibians, Birds, Reptiles, Fish, Insects and Butterflies. This sanctuary is also referred to as the "**state capital for the massive gaur**" by wildlife experts.



## 7. Sathyamangalam Wildlife Sanctuary

- **Location** - Tamil Nadu, India
- **Geography** - Sathyamangalam Tiger Reserve, the **Gateway to Eastern Ghats**, is a significant ecosystem and a wildlife corridor in the Nilgiri Biosphere Reserve between the Western Ghats and the rest of the Eastern Ghats and a genetic link between the five other protected areas which it adjoins, including the Biligiriranga Swamy Temple Wildlife Sanctuary, Sigur Plateau, Mudumalai National Park, Bandipur National Park and the Cauvery Wildlife Sanctuary.
- **Flora** - The Sathyamangalam forest is mostly tropical dry forest, part of the South Deccan Plateau dry deciduous forests ecoregion. There are five distinct forest types:

tropical evergreen (Shola), semi-evergreen, mixed-deciduous, dry deciduous and thorn forests.

- **Fauna** - The Sathyamangalam forests **link the Eastern Ghats and Western Ghats** allowing gene flow between diverse fauna populations of the two ecoregions.
  - Bengal tiger, Indian elephants, gaurs, and leopards, spotted deer, blackbucks, sambar deer, barking deer and four-horned antelopes, wild boars, and sloth bears. Herds of the famous feral buffaloes can also be spotted in places near the Moyar river.
  - Many bird species including treepeeps, bulbuls, babblers, mynahs and crows were noted.



## 8. Marine National Park

- **Location** - Gujarat, India
- **Geography** - Marine National Park in the Gulf of Kutch is situated on the southern shore of the Gulf of Kutch.
- **Important Facts** - There are 42 islands on the Jamnagar coast in the Marine National Park, most of them surrounded by reefs.
- **Fauna** - The fauna found here include: species of sponges, Coral species including hard coral , soft coral and species of birds.



## 9. Pin Valley National Park

- **Location** - Himachal Pradesh, India
- **Geography** - The park is located in the desert habitat of the Spiti Valley, within the Cold Desert Biosphere Reserve, in the Himalayas region. Spreading south of Dhankar Gompa near the Tibetan border, the park marks the border between the formerly separate districts of Lahaul and Spiti.
- **Important Facts** - Steeped in historical and present day Buddhist Tibetan culture, the area has many Tibetan Buddhist influences, evident architecturally in monasteries and stupas, and in the daily living of its residents and lamas.
- **Flora** - Because of the park's high altitude and extreme temperatures, the vegetation density is sparse, consisting mostly of alpine trees and groves of Himalayan cedar. Twenty-two rare and endangered medicinal plant species, have been discovered in and around Pin Valley National Park,
- **Fauna - Snow Leopards and Siberian Ibex**



## 10. Great Himalayan National Park

- **Location** - Himachal Pradesh, India
- **Geography** - The GHNP is at the junction of world's two major biogeographic realms: the Indomalayan realm to the south and the Palearctic realm to the north. the park has a diversity of zones with their representative flora and fauna, such as alpine, glacial, temperate, and subtropical forests.
- **Important Facts** - In June 2014, the Great Himalayan National Park was added to the **UNESCO list of World Heritage Sites.**
- **Flora** - pines and spruces, horse chestnuts
- **Fauna** - blue sheep, snow leopard, Himalayan brown bear, Himalayan tahr, and musk deer.



## 11. Dachigam National Park

- **Location** - Srinagar, Jammu and Kashmir, India
- **Geography** - Dachigam National park is located in the Zabarwan Range of the western Himalayas. The terrain ranges from gently sloping grasslands to sharp rocky outcrops and cliffs
- **Flora** - The mountain sides below the tree line are wooded. Most of this coniferous forest consists of broadleaf species. Interspersed between these are alpine pastures, meadows, waterfalls and scrub vegetation with deep gullies, locally known as Nars, running down the mountain face. Most of the grasslands and meadows, except in the harsh winters, are covered with coloured flowers.
- **Fauna** - The main animal species that Dachigam is known for is the **hangul, or the Kashmir stag.**



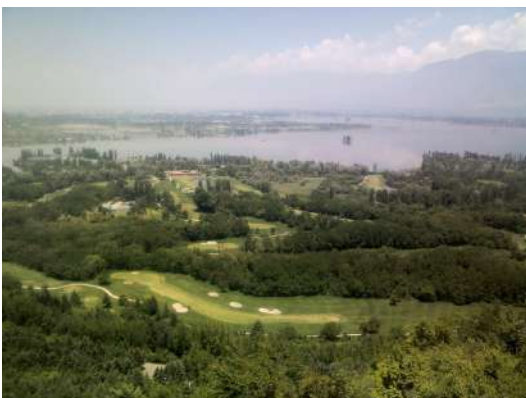
## 12. Hemis National Park

- **Location** - Ladakh, India
- **Geography** - The park is bounded on the north by the banks of the Indus River, and includes the catchments of Markha, Sumdah and Rumbak, and parts of the Zaskar Range. The park lies within the Karakoram-West Tibetan Plateau alpine steppe ecoregion, and contains pine forests, alpine shrublands and meadows, and alpine tundra vegetation.
- **Important Facts** - It is the **only national park in India that is north of the Himalayas**, the largest notified protected area in India (largest National park) and is the second largest contiguous protected area, after the Nanda Devi Biosphere Reserve.
- **Flora** - This region is in the rain shadow of the Himalayas, and does not receive much precipitation. Hence, dry forests of juniper, *Populus* - *Salix* forests, subalpine dry birch - fir are present at lower altitudes. Alpine and steppe trees are predominantly found here. These trees and shrubs are spread across the valley bottoms.
- **Fauna** - snow leopards, Great Tibetan Sheep, Blue Sheep, Ladakhi Urial, Tibetan wolf, the Eurasian brown bear



### 13. Salim Ali National Park

- **Location** - Srinagar, Jammu and Kashmir, India
- **Important Facts** - The park was converted into the **Royal Springs Golf Course**
- **Fauna** - The park featured a wildlife species such as the hangul, musk deer, Himalayan black bear, leopard, Himalayan serow and 70 species of birds, including the paradise flycatcher, Himalayan monal, and **Himalayan snowcock**



#### 14. Nagarhole National Park

- **Location** - Karnataka, India
- **Geography** - The park ranges the foothills of the Western Ghats spreading down the Brahmagiri hills and south towards Kerala state.
- **Flora** - The vegetation here consists mainly of North Western Ghats moist deciduous forests with *teak* and *rosewood* predominating in the southern parts. There is Central Deccan Plateau dry deciduous forests with Pala indigo and thorny wattle towards the east. There are some sub-montane valley swamp forests with several species of the *Eugenia* genus.
- **Fauna** - Bengal tiger, Indian leopard, **dhole**, sloth bear and the striped hyena. Recognised as an **Important Bird Area** the park has over 270 species of birds including the '**critically endangered**' **Oriental white-backed vulture**



## 15. Bandipur National Park

- **Location** - Karnataka, India
- **Geography** - It is part of the Nilgiri Biosphere Reserve. The park has a variety of biomes including dry deciduous forests, moist deciduous forests and shrublands. The wide range of habitats help support a diverse range of organisms. The park is flanked by the Kabini river in the north and the Moyar river in the south.
- **Flora** - Bandipur supports a wide range of timber trees including: teak , rosewood, sandalwood, Indian-laurel, Indian kino tree, giant clumping bamboo, clumping bamboo and *Grewia tiliaefolia*.
- **Fauna** - Bandipur supports a good population of endangered and vulnerable species like Indian elephants, gaurs, tigers, sloth bears, muggers, Indian rock pythons, four-horned antelopes, jackals, Gray Langur and dholes



## 16. Bannerghatta National Park

- **Location** - Karnataka, India
- **Geography** - The park has a hilly terrain of granite sheets under moist deciduous forest valleys and scrubland on higher areas. Sixteen villages border the park. Most of the national park consists of artificial forest, and some animals have been introduced. The park is part of a wildlife corridor for elephants which connects the BR Hills and the Sathyamangalam forest.
- **Important Facts** - The park's rainfall is 700 mm per year. The Suvarnamukhi stream runs through the national park.
- **Flora** - Sandalwood, Tamarind, Bamboo, Neem
- **Fauna** - Royal Bengal tiger, Russell's viper, Blackbuck, Pelican



## 17. Silent Valley National Park

- **Location** - Kerala, India
- **Geography** - It is located in the Nilgiri hills, The national park is one of the last undisturbed tracts of South Western Ghats mountain rain forests and tropical moist evergreen forest in India. It is the **core of the Nilgiri Biosphere Reserve**. Silent Valley gets copious amounts of rainfall during the monsoons, but the actual amount varies within the region due to the varied topography
- **Important Facts** - The **Kunthipuzha River** drains the entire 15 km length of the park from north to south into the Bharathapuzha River. the Mudugar and Irula tribal people are indigenous to the area
- **Flora** - The valley areas of the park are in a **Tropical and subtropical moist broadleaf forests** Ecoregion. the evergreen forests begin to give way to stunted forests, called sholas, interspersed with open grassland
- **Fauna** - threatened lion-tailed macaque, Nilgiri langur, Malabar giant squirrel, Nilgiri tahr, Peshwa's bat and hairy-winged bat. Nilgiri wood-pigeon, Malabar parakeet, Malabar grey hornbill, white-bellied treepie, grey-headed bulbul, broad-tailed grassbird.



## 18. Periyar National Park

- **Location** - Kerala, India
- **Geography** - The park is a repository of rare, endemic, and endangered flora and fauna and forms the major watershed of two important rivers of Kerala: the Periyar and the Pamba. The park is located high in the Cardamom Hills and Pandalam Hills of the south Western Ghats along the border with Tamil Nadu.
- **Important Facts** - It is notable as an elephant reserve and a tiger reserve.

- **Flora** - The park is made up of tropical evergreen and moist deciduous forests, grasslands, stands of eucalyptus, and lake and river ecosystems. The forests contain deciduous and semi evergreen trees like teak, rosewoods, terminalias, sandalwoods.
  - The park is surrounded by agricultural regions, especially plantations of such crops as tea, cardamom, and coffee.
- **Fauna** - Bengal tigers, white tigers, Salim Ali's fruit bat, stripe-necked mongoose, and Nilgiri marten. Malabar grey hornbill, Nilgiri wood pigeon, blue-winged parakeet, Nilgiri flycatcher, crimson-backed sunbird, and white-bellied blue flycatcher.



## 19. Bandhavgarh National Park

- **Location** - Madhya Pradesh, India
- **Important Facts** - The density of the tiger population at Bandhavgarh (8 tigers per km<sup>2</sup>) is one of the highest known in India.

- **Flora** - The richness and tranquility of grasslands invites pairs of sarus cranes to breed in the rainy season
- **Fauna** - Tiger, Gaur, Nilgai, Indian Wolf



## 20. Panna National Park

- **Location** - Madhya Pradesh, India
- **Geography** - The National Park is situated at a point where the continuity of the Tropical and subtropical dry broadleaf forests belt, which starts from Cape Comorin in South India, is broken and beyond this the Upper Gangetic Plains moist deciduous forests of the great Indo-Gangetic Plain begins

- **Fauna** - Among the animals found here are the tiger, leopard, chital, chinkara, nilgai, sambhar and sloth bear. The park is home to more than 200 species of birds including the bar-headed goose, honey buzzard, king vulture, blossom-headed parakeet, changeable hawk-eagle and Indian vulture.



## 21. Pench National Park

- **Location** - Madhya Pradesh, India
- **Geography** - It includes Pench Tiger Reserve and derives its name from the Pench River that flows through the park from north to south dividing the park into almost equal western and eastern halves. The protected area is covered with small hills and well-stocked teak mixed forest in the southern reaches of the Satpura Ranges.

- **Flora** - The forest cover in the park area includes teak mixed with other species like saja, bijiayasal, lendia, haldu, dhaora, salai, amla, amaltas. The ground is covered with a maze of grasses, plants, bushes and saplings. Bamboo is also found at places
- **Fauna** - Bengal tiger, Indian leopard, sloth bear, indian wolf, wild dog, porcupine, monkey, jungle cat, fox, striped hyena, gaur, four-horned antelope and barking deer live in the park.
  - Birds include Crimson-breasted barbet, red-vented bulbul, racket-tailed drongo, Indian roller, magpie robin



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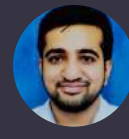
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# VALUE ADDITION NOTES

**NATIONAL PARK - 3**



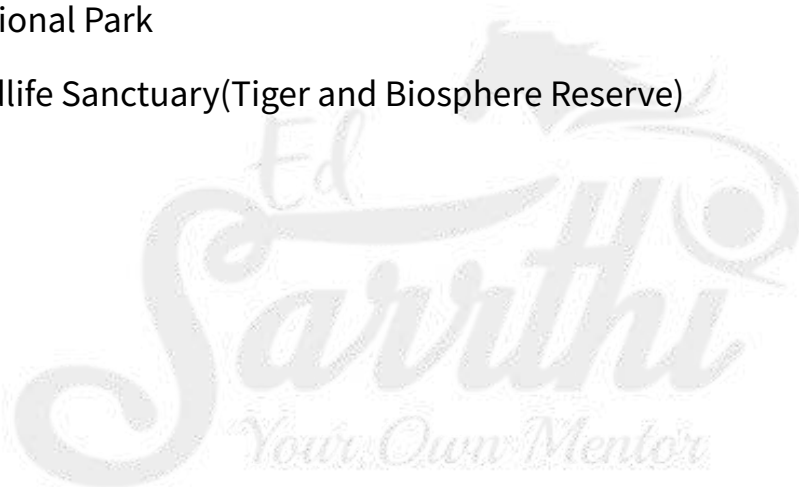
## **VALUE ADDITION NOTES**

### **NATIONAL PARK**

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## 1. Kanha National Park(Tiger Reserve)

- **Location** - Madhya Pradesh, India
- **Important Facts** - An exciting conservation effort in this national park is the reintroduction of barasingha
- **Flora** - The lowland forest is a mixture of sal and other mixed-forest trees, interspersed with meadows. The highland forests are tropical moist, dry deciduous type and of a completely different nature from bamboo on slopes.
- **Fauna** - Kanha Tiger Reserve hosts populations of tiger, leopard, wild dog, sloth bear, foxes and jackals.



## 2. Sanjay National Park

- **Location** - Madhya Pradesh and Chhattisgarh, India
- **Geography** - It is located in the Narmada Valley dry deciduous forests ecoregion.
- **Important Facts** - It is a part of the **Sanjay-Dubri Tiger Reserve**.
- **Flora** - The national park is mostly composed of **tropical forests of Sakhua trees** (sal trees).
- **Fauna** - The Bengal tiger, Indian leopard, Spotted deer, Sambar deer, wild boar, Nilgai, Chinkara, Civet, Porcupine, Monitor lizard, and 309 species of birds are found here. Among the many birds here are the Golden Hooded Oriole, Racket-tailed Drongo, Indian pitta, Rufous treepie, Lesser adjutant, Red-headed vulture, Cinereous vulture, White-rumped vulture, Egyptian vulture and Nightjar.



### 3. Satpura National Park(Tiger Reserve)

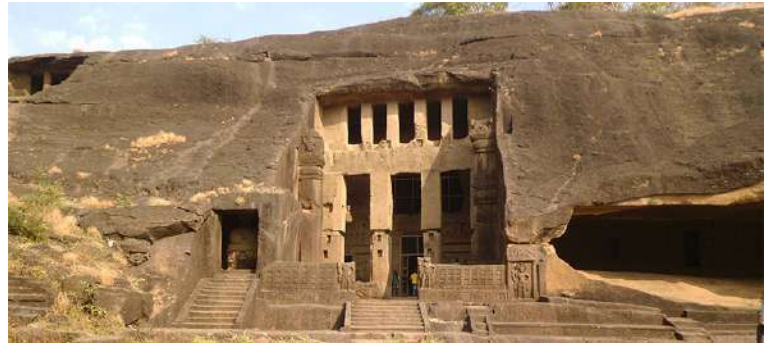
- **Location** - Madhya Pradesh, India
- **Geography** - The terrain of the national park is extremely rugged and consists of sandstone peaks, narrow gorges, ravines and dense forests.
- **Important Facts** - Its name is derived from the **Satpura Range**
- **Flora** - Saal is the most prominent flora of the National Park which is mainly found in the **Pachmarhi plateau** area of tiger reserve whereas lower plains on the northern side of the park mainly consist of teak forest on basaltic traps.
  - It has about 10% of the area which is classified as grasslands
- **Fauna** - leopard, sambar, chital, Indian muntjac, nilgai, four-horned antelope, Chinkara, wild boar, bear, black buck, fox, porcupine, flying squirrel, mouse deer, and Indian giant squirrel. Hornbills and peafowl are common birds found here.



#### 4. Sanjay Gandhi National Park

- **Location** - Mumbai, Maharashtra, India
- **Important Facts** - It is notable as one of the major national parks **existing within a metropolis limit** and is one of the most visited parks in the world.
  - The **Kanheri Caves** in the centre of the park were an important Buddhist learning centre and pilgrimage site
- **Flora** - Kadamba, teak, karanj, shisham, and species of acacia, ziziphus, euphorbia, flame of the forest, red silk cotton and a number of other varieties of flowers.
  - The *karvi* shrub, as it is locally called in the Marathi language, only blooms once in eight years in a mass flowering covering the forest floor in a lavender blush. It grows in abundance in the Western Ghats hills near Mumbai and throughout the Sanjay Gandhi National Park as in other parts of its natural range.
- **Fauna** - Chital, rhesus macaque and bonnet macaque are some of the wild mammals often spotted inside the park. Other large mammals found in the park include black-naped or Indian hare, muntjac, porcupine, Asian palm civet, chevrotain, Hanuman or grey langur, Indian flying fox, sambar deer and leopard. One can also spot striped hyena or four-horned antelope.

- Some of the birds found in the park are jungle owlets, golden orioles, racket-tailed drongos, minivets, magpies, robins, hornbills, bulbuls, sunbirds, peacock, and woodpeckers.



## 5. Keibul Lamjao National Park

- **Location** - Manipur, India
- **Geography** - The park is a swamp with **floating mass of vegetation** created by accrual of organic garbage and biomass with soil particles that has been thickened into a solid form called **phumdis**, at the south-eastern side of the Loktak Lake, which has been declared a Ramsar site.
  - The distinctive nature of the park is that it is "too deep to be a marsh, too shallow to be a lake".

- **Important Facts** - The **only floating park in the world**, located in North East India, and an integral part of Loktak Lake.
  - The national park is characterized by floating decomposed plant material locally called phumdi
- **Flora** - The park, primarily composed of moist semi-evergreen forests, has a rich amalgam of aquatic, wetland and terrestrial ecosystem
- **Fauna** - Mammals reported are the hog deer , wild boar, large Indian civet, common otter , fox, jungle cat , Asian golden cat, bay bamboo rat, musk shrew, common shrew, flying fox and sambar
  - Kingfisher, black kite, lesser sky-lark, northern hill myna, Burmese pied myna, North Indian black drongos, lesser eastern jungle crow, yellow headed wagtail, spotbill duck, blue-winged teal, ruddy shelduck, threatened hooded crane, Burmese sarus sarus crane, Indian white-breasted waterhen and crimson-breasted pied woodpecker.



## 6. Nokrek National Park

- **Location** - Meghalaya, India
- **Geography** - The entire Biosphere Reserve is hilly. In most of the Biosphere Reserve area the soil is red loam
- **Important Facts** - UNESCO added this National park to its list of Biosphere Reserves in May 2009. Along with Balpakram national park, Nokrek is a hotspot of biodiversity in Meghalaya.
- **Flora** - There is a vast range of plants found in the park. A huge canopy of thick, tall and lush green forests cover Nokrek and its environment
- **Fauna** - red panda, Asian elephant, marbled cat, stump-tailed macaque, pig-tailed macaque, Hoolocks



## 7. Simlipal National Park(Tiger Reserve)

- **Location** - Odisha, India
- **Geography** - With a wide range of rainfall and edaphic variations, from dry deciduous to moist green forests, it is suitable to many species of flora and fauna. Some of the beautiful waterfalls like Joranda and Barehipani Falls.
- **Important Facts** - It is part of the **Mayurbhanj Elephant Reserve**. Simlipal National Park derives its name from the abundance of red silk cotton trees growing in the area.
  - This protected area is part of the **UNESCO World Network of Biosphere Reserves**
- **Flora** - It lies in the Eastern Highlands moist deciduous forests ecoregion, with tropical moist broadleaf forest and tropical moist deciduous forests with dry deciduous hill forest and high level Sal forests. The grasslands and the savannas provide grazing grounds for the herbivores and hiding place to the carnivores.
- **Fauna** - The major mammals include tiger, leopard, Asian elephant, sambar, barking deer, gaur, jungle cat, wild boar, chausingha (four



horned antelope), giant squirrel and common langur. The grey hornbill, Indian pied hornbill, Malabar pied hornbill and Indian trogon are also found in the reserve.

- **Mugger crocodile**

## 8. Bhitarkanika National Park

- **Location** - Odisha, India
- **Geography** - It is surrounded by Bhitarkanika Wildlife Sanctuary. Gahirmatha Beach and Marine Sanctuary are to the east, separating swamp region and mangroves from the Bay of Bengal. The national park and wildlife sanctuary is inundated by the rivers Brahmani, Baitarani, Dhamra, Pathsala. It hosts many mangrove species, and is the **second largest mangrove ecosystem in India.**
- **Important Facts** - The area has been designated as the second **Ramsar site of the State after the Chilika Lake.**
- **Flora** - Mangroves are salt tolerant, complex and dynamic ecosystems that occur in tropical and subtropical intertidal regions. Bhitarkanika is one such location of a rich, lush green vibrant ecosystem lying in the estuarine region of Brahmani- Baitarani in the North-Eastern corner of Kendrapara district of Odisha.

- **Fauna** - The park is home to the saltwater crocodile, Indian python, black ibis, wild boar, rhesus monkey, chital, darter, cobra, monitor lizard. Olive ridley turtles nest on Gahirmatha and other nearby beaches.



## 9. Keoladio National Park

- **Location** - Rajasthan, India
- **Geography** - One third of the Keoladeo National Park is wetland with mounds, dykes and open water with or without submerged or emergent plants. The uplands have grasslands with tall grass species together with scattered trees and shrubs present in varying density.
  - Water remains only in some depressions. This alternate wetting and drying helps to maintain the ecology of the freshwater swamp, ideal for water-fowl and resident water birds.

- **Important Facts** - It is a famous **avifauna sanctuary** that hosts thousands of birds, especially during the winter season. It is also a World Heritage Site.
  - Keoladeo Ghana National Park is a **man-made and man-managed** wetland and one of the national parks of India.
- **Flora** - In an area characterized by sparse vegetation, the park is the only spot which has dense vegetation and trees. The principal vegetation types are tropical dry deciduous forests intermixed with dry grasslands. Where the forest has degraded, the greater part of the area is covered with shrubs and medium-sized trees. The park is a freshwater swamp and is flooded during the monsoon. The rest of the area remains dry
- **Fauna** - Macro invertebrates such as worms, insects, and mollusks, though more abundant in variety and numbers than any other group of organisms, are present mostly in aquatic habitats.



## 10. Sariska National Park(Tiger Reserve)

- **Location** - Rajasthan, India
- **Geography** - It is a part of the Aravalli Range and the Khathiar-Gir dry **deciduous forests ecoregion**. It is rich in mineral resources, such as copper.
- **Important Facts** - It is the first reserve in the world with successfully relocated tigers.
- **Flora** - comprising scrub-thorn arid forests, dry deciduous forests, grasslands, and rocky hills. The dominant tree in the forests is dhok. Other trees include the salar , kadaya, dhak , gol , ber and khair. Bargad, arjun , gugal or bamboo. Shrubs are numerous, such as kair , adusta and jhar ber .
- **Fauna** - Bengal tiger, Indian leopard, jungle cat, caracal, striped hyena, golden jackal, chital, sambar deer, nilgai, wild boar, small Indian civet, Javan mongoose, ruddy mongoose, honey badger, Rhesus macaque and Northern plains gray langur and Indian hare
  - Rarest feathered species like **grey partridge, white-throated** kingfisher, Indian peafowl, bush quail, sandgrouse, treepie, golden-backed woodpecker, crested serpent eagle and the Indian eagle-owl.



## 11. Ranthambore National Park

- **Location** - Rajasthan, India
- **Geography** - It is bounded to the north by the **Banas River** and to the south by the **Chambal River**. It is named after the historic Ranthambore Fort, which lies within the park.
- **Important Facts** - **Padam Talao** is the largest of the many lakes in the park. A red sandstone Jogi Mahal is at the edge of the lake. A gigantic Banyan tree, considered to be India's second largest, is also near the lake.
- **Flora** - Ranthambore National Park harbours **dry deciduous forests** and open grassy meadow.

- **Fauna** - Bengal tiger, Indian leopard, nilgai, wild boar, sambar, striped hyena, sloth bear, southern plains gray langur, rhesus macaque, mugger crocodile and chital.



## 12. Desert National Park

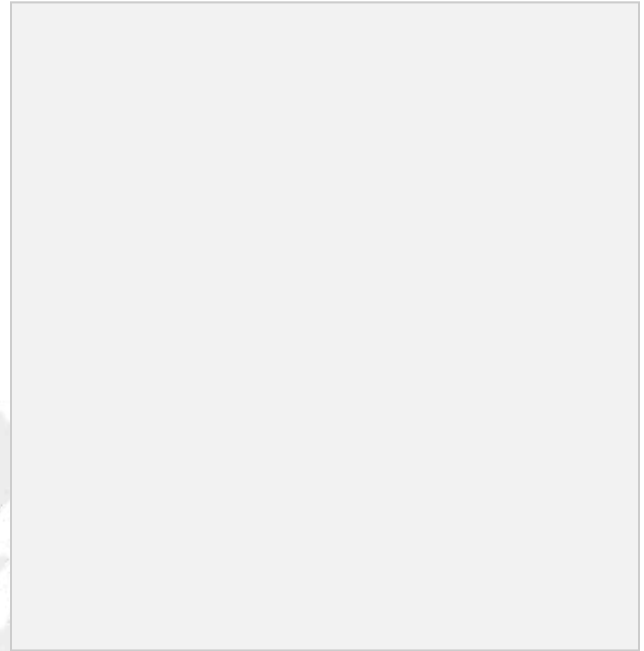
- **Location** - Rajasthan, India
- **Geography** - The Desert National Park is an excellent example of the ecosystem of the Thar Desert. Sand dunes form around 20% of the Park. The major landform consists of craggy rocks and compact salt lake bottoms, intermedial areas and fixed dunes.
- **Important Facts** - Great Indian Bustard

- **Flora** - The vegetation is sparse, and patches of sewan grass and aak shrub can be seen. The landscape includes craggy rocks and compact salt lake bottoms, as well as intermediate areas and both fixed and shifting dunes. Around 20 percent of the vast expanse is covered with sand dunes.
- **Fauna** - Many eagles, harriers, falcons, buzzards, kestrel and vultures are spotted here. Short-toed eagles, tawny eagles, spotted eagles, laggar falcons and kestrels are the most common among these. Sand grouses are spotted near small ponds or lakes. The endangered great Indian bustard is a magnificent bird found in relatively fair numbers.



### 13. Kanchendzonga National Park

- **Location** - Sikkim, India
- **Important Facts** - It was inscribed to the UNESCO World Heritage Sites list in July 2016, becoming the first "**Mixed Heritage**" site of India. It was included in the UNESCO Man and the Biosphere Programme.
  - The park contains Tholung Monastery, a gompa located in the park's buffer zone. It is considered one of the most sacred monasteries in Sikkim.
- **Flora** - The vegetation of the park include temperate broadleaf and mixed forests consisting of oaks, fir, birch, maple, willow. The vegetation of the park also includes Alpine grasses and shrubs at higher altitudes along with many medicinal plants and herbs.
- **Fauna** - The park contains many mammal species including musk deer, snow leopard, Himalayan tahr, dhole, sloth bear, viverrids, Himalayan black bear, red panda, Tibetan wild ass, Himalayan blue sheep, serow, goral and takin, as well as reptiles including rat snake and Russell's viper.
  - species of birds are found inside the park including blood pheasant, satyr tragopan, osprey, Himalayan griffon, lammergeier, Western tragopan, green pigeon, Tibetan snowcock, snow pigeon, impeyan pheasant, Asian emerald cuckoo, sunbird and eagle



#### 14. Mudumalai national Park(Tiger Reserve)

- **Location** - Tamil Nadu, India
- **Geography** - It shares its boundaries with the states of Karnataka and Kerala.
- **Flora** - The dominant type of habitat found in the Mudumalai National Park is **Tropical moist forest**. Tropical dry forest and scrub forests are also found here. Moist bamboo brakes are found amidst dry deciduous, moist deciduous and semi-evergreen forests and along the fringes of riparian forests and swamps.

- **Fauna** - home to several endangered and vulnerable species including Indian elephant, Bengal tiger, gaur and Indian leopard. There are at least 266 species of birds in the sanctuary, including critically endangered Indian white-rumped vulture and long-billed vulture



## 15. Mukurthi National Park

- **Location** - Tamil Nadu, India
- **Geography** - The park is characterised by montane grasslands and shrublands interspersed with sholas in a high altitude area of high rainfall, near-freezing temperatures and high winds.

- **Important Facts** - The park was created to protect its keystone species, the **Nilgiri tahr**.
- **Fauna** - Several threatened mammal species live here including Nilgiri tahr, Indian elephant, Bengal tiger, Nilgiri marten, Nilgiri langur and Bonhote's mouse.



#### **16. Indira Gandhi National Park(Anamalai Tiger Reserve)**

- **Location** - Tamil Nadu, India
- **Important Facts** - It has significant anthropological diversity with more than 4600 Adivasi people from six tribes of indigenous people living in 34 settlements. The tribes

are the Kadars, Malasars, Malaimalasar s, Pulaiyars, Muduvars and the Eravallan (Eravalar)

- **Flora** - wet evergreen forest and semi-evergreen forest, montane shola-grassland, moist deciduous, dry deciduous, thorn forests and marshes. Tropical montane forests occur at higher elevations and are interspersed with montane grasslands, forming the shola-grassland complex.
- **Fauna** - Threatened species of mammals in the sanctuary include: the endangered Bengal tiger, Indian elephant, Indian leopard, **Nilgiri tahr and lion-tailed macaque**,
  - the vulnerable brown mongoose, gaur, Malabar spiny dormouse, Nilgiri langur, rusty-spotted cat, sambar deer, sloth bear and smooth-coated otter,
  - the near threatened Indian giant squirrel, Indian leopard and Indian pangolin.



## 17. Gulf of Mannar Marine National Park

- **Location** - Tamil Nadu, India
- **Important Facts** - **Mangroves** dominate the intertidal zones of the park islands
- **Flora** - The park includes estuaries, mudflats, beaches and forests of the near shore environment. It also includes marine components such as coral reefs, seaweed communities, sea grasses, salt marshes and mangroves.
- **Fauna** - The dugong, Indo-Pacific bottlenose dolphin, finless porpoise, spinner dolphin, common dolphin, Risso's dolphin, melon-headed whale, and dwarf sperm whale. Larger whales include sperm whale, minke whale, Bryde's whale, sei whale, and critically endangered species including humpback whale, fin whale, and blue whale.



## 18. Clouded Leopard National Park

- **Location** - Tripura, India
- **Important Facts** - It is home to **four species of primate monkey including Phayre's langur**



## 19. Rajaji National Park(Tiger Reserve)

- **Location** - Uttarakhand, India
- **Important Facts** - Rajaji National Park has been named after C. Rajagopalachari (*Rajaji*), a prominent leader of the Freedom Struggle, the second and last Governor-General of independent India and one of the first recipients of India's highest civilian award Bharat Ratna (in 1954).

- **Flora** - Rajaji National Park of India is nestled between the **Shivalik ranges and the Indo-Gangetic plains**. Broadleaved deciduous forests, riverine vegetation, scrubland, grasslands and pine forests form the range of flora in this park.
- **Fauna** - Asian elephant, Bengal tiger, Indian Leopard, Striped hyena, Jungle cat, Goral Indian hare, Sloth bear, Himalayan black bear, King cobra, Jackal, Barking deer, Sambar, Wild boar, Rhesus macaque, Indian langur, Indian porcupine, Monitor lizard, Python



## 20. Jim Corbett National Park

- **Location** - Uttarakhand, India
- **Geography** - Corbett National Park comprises area of hills, riverine belts, marshy depressions, grasslands and a large lake. The elevation ranges from 1,300 to 4,000 ft (400 to 1,220 m). Winter nights are cold but the days are bright and sunny. It rains from

July to September. The park has sub-Himalayan belt geographical and ecological characteristics

- **The Ramganga river Passes** through it.
- **Important Facts** - The park was the **first to come under the Project Tiger initiative**. It was named after hunter and naturalist Jim Corbett.
- **Flora** - It has numerous ravines, ridges, minor streams and small plateaus with varying aspects and degrees of slope.
- **Fauna** - Bengal tigers, Leopards, jungle cat, fishing cat and leopard cat, barking deer, sambar deer, hog deer and chital, sloth and Himalayan black bears, Indian grey mongoose, otters, yellow-throated martens, Himalayan goral, Indian pangolins, and langur and rhesus macaques, Owls and nightjars



## 21. Nanda Devi National Park(Nanda Devi Biosphere Reserve)

- **Location** - Uttarakhand, India
- **Geography** - Within the National Park lies the **Nanda Devi Sanctuary**, a glacial basin surrounded by a ring of peaks between 6,000 metres (19,700 ft) and 7,500 m (24,600 ft) high, and drained by the Rishi Ganga through the Rishi Ganga Gorge.
- **Important Facts** - The National Park was inscribed a **World Heritage Site by UNESCO**
- **Flora** - Vegetation is scarce In the inner sanctuary due to the dryness of the conditions. One will not find vegetation near Nanda Devi Glacier.
- **Fauna** - Himalayan musk deer, mainland serow, Himalayan tahr, Goral, snow leopard, Himalayan black bear and perhaps also brown bear



## 22. Valley of Flowers National Park

- **Location** - Uttarakhand, India
- **Geography** - The Valley of Flowers is a high-altitude Himalayan valley. region containing a diversity of alpine flora, representative of the Western Himalayan alpine shrub and meadows ecoregion.
- **Flora** - The valley has three sub-alpine between 3,200 m and 3,500 m which is the limit for trees, lower alpine between 3,500 m and 3,700 m, and higher alpine above 3,700 m. The habitats include valley bottom, river bed, small forests, meadows, eroded, scrubby and stable slopes, moraine, plateau, bogs, stone desert and caves.
- **Fauna** - Asiatic black bear, snow leopard, musk deer, brown bear, red fox and blue sheep.



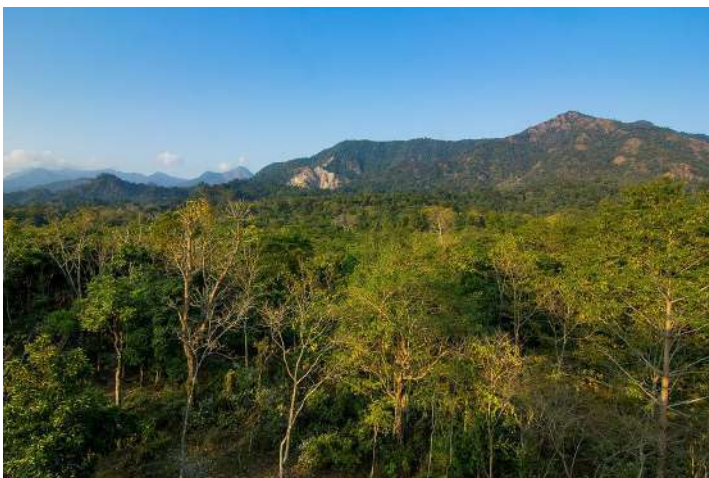
## 23. Dudhwa National Park

- **Location** - Uttar Pradesh, India
- **Geography** - It represents one of the few remaining areas of the diverse and productive Terai ecosystem.
- **Important Facts** - Indian rhinoceros was reintroduced into Dudhwa from Pobitora Sanctuary in Assam and Nepal.
- **Flora** - The area of the park falls within the Upper Gangetic plains and is a vast alluvial plain. The forests, especially the sal forests, have always been very dense and can be categorized into northern tropical semi-evergreen forest, northern Indian moist deciduous forest, tropical seasonal swamp forest and northern tropical dry deciduous forest.
- **Fauna** - Tigers, Leopard, swamp deer, sambar deer, barking deer, spotted deer, hog deer, sloth bear, ratel, jackal, civets, jungle cat, fishing cat, leopard cat.



## 24. Buxa National Park(Tiger Reserve)

- **Location** - West Bengal, India
- **Important Facts** - **Rajabhatkhawa Vulture Breeding Centre** at Buxa Tiger Reserve for the breeding and conservation of endangered Indian vultures
- **Flora** - Forest types include, Northern dry deciduous, Eastern Bhabar and Terai sal, East Himalayan moist mixed deciduous forest, Sub-Himalayan secondary wet mixed forest, Eastern sub-montane semi-evergreen forest, Northern tropical evergreen forest, East Himalayan subtropical wet hill forest, Moist sal savannah, Low alluvium, Savannah woodland
- **Fauna** - Mammals present include Asian elephant, gaur, Sambar deer, clouded leopard, Indian leopard.



## 25. Neora Valley National Park

- **Location** - Darjeeling, West Bengal, India
- **Geography** - pristine undisturbed natural habitat with rugged inaccessible hilly terrain
- **Flora** - Neora Valley sustains a unique ecosystem where tropical, subtropical, sub-temperate, and temperate vegetative systems still harbour a wealth of flora and fauna. The forest consists of mixed species like rhododendron, bamboo, oak, ferns, sal, etc. The valley also has numerous species of orchids.
- **Fauna** - Mammals reported from this area are Indian leopard, five viverrid species, Asiatic black bear, sloth bear, Asian golden cat, wild boar, leopard cat, goral, serow, barking deer, sambar deer, flying squirrel and tahr, red panda, clouded leopard.



## 26. Sunderbans Wildlife Sanctuary(Tiger and Biosphere Reserve)

- **Location** - West Bengal, India
- **Geography** - It is part of the Sundarbans on the Ganges Delta, and adjacent to the Sundarban Reserve Forest in Bangladesh. The delta is densely covered by mangrove forests, and is **one of the largest reserves for the Bengal tiger.**
- **Important Facts** - It is a **UNESCO World Heritage Site inscribed in 1987**, and it has been designated as a **Ramsar site since 2019**. It is considered as a **World Network of Biosphere Reserve (Man and Biosphere Reserve) from 1989.**
- **Flora** - Vast area of mangrove forests with a variety of flora and diverse fauna in a unique ecosystem. Sundarban has achieved its name from the **Sundari tree.**
- **Fauna** - Royal Bengal tigers, fishing cats, leopard cats, macaques, wild boar, Indian grey mongoose, fox, jungle cat, flying fox, chital.
  - Some of the birds commonly found in this region are openbill storks, black-capped kingfishers, black-headed ibis, water hens, coots, pheasant-tailed jacanas, pariah kites, brahminy kite, marsh harriers, swamp partridges, red junglefowl, spotted doves.

# EdSarrthi

Mentorship Platform for  
UPSC CSE Preparation



$$A^2 = B^2 + C^2$$



## Our Past Results 2021



# 41 Selections in UPSC CSE 2021 09 in Top 100

**Rank 08**



Ishita Rathi

**Rank 11**



Shubhankar Pathak

**Rank 16**



Anshu Priya

**Rank 29**



Bhavishya

**Rank 32**



Anay Nitin

**Rank 44**



Anjali Shrotriya

**Rank 75**



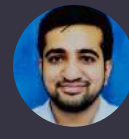
Surbhi Singla

**Rank 81**



Ishu Agrawal

**Rank 84**



Sujawal  
Jagga



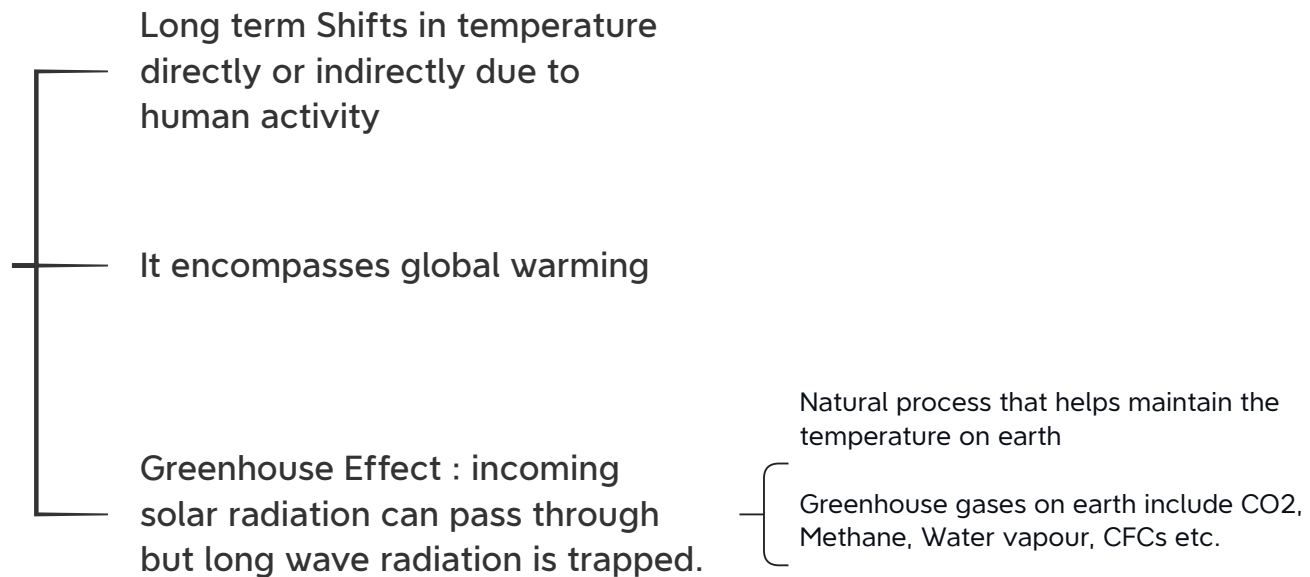
✉ [ask@sarrthi.com](mailto:ask@sarrthi.com)

📺 EdSarrthi IAS

🌐 [www.sarrthi.com](http://www.sarrthi.com)

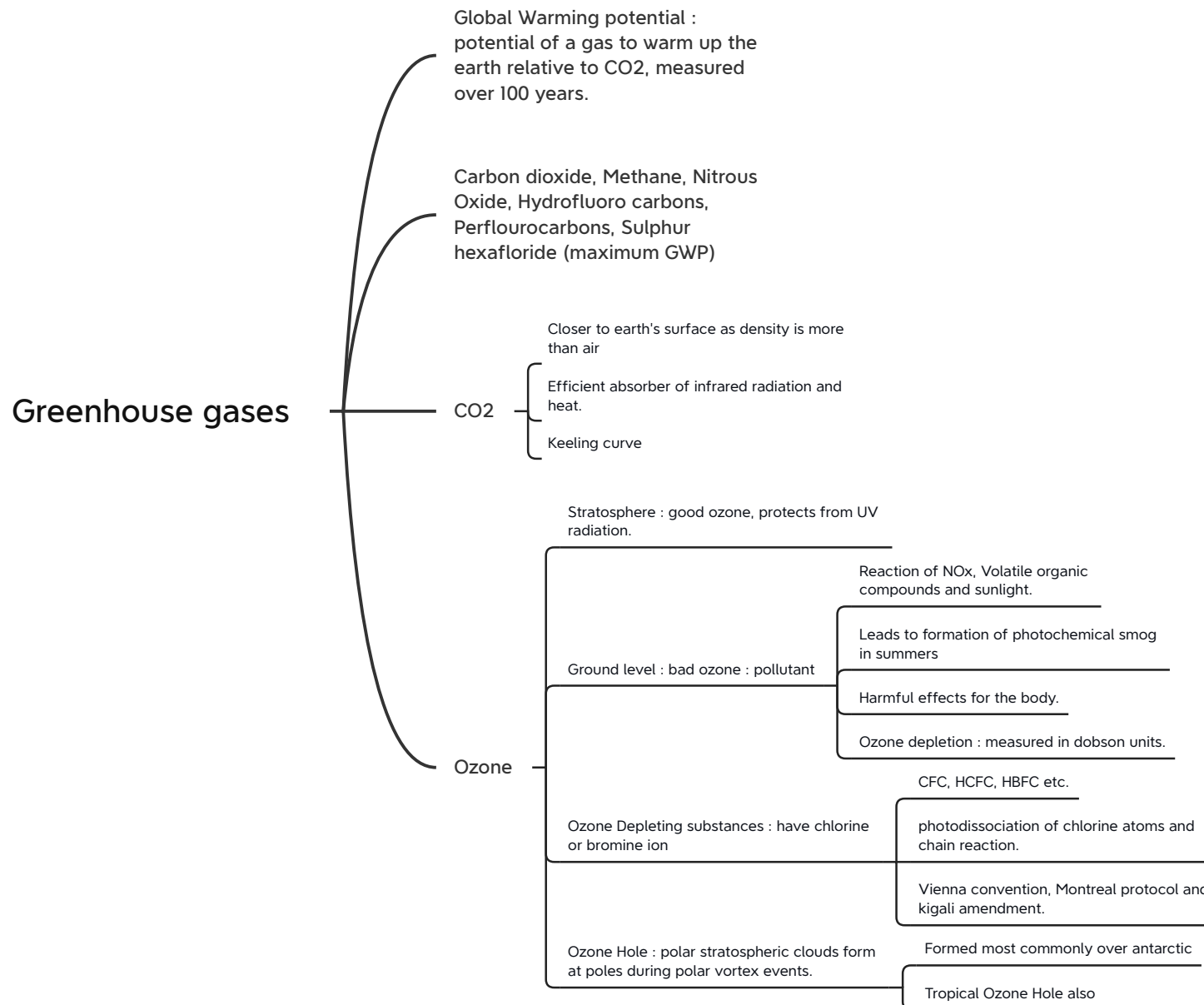
☎ 9569093856

## Basics of Climate Change



Presented with xmind





## Short Lived Climate Pollutants

Stay in the atmosphere for a much shorter time than carbon dioxide.

Climate and Clean Air Coalition :  
UNEP and many countries  
founded it, India is a member (not  
founding member)

**Methane**

- Component of natural gas, lifespan is 20 years, GWP is 86 times that of CO<sub>2</sub>.
- Sources : wetlands, rice, fossil fuels etc.
- Sink : chemical reactions in atmosphere.
- Important Reports
  - Global Methane Assessment : UNEP and CCAC
  - Global Methane Budget : Global Carbon Project
- Global Methane Pledge : India did not join.

**Trophospheric Ozone**

**Black Carbon**

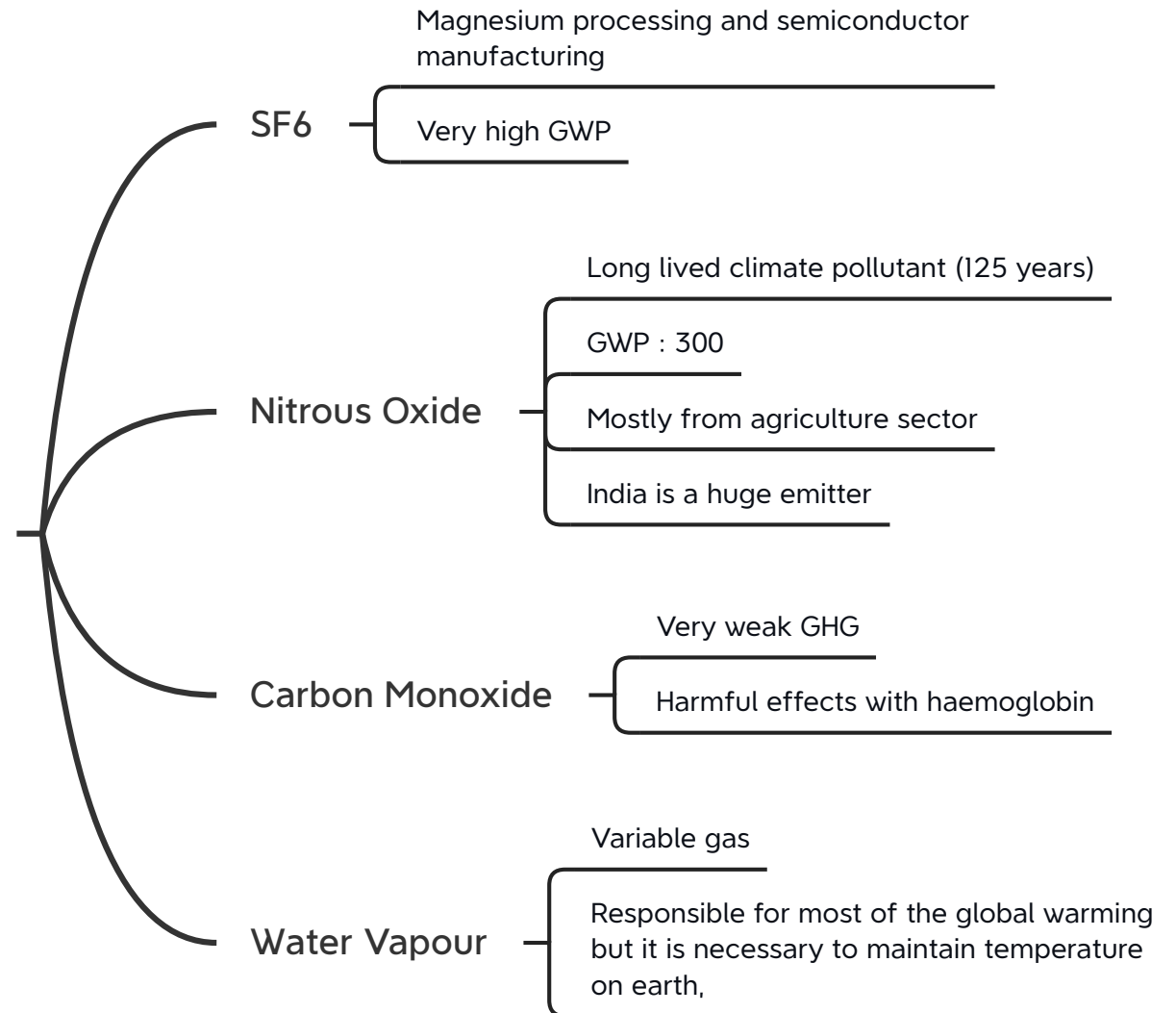
- Reduces albedo and positive feedback cycle
- strongest absorber of sunlight (black)
- Linked with Covid 19 in a study

**HFCs**

- through kigali amendment
- Meant to replace CFCs



## Other Greenhouse Gases



## Important Phenomenon

### Carbon Footprint

Impact that one's activities have on Carbon emissions.

### Carbon Bomb

Projects that are likely to emit over a gigaton of CO<sub>2</sub> over their lifetime.

### Carbon Sensitivity

Sensitivity of the earth's climate to changes in carbon dioxide

### Climate Feedback

Earth's climate either amplifying (positive) or dampening (negative) the effects of an initial carbon forcing.

Ice albedo, permafrost, water vapour : positive feedback.

### Clouds and Climate Change

High clouds trap terrestrial heat.

Low cloud reflect sunlight.

### Carbon fertilisation

Greening due to increased CO<sub>2</sub> (temporary)

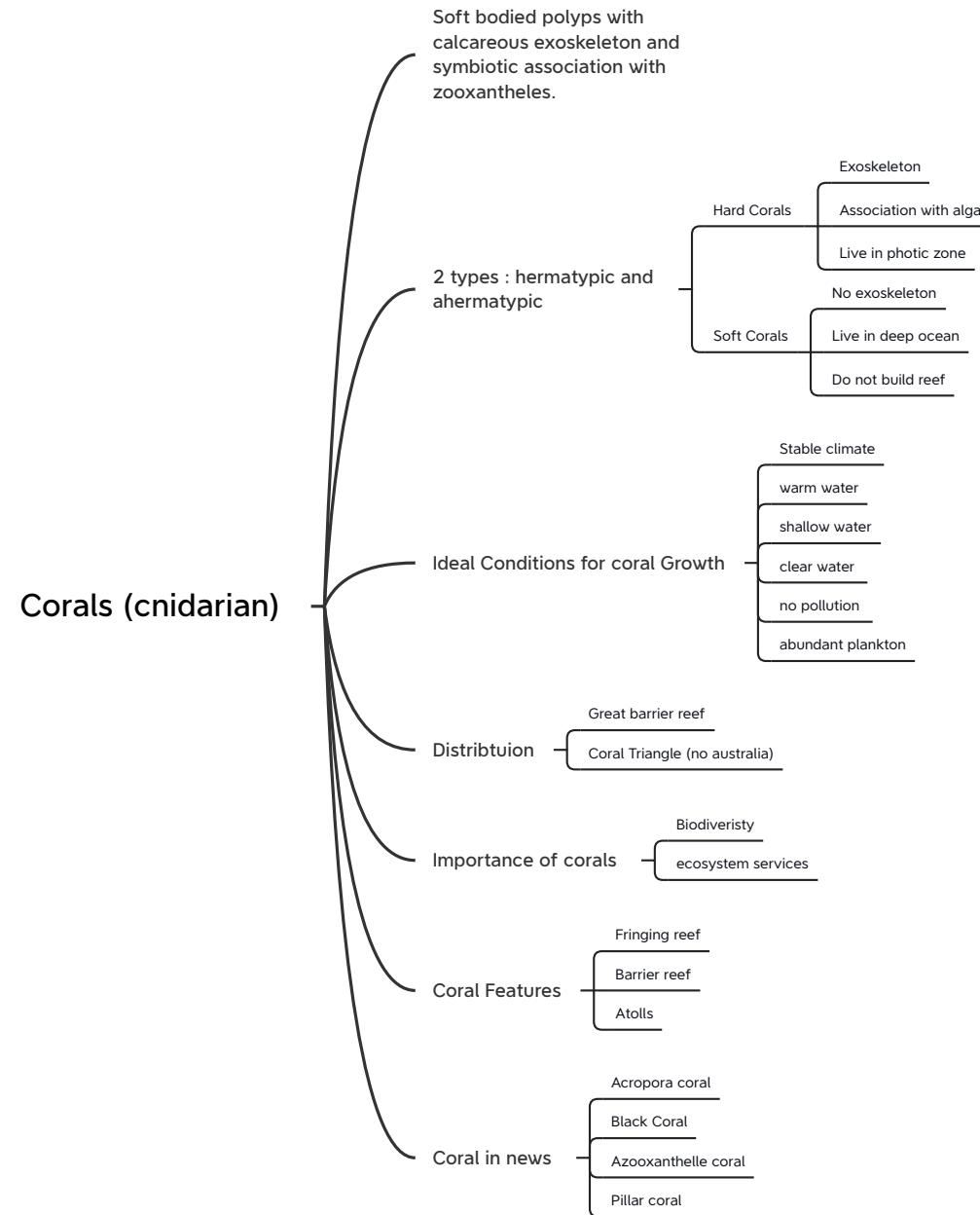
Climate Tipping points : 9 irreversible points of change

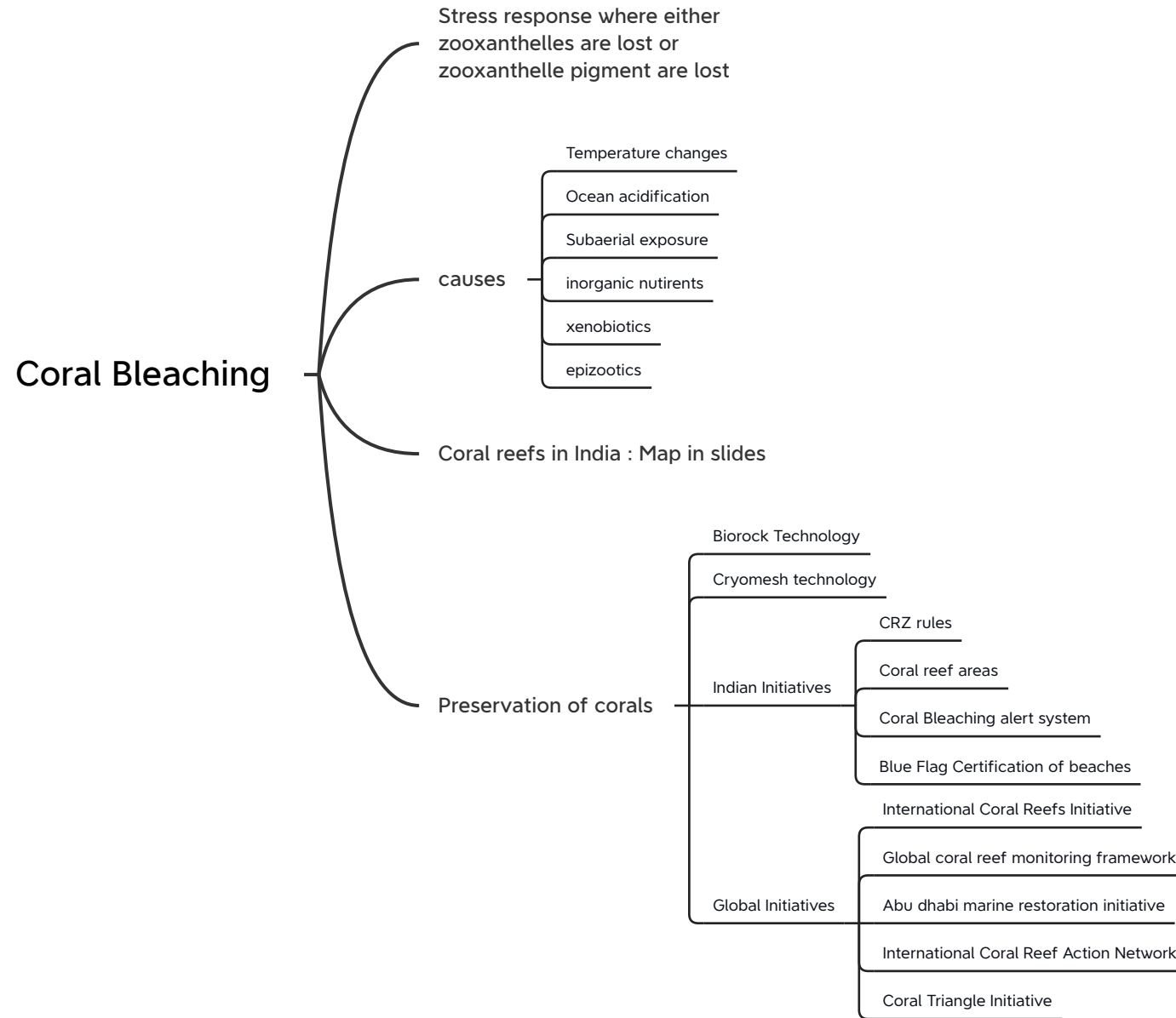


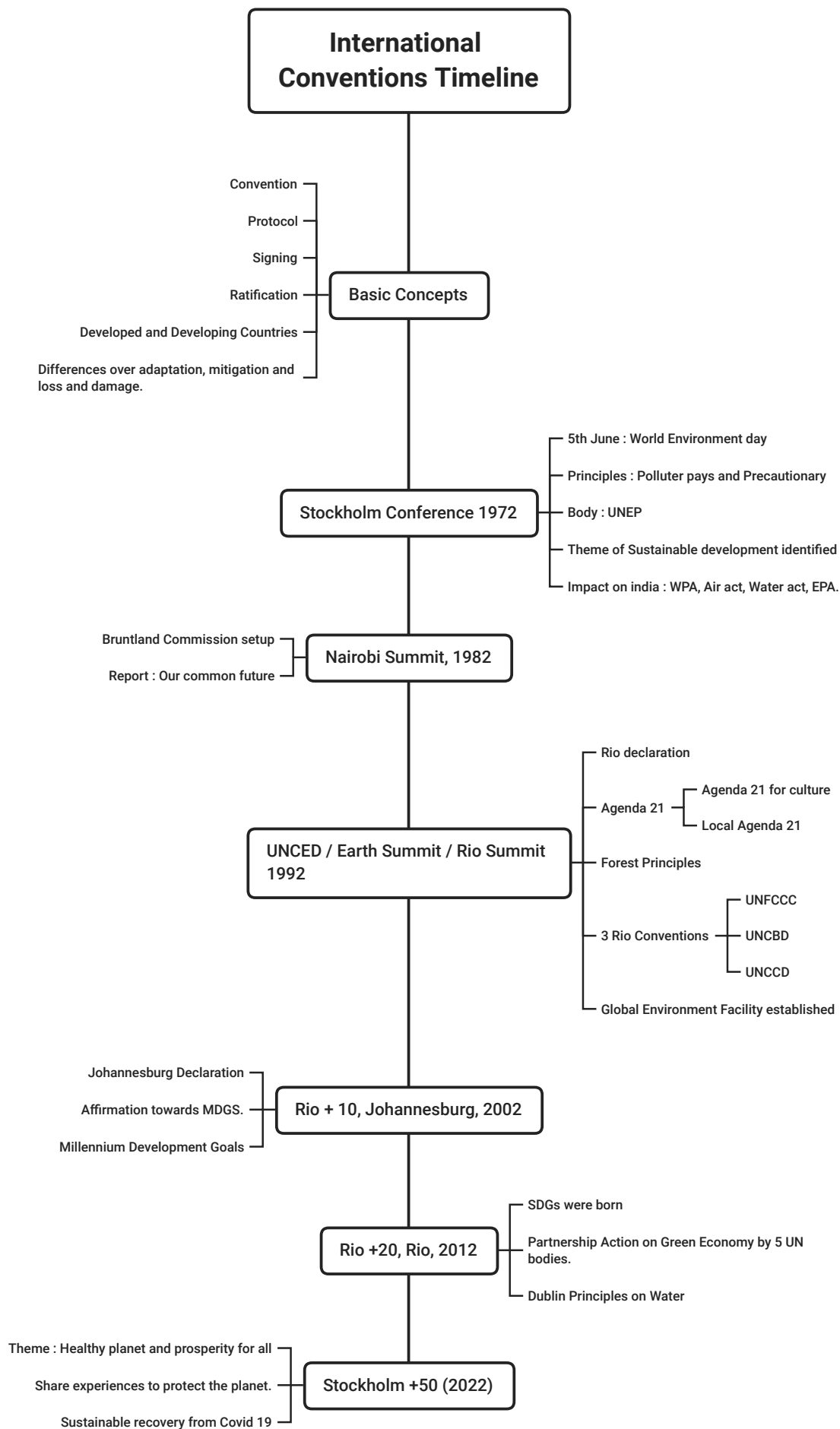
## Impacts of Global Warming

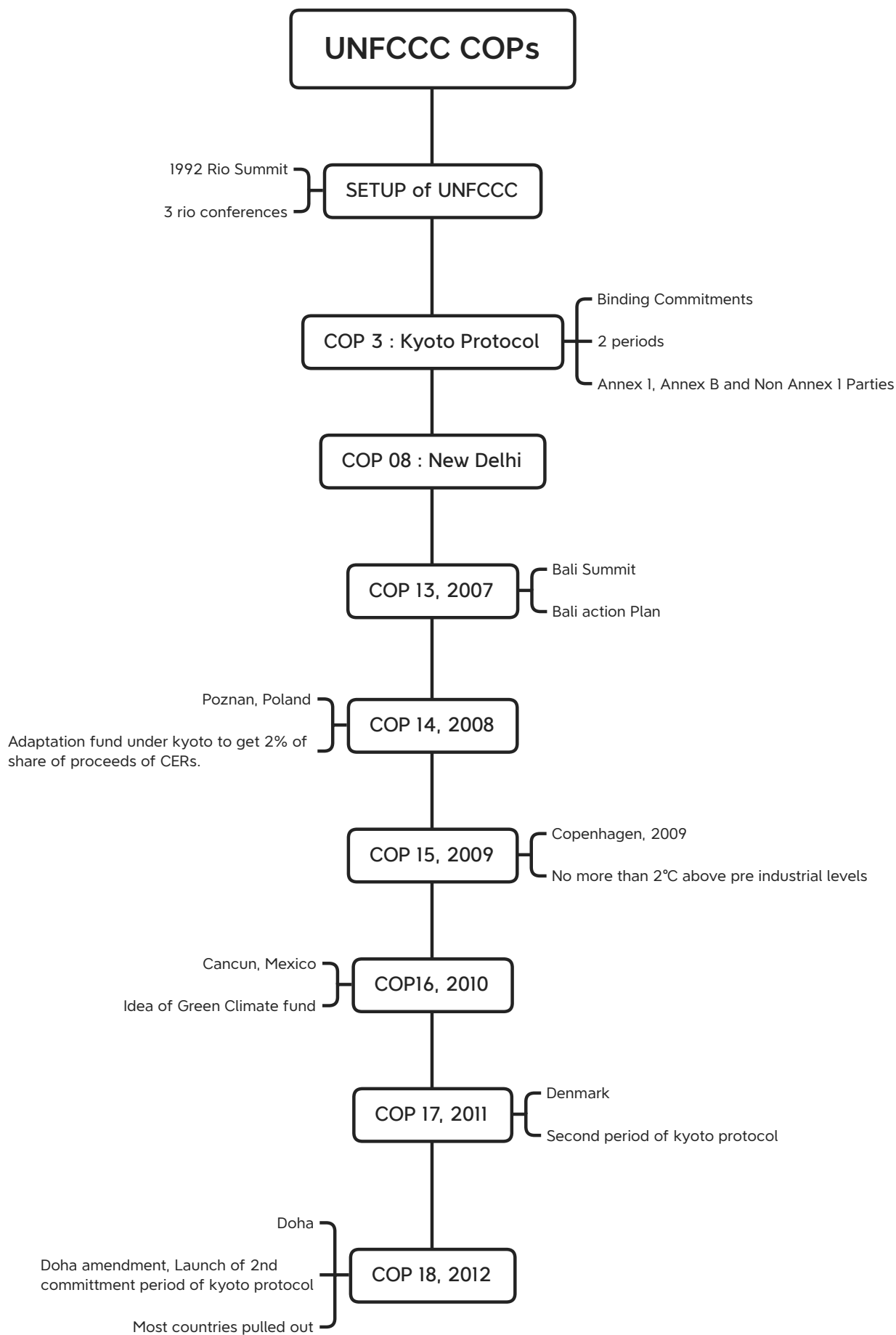
- Marine Heat Waves
- Coral Bleaching
- Weather Patterns change
- Severe Cyclones
- Urban Heat islands
- Increased Wildfires
- Shrinking Cryosphere
- Arctic Amplification
- Zombie ice melting
- Sea Level Changes (eustatic)
- Coastal Flooding SIDS : barbados plan of action and mauritius strategy
- Deterioration of carbon sinks (tundra)
- Climate Migrants
- Economic Losses
- Social Cost of Carbon
- Oxygen Deoxygenation.

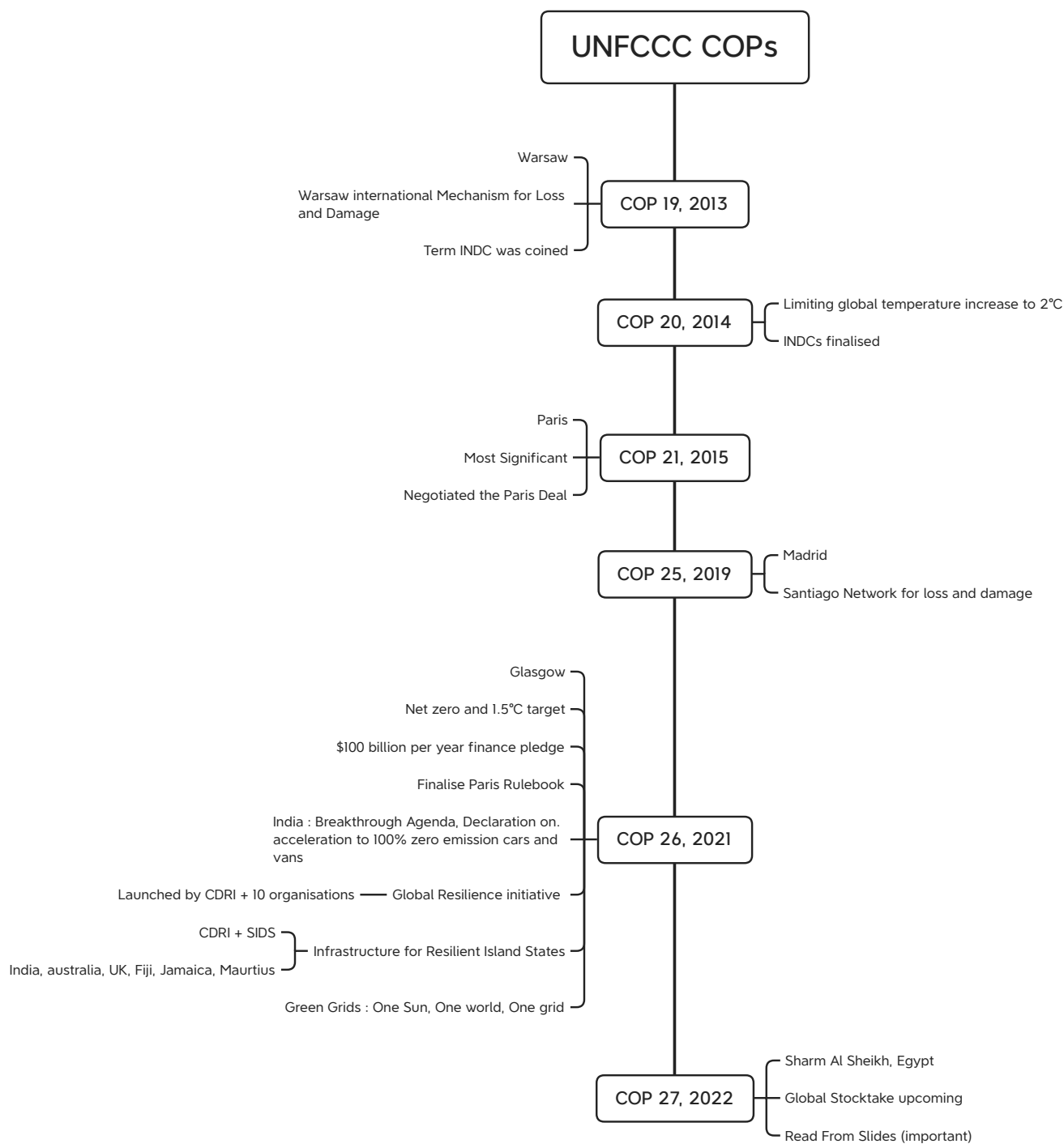


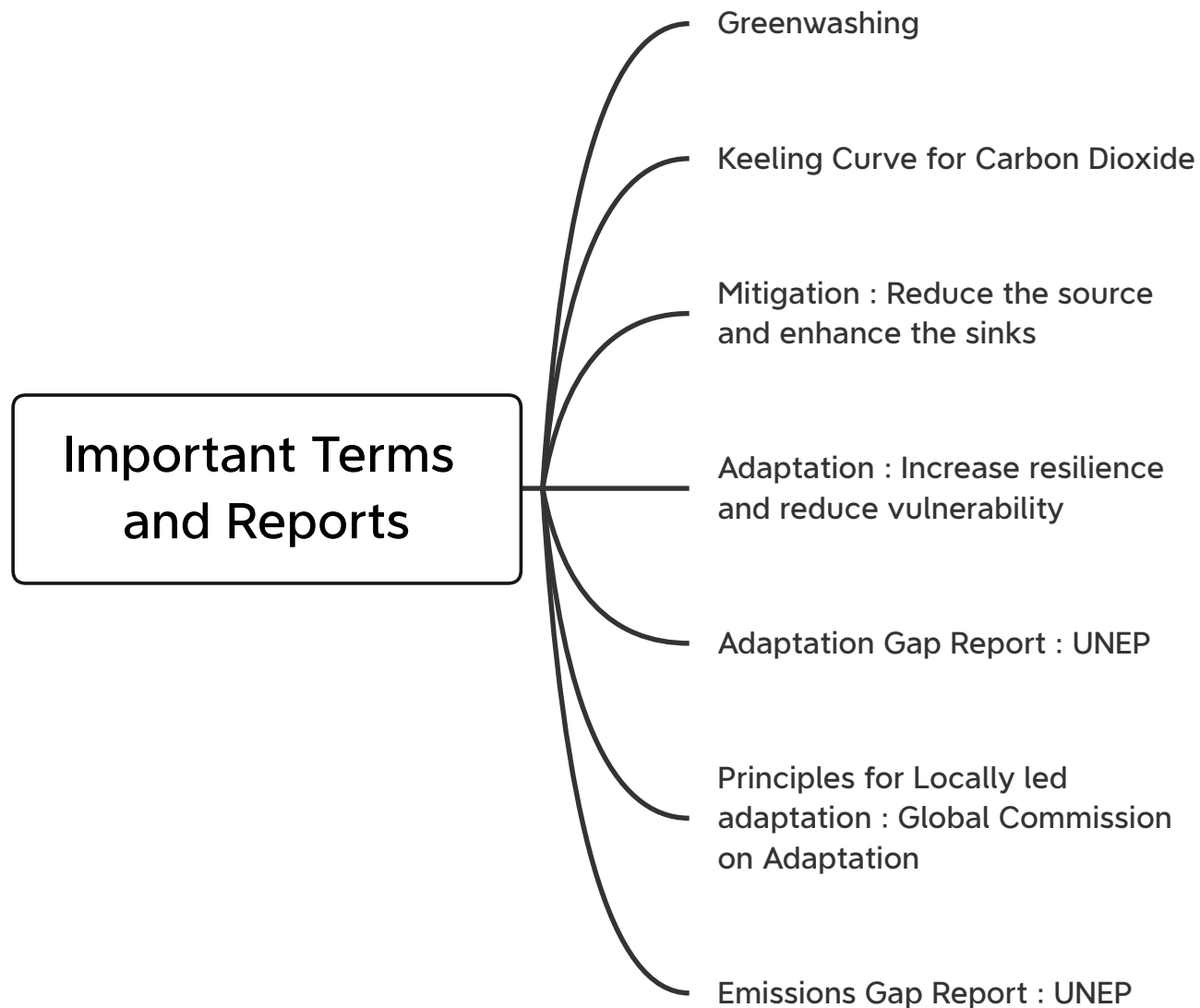


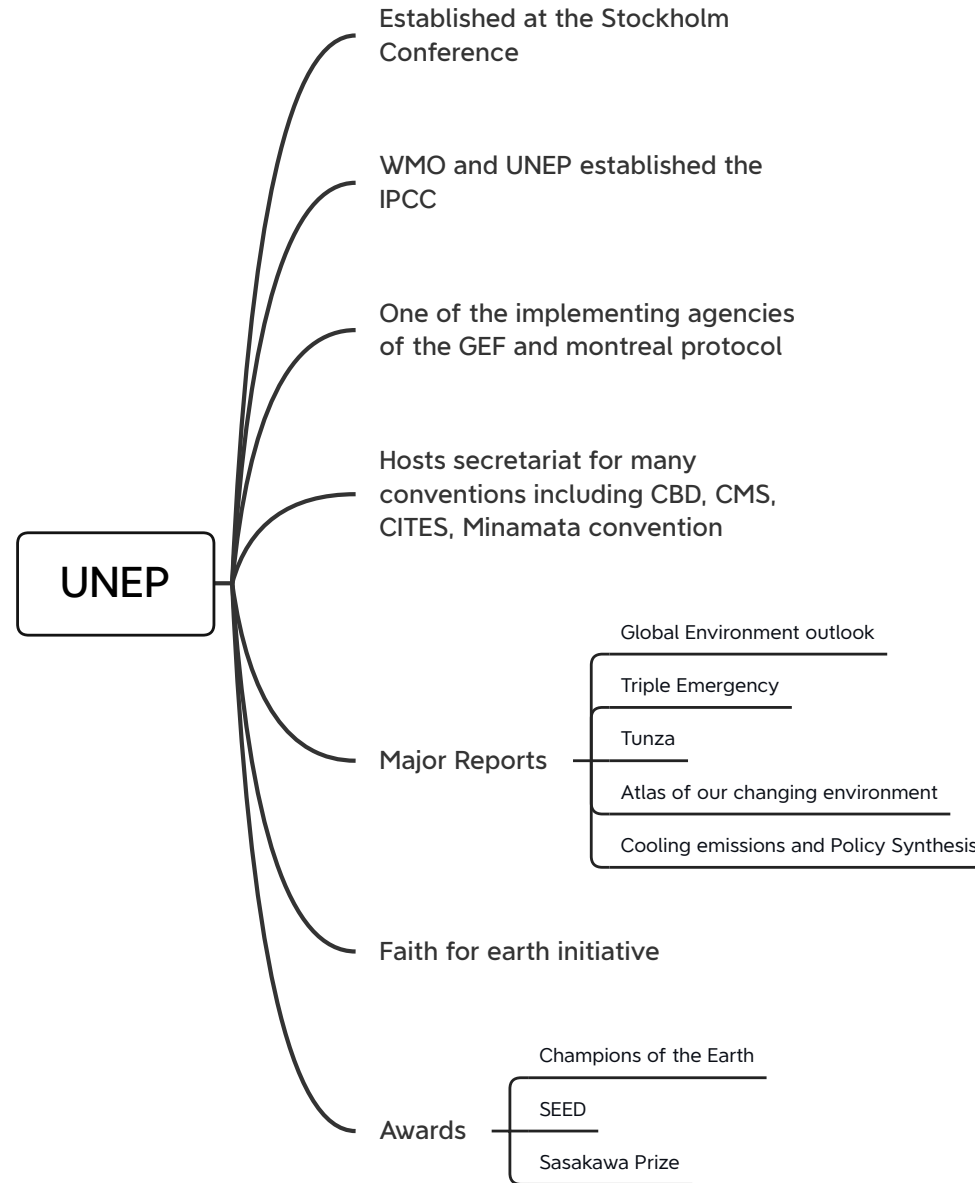


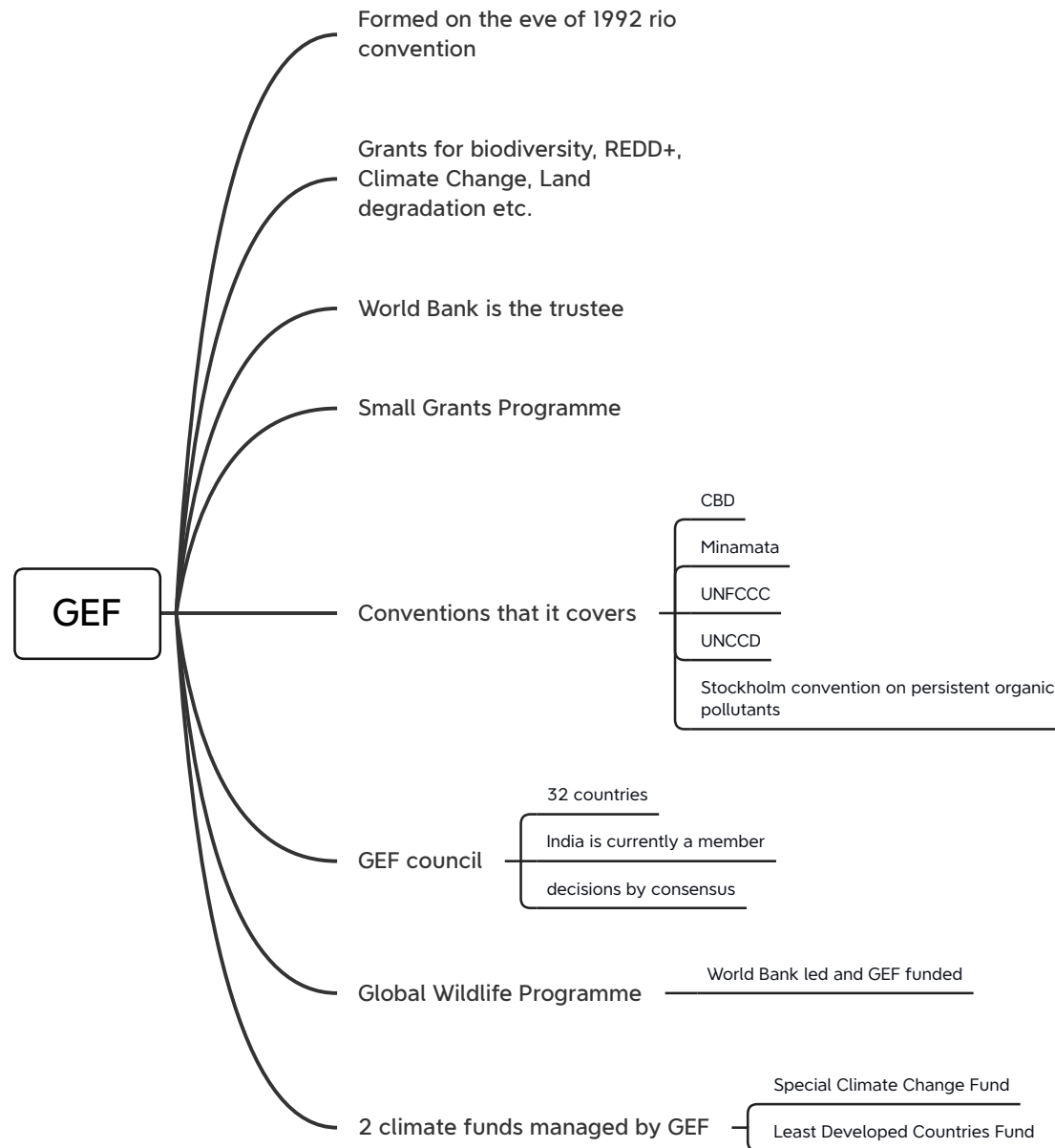


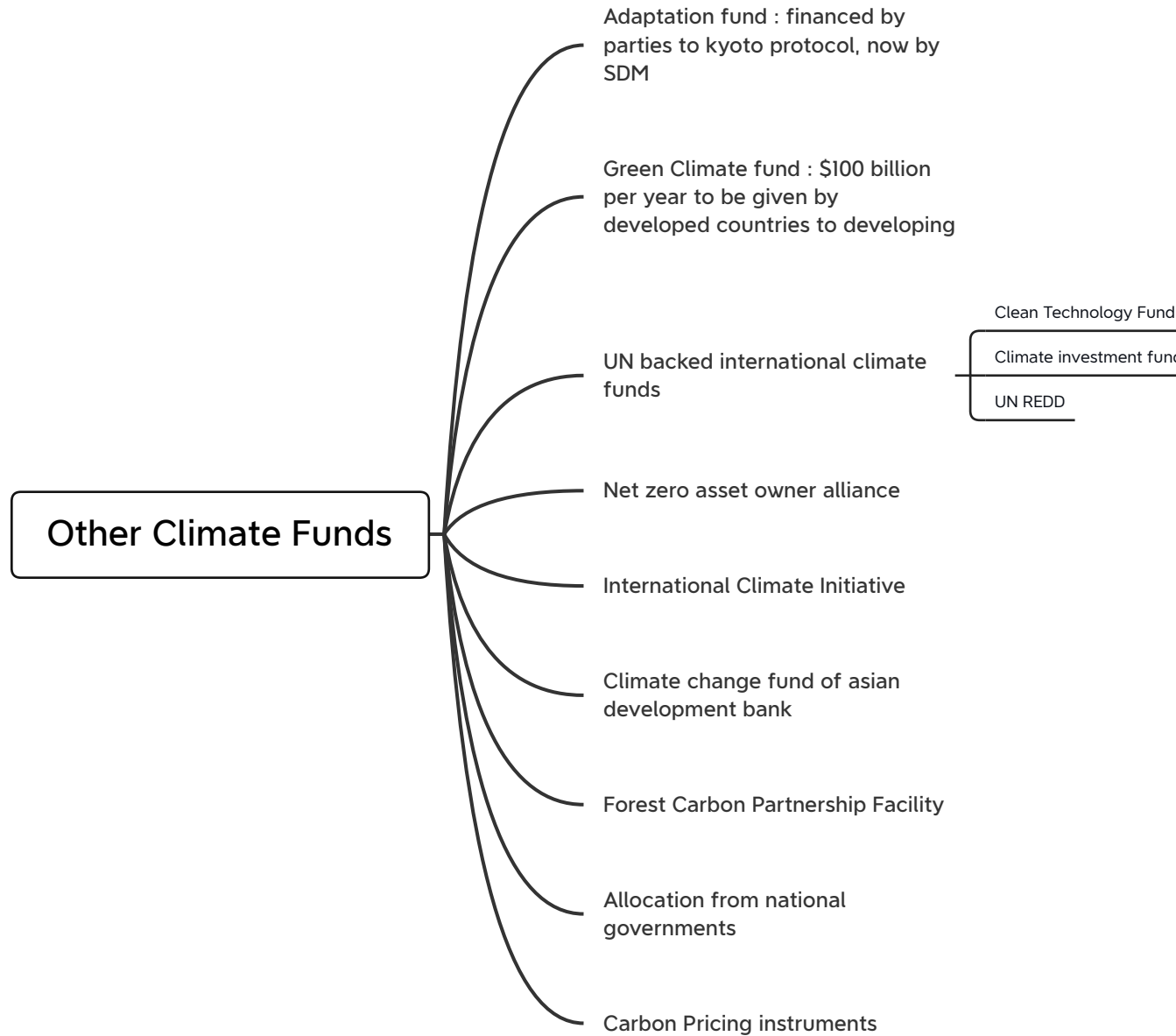


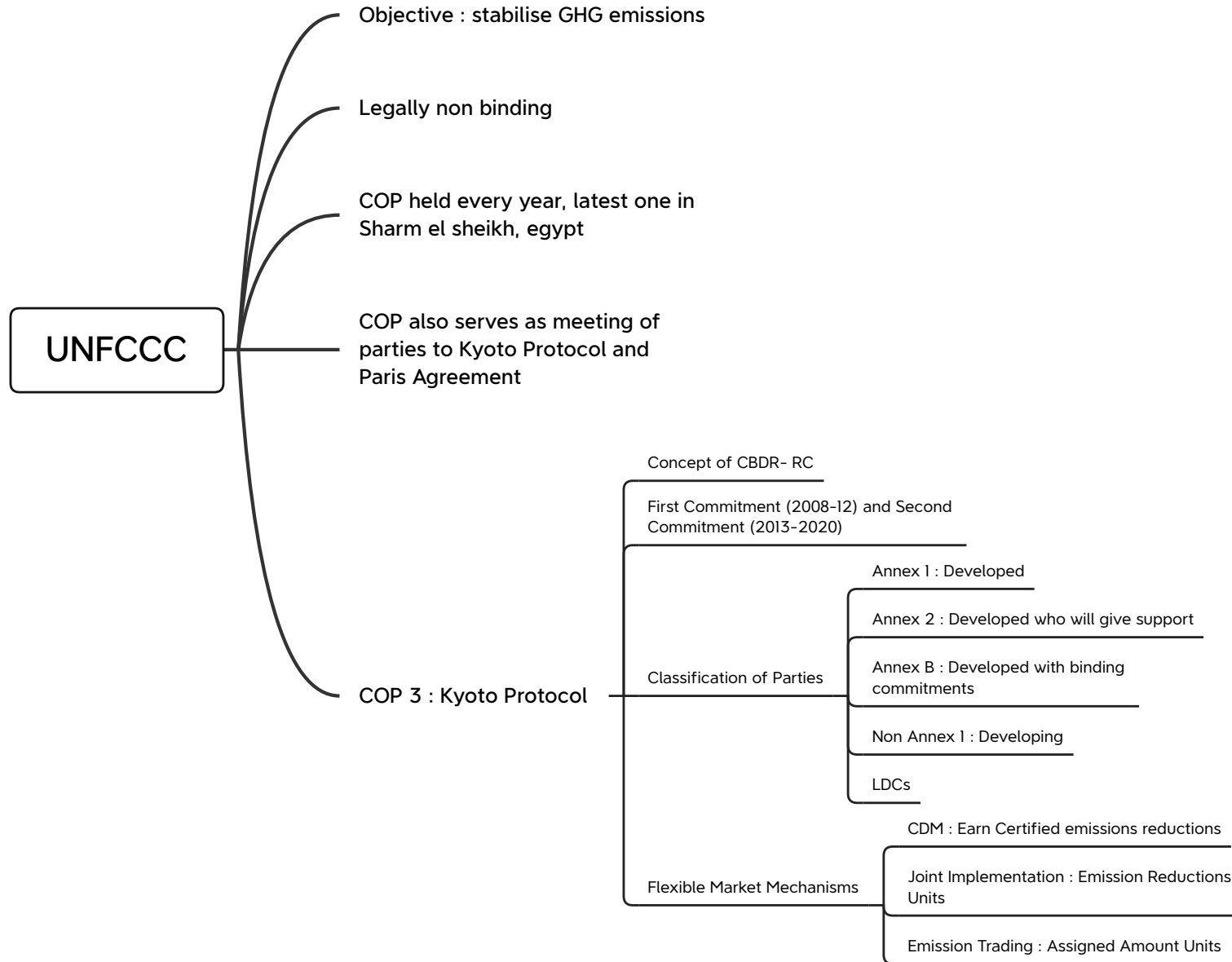


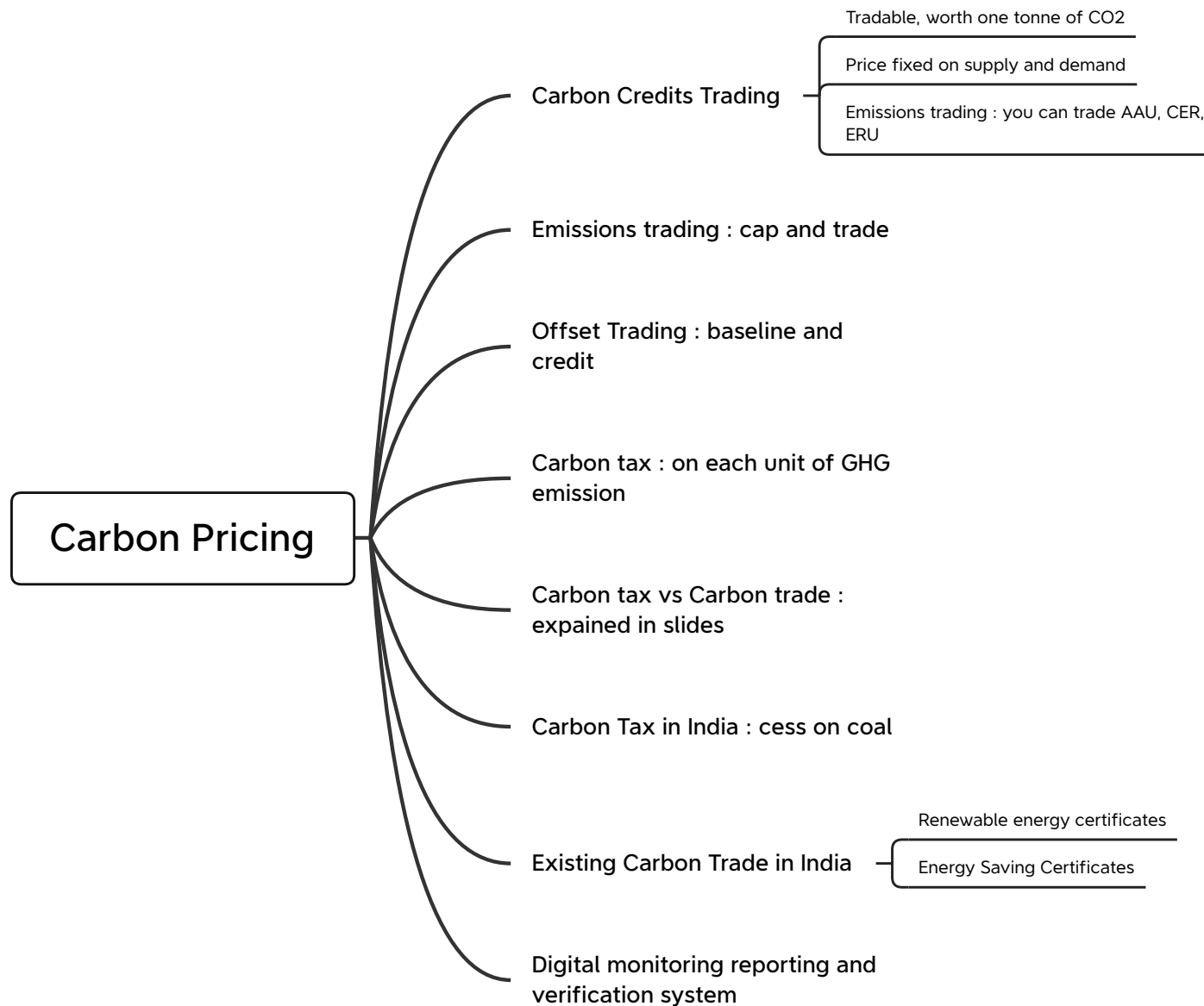












## Paris Climate Deal

most crucial conference post  
kyoto

INDCs by 146 countries

No timetable, unlike kyoto

180 countries have ratified till now

Target : well below 2°C, less than  
1.5°C

\$100 billion a year in finance

US pulls out (shri donald trump)

Paris rulebook

Legally binding international treaty

Governs reporting on progress

Global Stocktake : every 5 years  
INDCs and progress to be  
reviewed to increase ambition

Article 6 : voluntary cooperation

Internationally Traded Mitigation Outcomes, 6.  
2

Sustainable Development Mechanism, 6.4

Non Market Based Mechanism, 6.8

Resolution of issue of double  
counting



## Important Terms

Net Zero and carbon neutrality

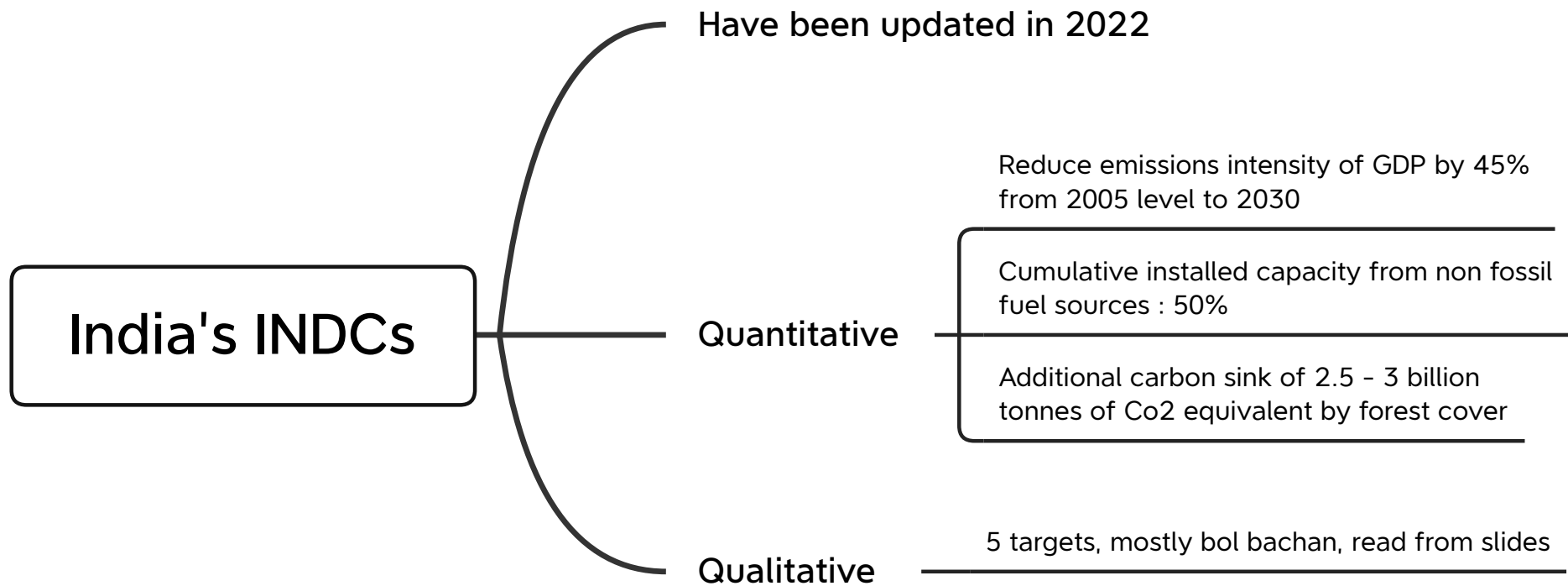
Net zero : different commitments by different countries

Carbon Neutral Panchayat : Jammu and Kashmir

Climate Neutral Now : UNFCCC secretariat

Carbon Border adjustment mechanism : EU





Presented with xmind



## India's LTLEDS

Broader than INDCs.

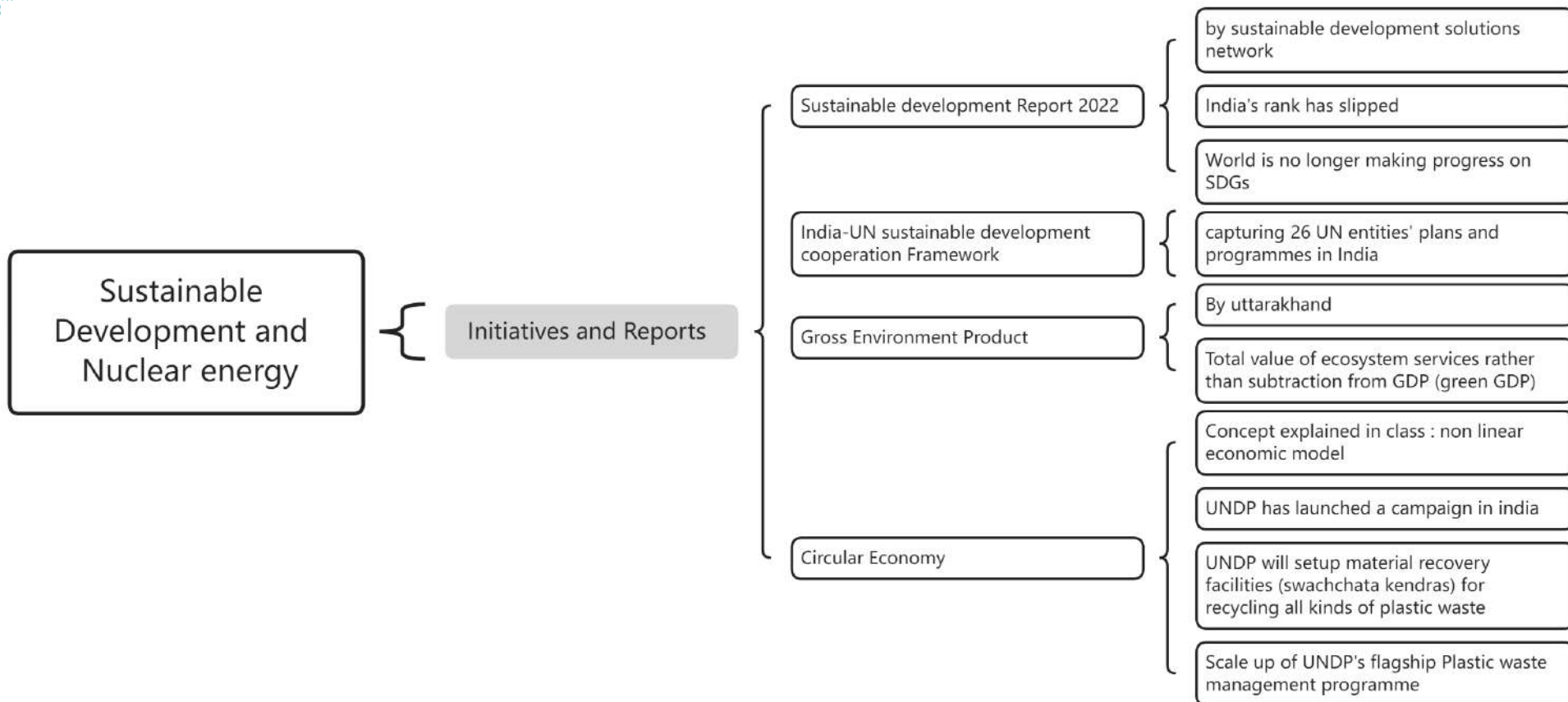
Non Legally binding

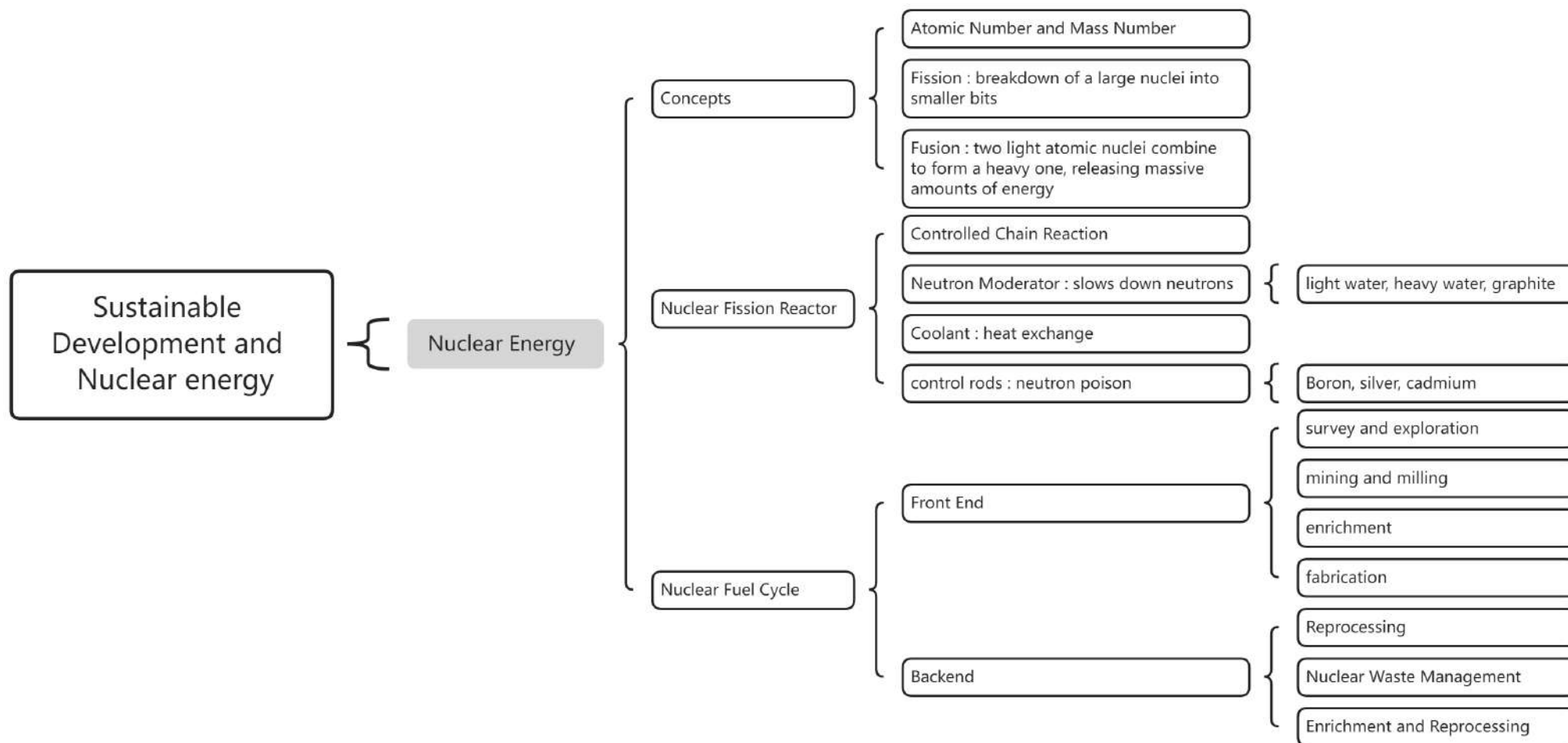
Under Paris agreement 4.19

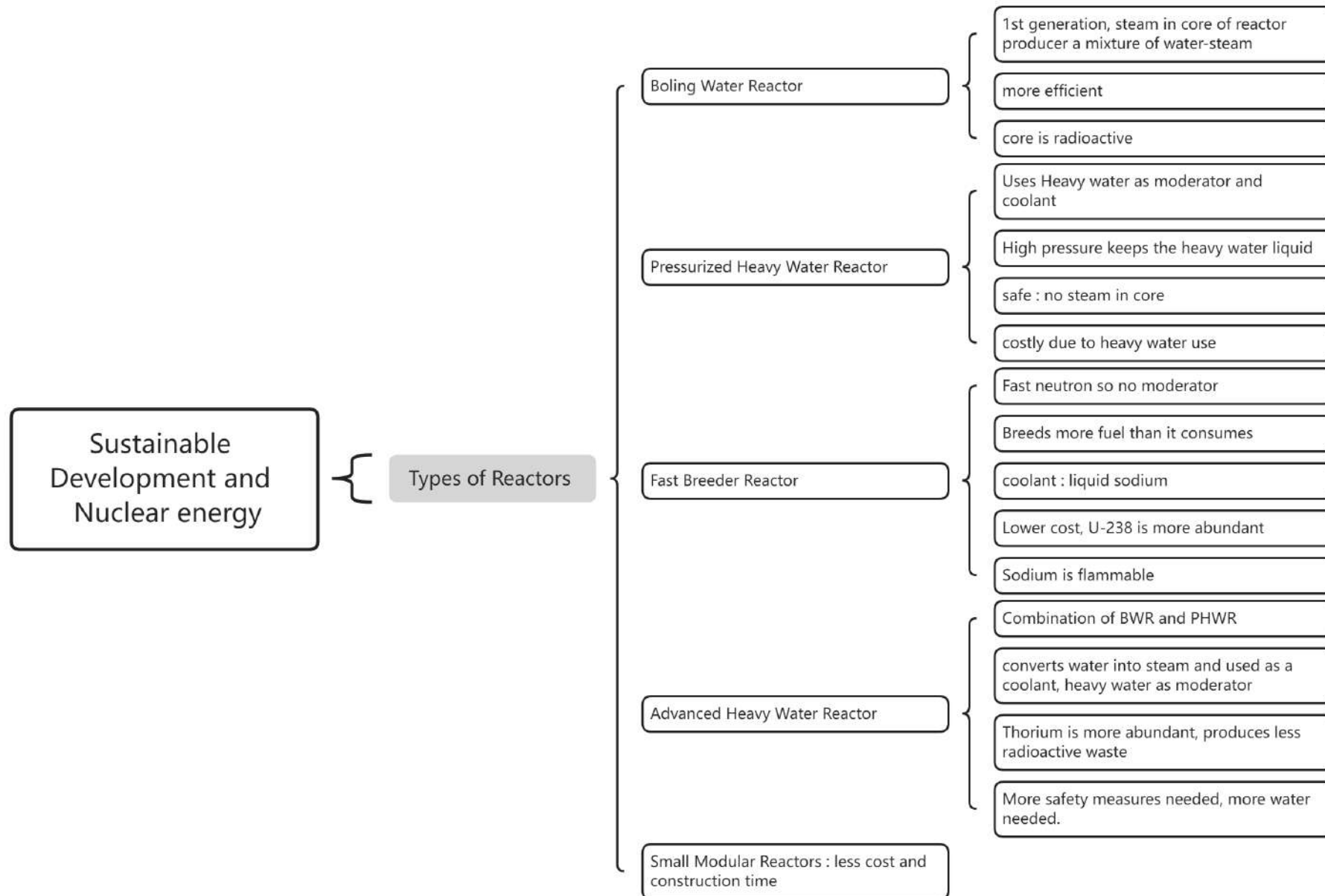
7 principles

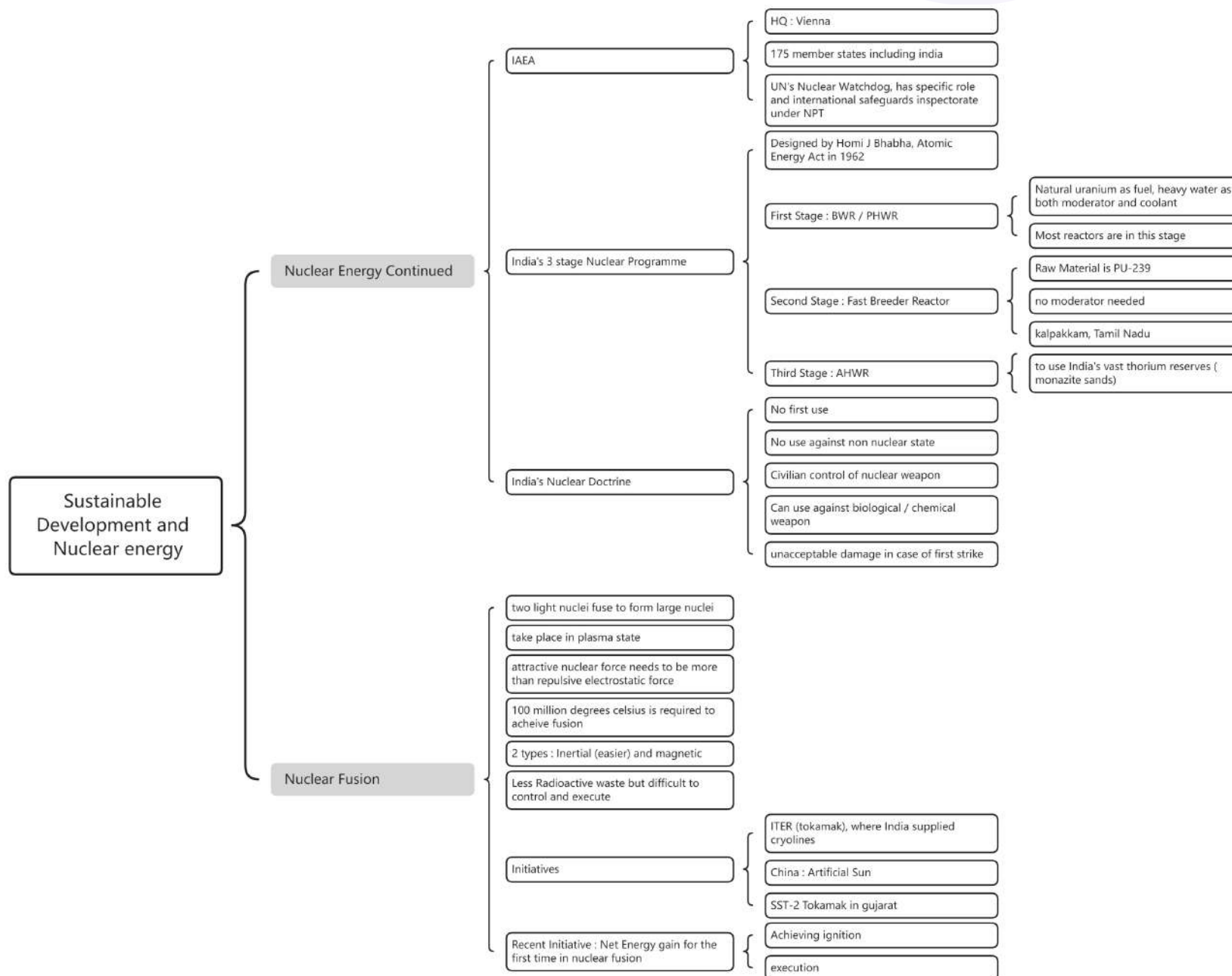
Read from slides (important)

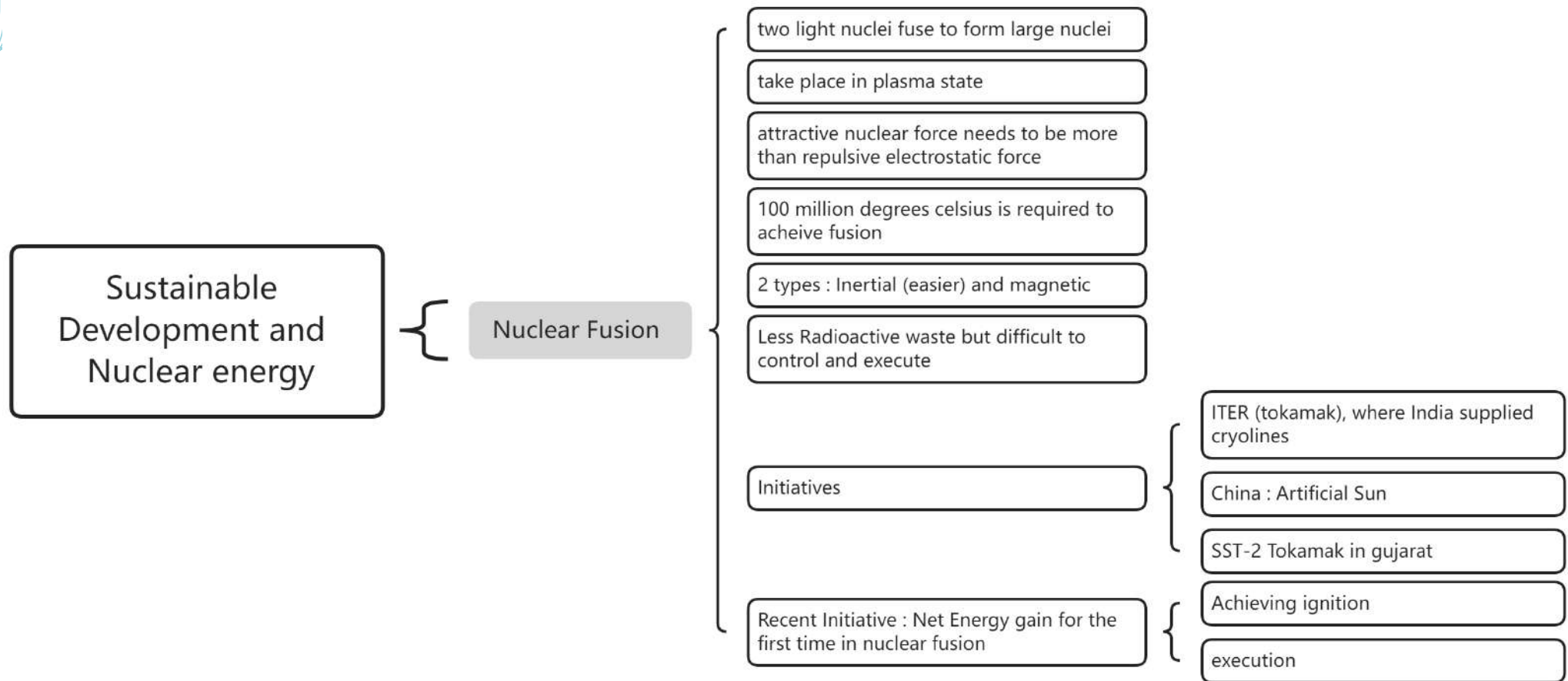


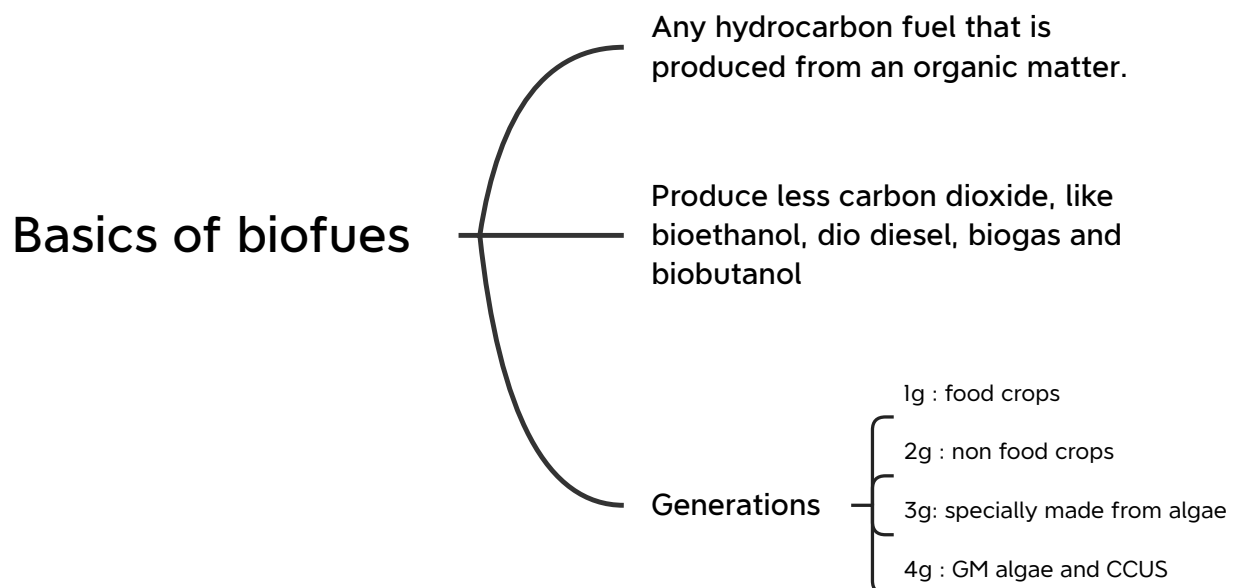












## National Policy on Biofuels

implemented by NBCC, chaired by minister of petroleum and natural gas.

Target of E20 blending in petrol by 2025 and 5% biodiesel in diesel by 2030

Classifies as basic, advanced and drop in fuels

Fodder for biofuels : sugarcane, damaged food grains, surplus food grains



## Bioethanol

ethyl alcohol

Can be produced from sugarcane, sugar beat, starch containing products and cellulosic materials like bagasse

Linkages between sugar production and ethanol : explained in class

Ethanol blending in petrol : to be made from sugarcane based raw materials like C and B heavy molasses, sugarcane juice, sugar, sugar syrup etc.

### Recent Initiatives

PM JIVAN for 2g biofuels

CCEA approved interest subvention scheme

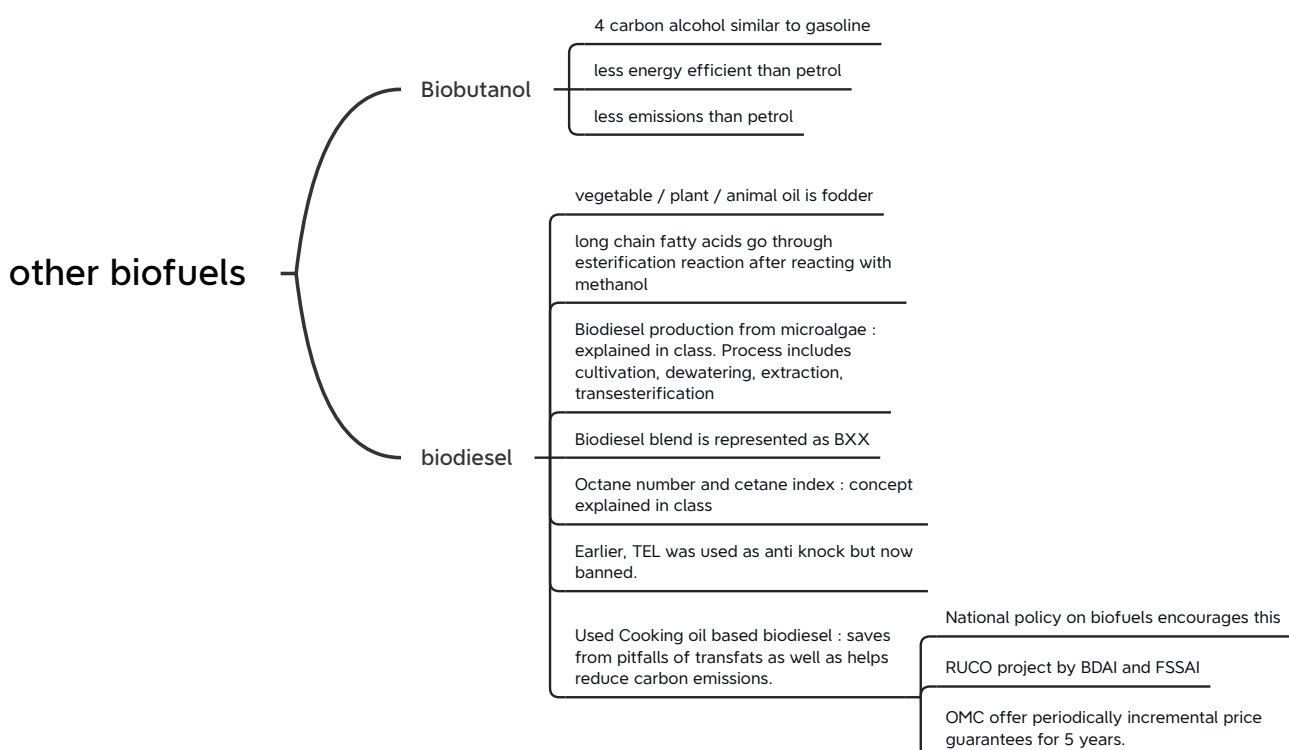
E100 pilot project

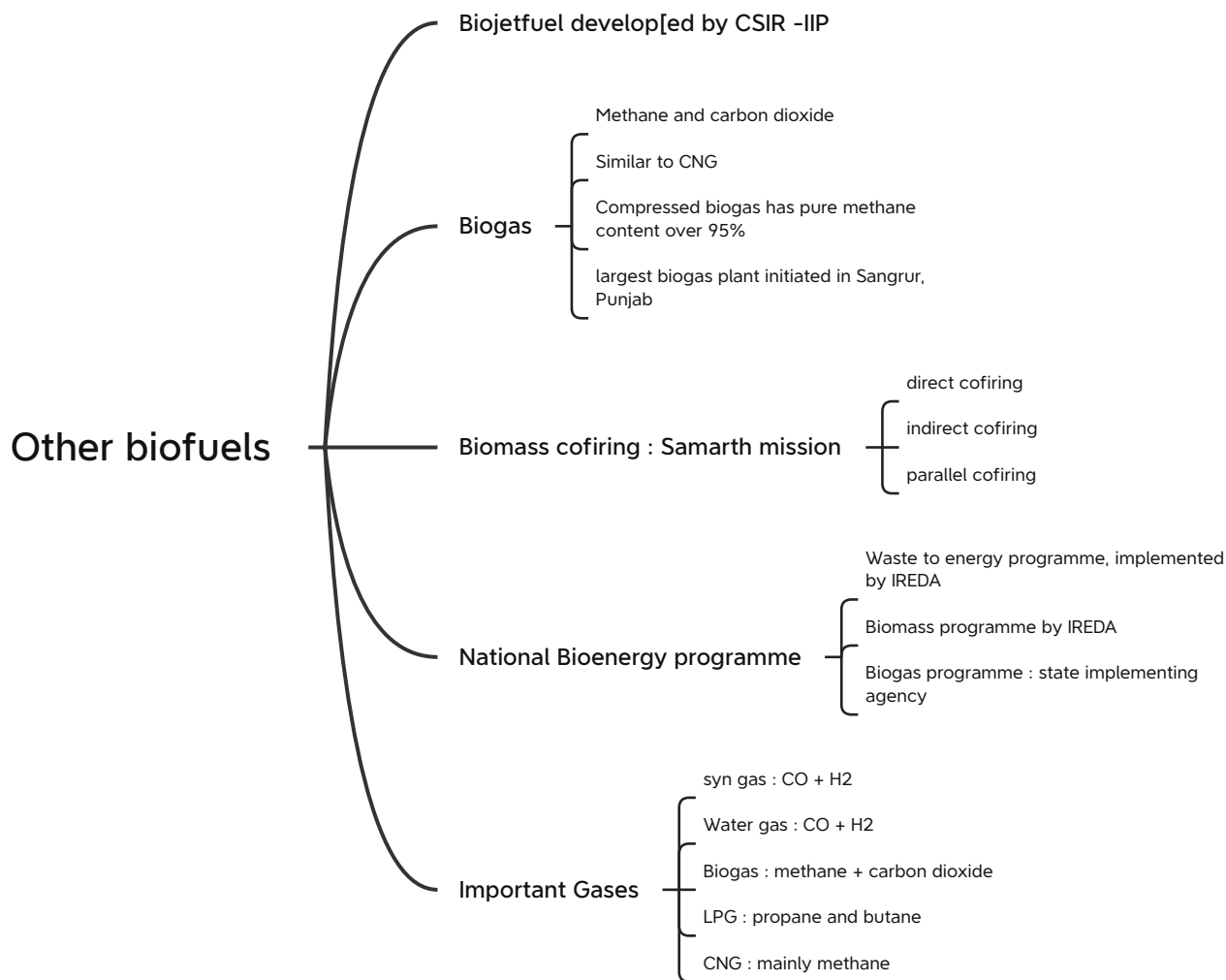
Mass emission standards for E12 and E15

Cassava : ICAR is considering it for 2G biofuel production

Flexi fuel vehicles : can run on blended fuel - petrol with ethanol and methanol







Presented with xmind



## Fuel Cell and other important initiatives

How fuel cell works : produces water and heat, explained in class

hydrogen enters at anode, positively charged particles cross the membrane to reach the cathode, electrons travel through another circuit to produce electricity. Water and heat are byproducts

Roshini : first saline water lantern which uses sea water, by NIOS



## Wind energy

Concept is same : turning the turbine to generate AC power

locations : coastal, hilly, islands, deserts

Wind aero generator : small wind electric generator having capacity of 30kw

Wind solar hybrid system : MoNRE has issued a policy in 2018

India is currently 4th in the world, target is 140GW by 2030, 29% has been met

Important wind farms : Muppandal, Jaisalmer

Offshore wind energy policy by MoNRE

Global Offshore wind alliance : IRENA and GWEC



## Small Hydropower

capacity of 25MW or below

earlier ministry of power, now  
MNRE

Penstock : pipe through which  
water flows and turns the turbine

Only local application, less power  
generation and seasonal



## Tidal Energy

using water difference in tides to move water across a turbine

requires high and low tide at regular intervals

Minimum height difference must be 5 meters

In india : gulf of kathch, gulf of khambat and sundarbans delta (durgaduani creek project was stopped)



## OTEC plant

temperature gradient between the surface and depth of the ocean is utilised to generate electricity and desalinate ocean water

difference must be atleast 20°C

open and closed cycle

India has relatively high potential of OTEC

NIOT (under MoES) to setup OTEC plant at kavaratti, lakshadweep with indigenous technology

Sindhuja : ocean wave energy converter by IIT madras



## Geothermal energy

ONGC started drilling in Puga (ladakh) for geothermal energy

Uses hydrothermal resources.

India has high potential

India's geothermal provinces and Sites : map in slides



## Deep Sea Mining

Remember UNCLOS limits

54% of the world's ocean is deep sea

permission must be taken from international seabed authority

Deep sea has critical minerals like Polymetallic nodules, massive sulphides and cobalt rich ferromanganese crusts

ISA

formed under UNCLOS

India is a member

effective protection of marine environment

UNCLOS has also established UNCLCS and ITLOS



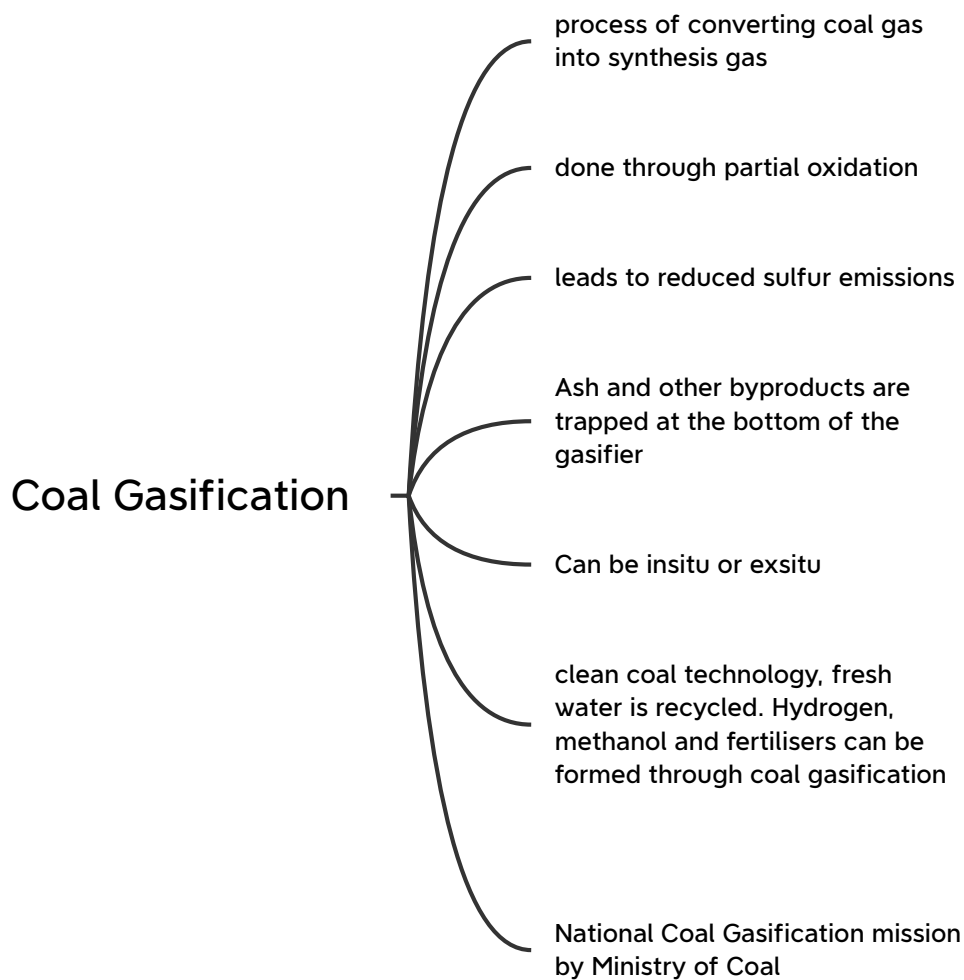
## Coal bed Methane

form of natural gas found in coal seams

extraction through use of water :  
Explained in class

change in pressure alters the porosity of coal and methane is released, however there is a risk of leakage



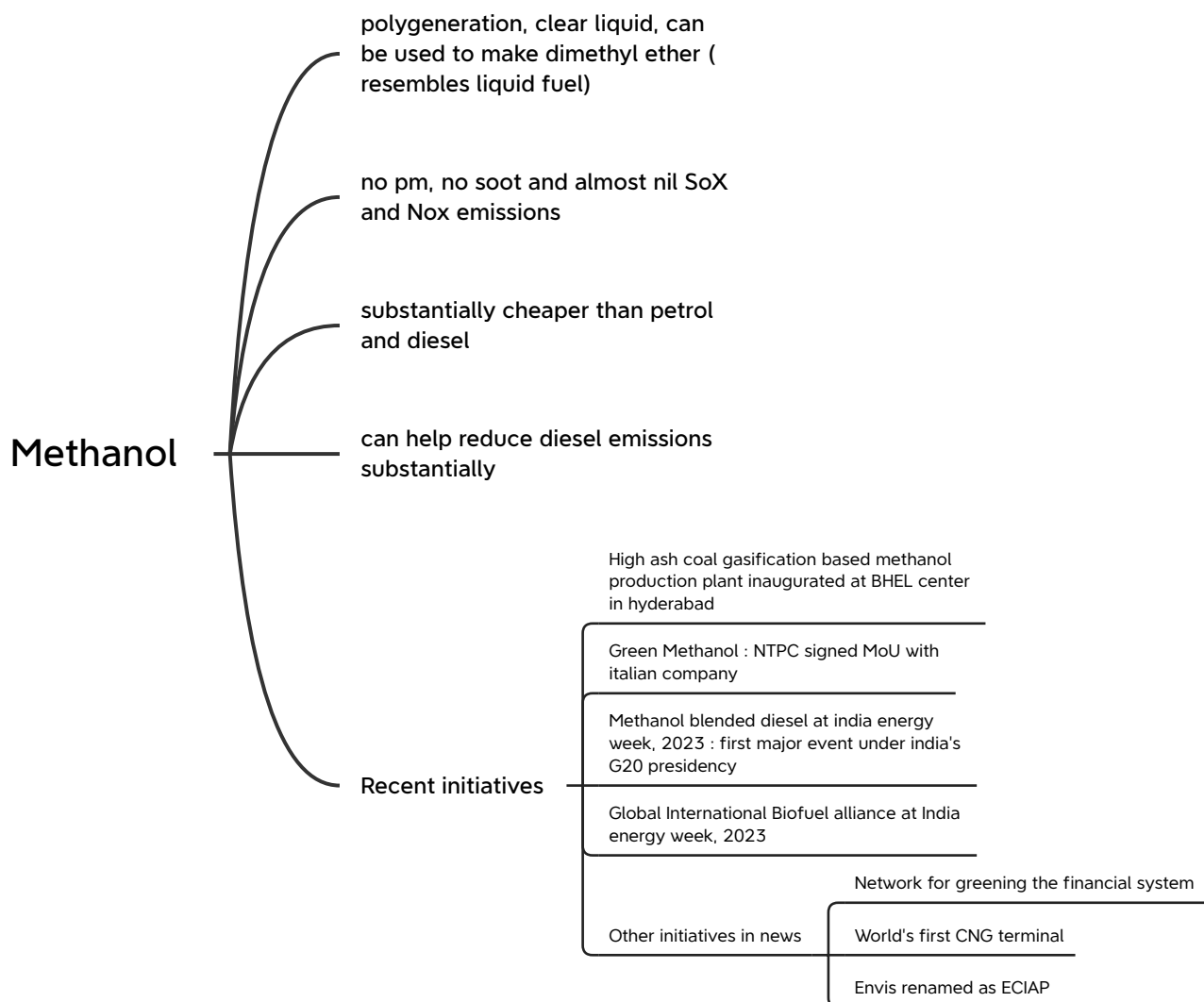


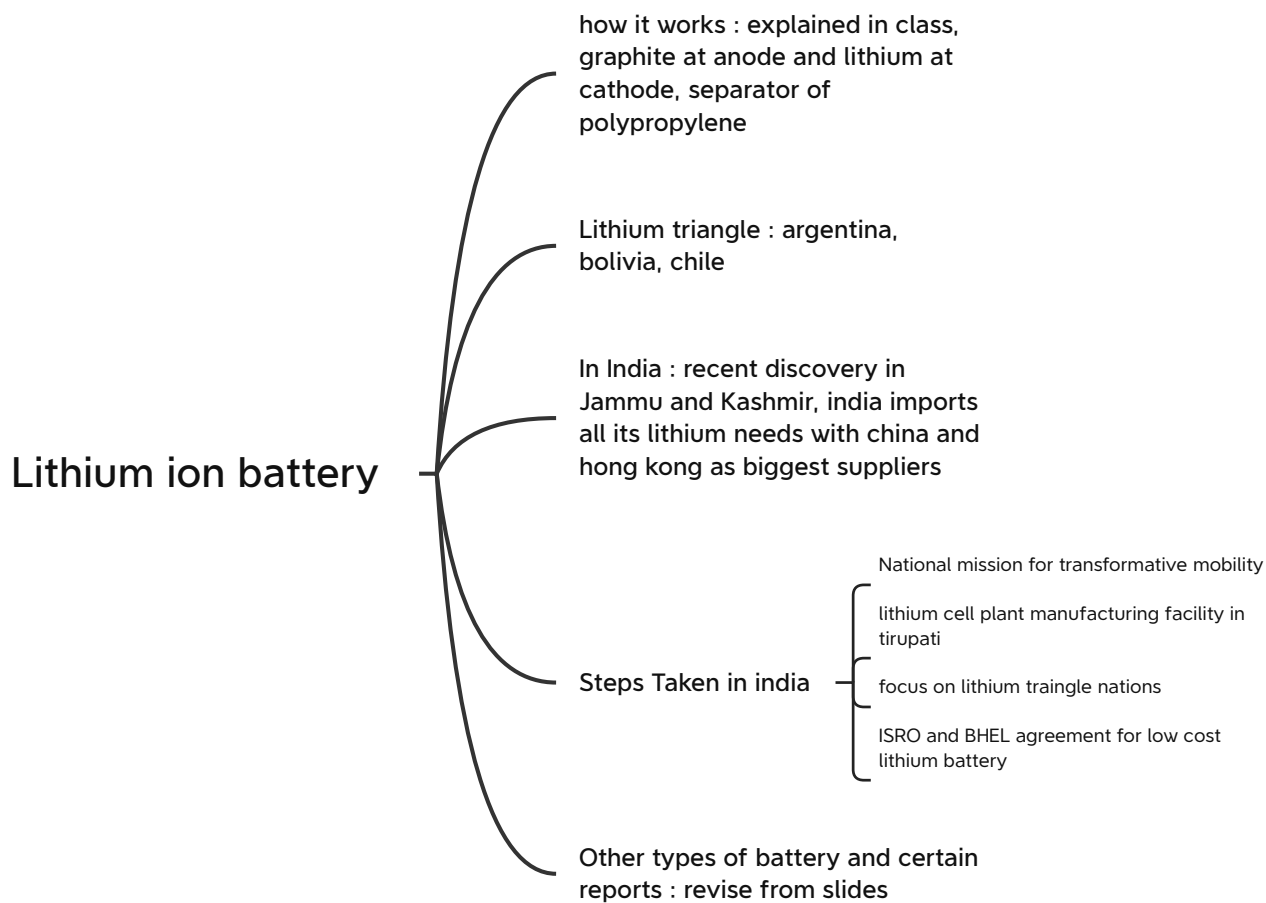
Interest rate subvention

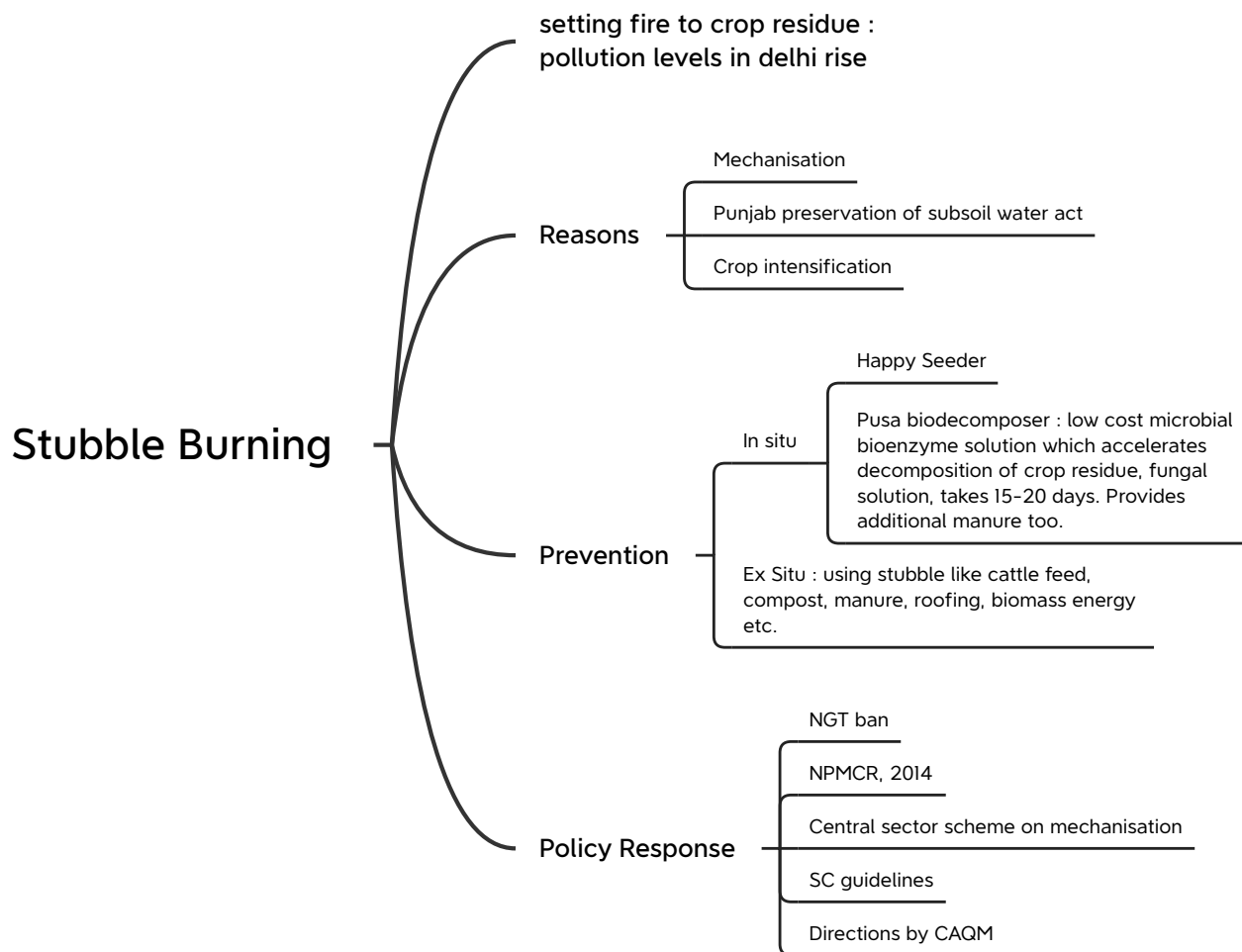
Import duty exemption

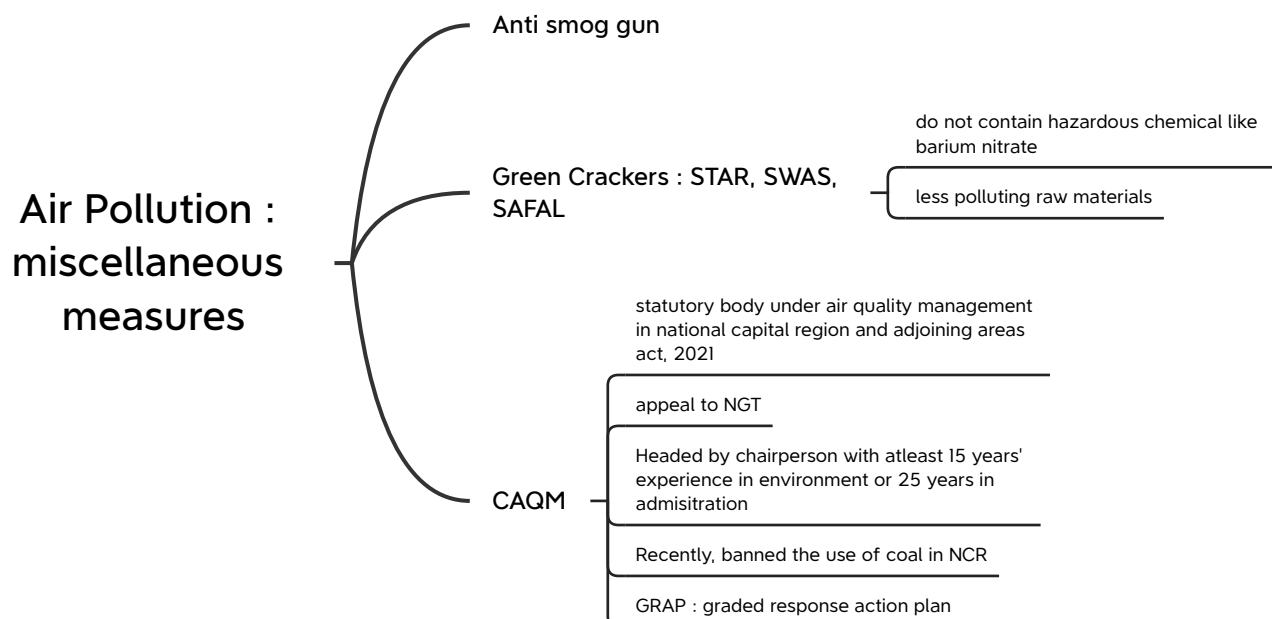
Subtopic 3



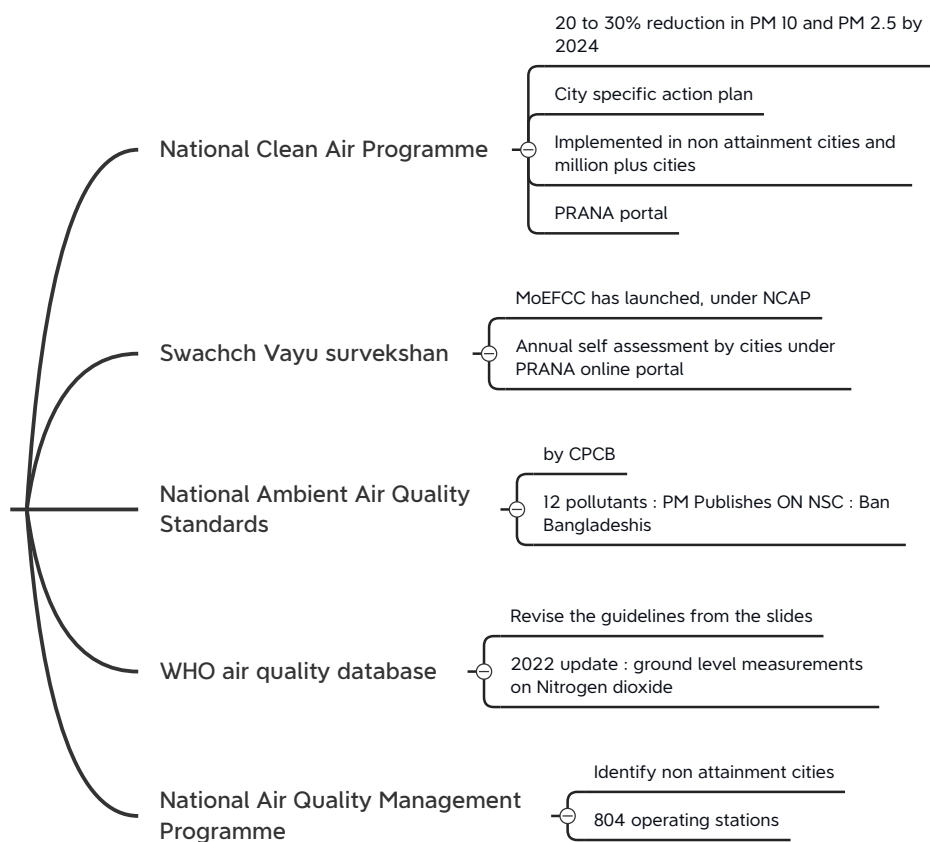


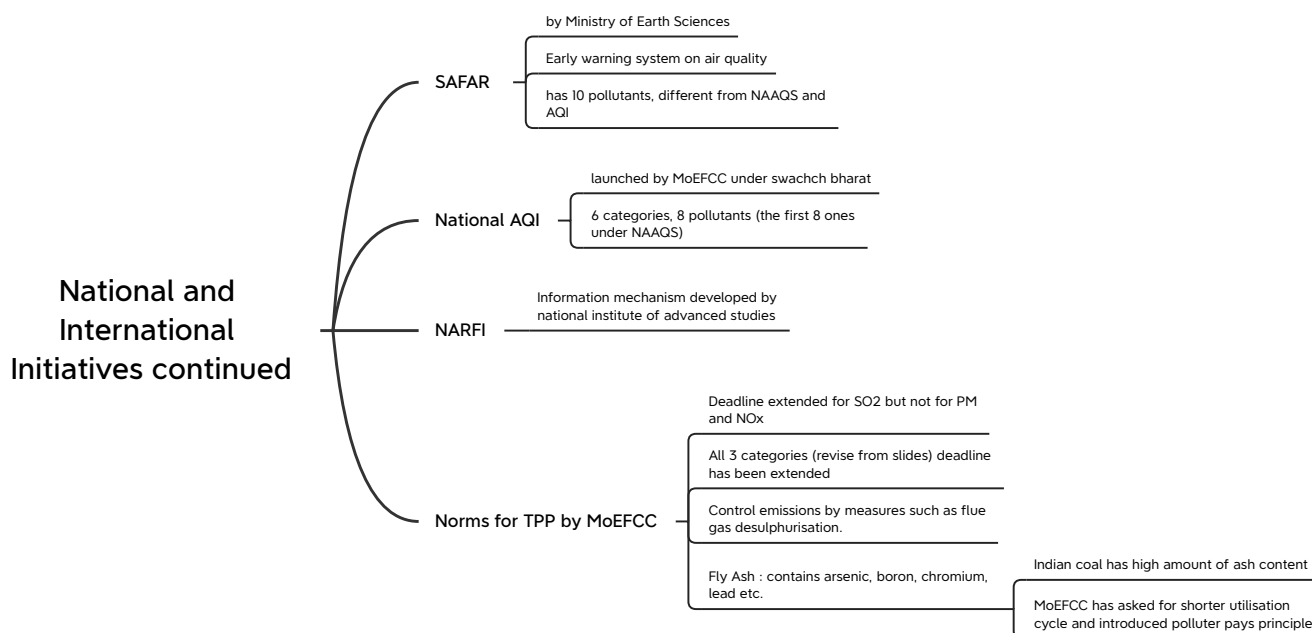


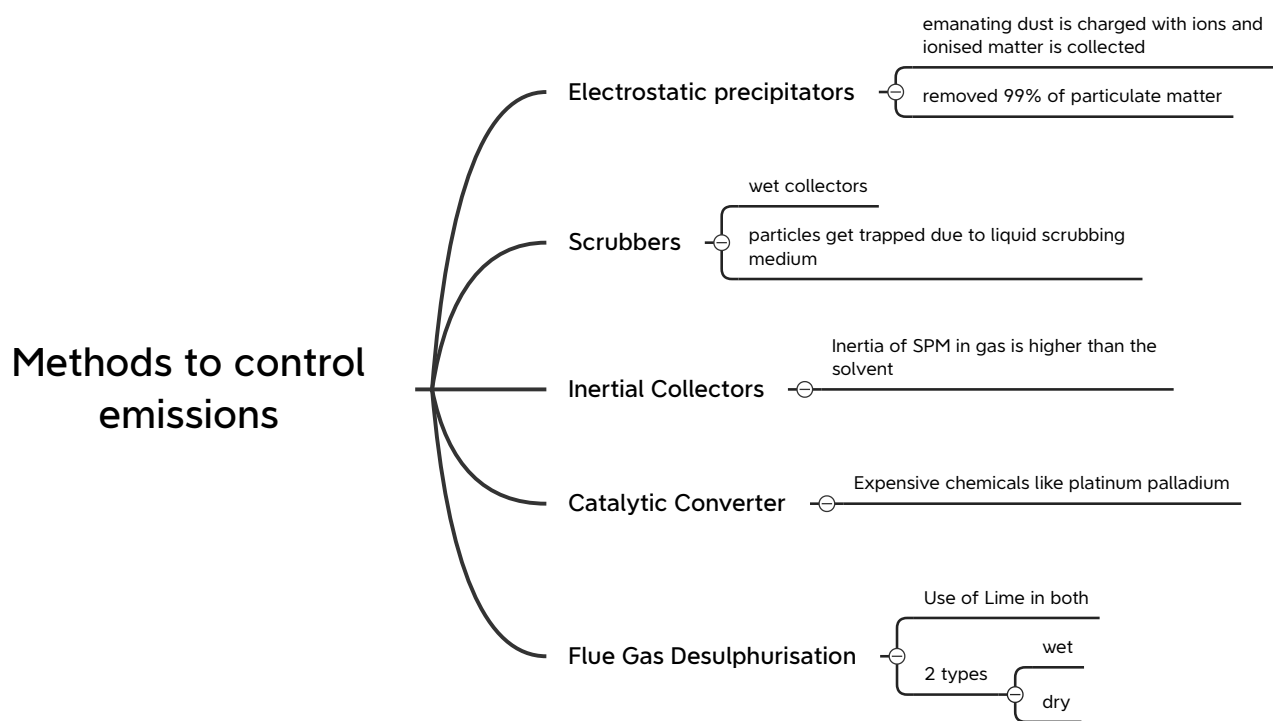


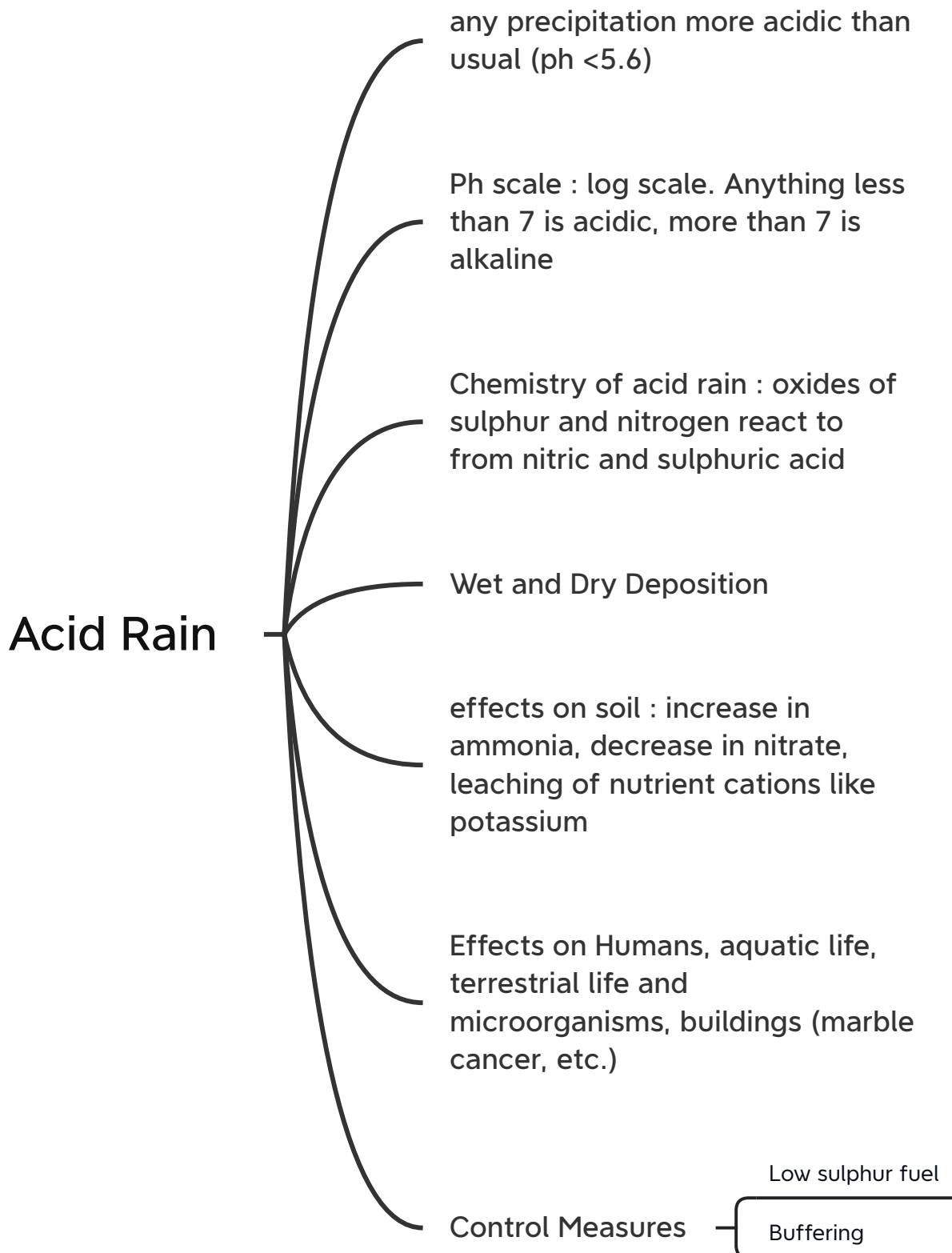


## National and International Initiatives for Air Pollution

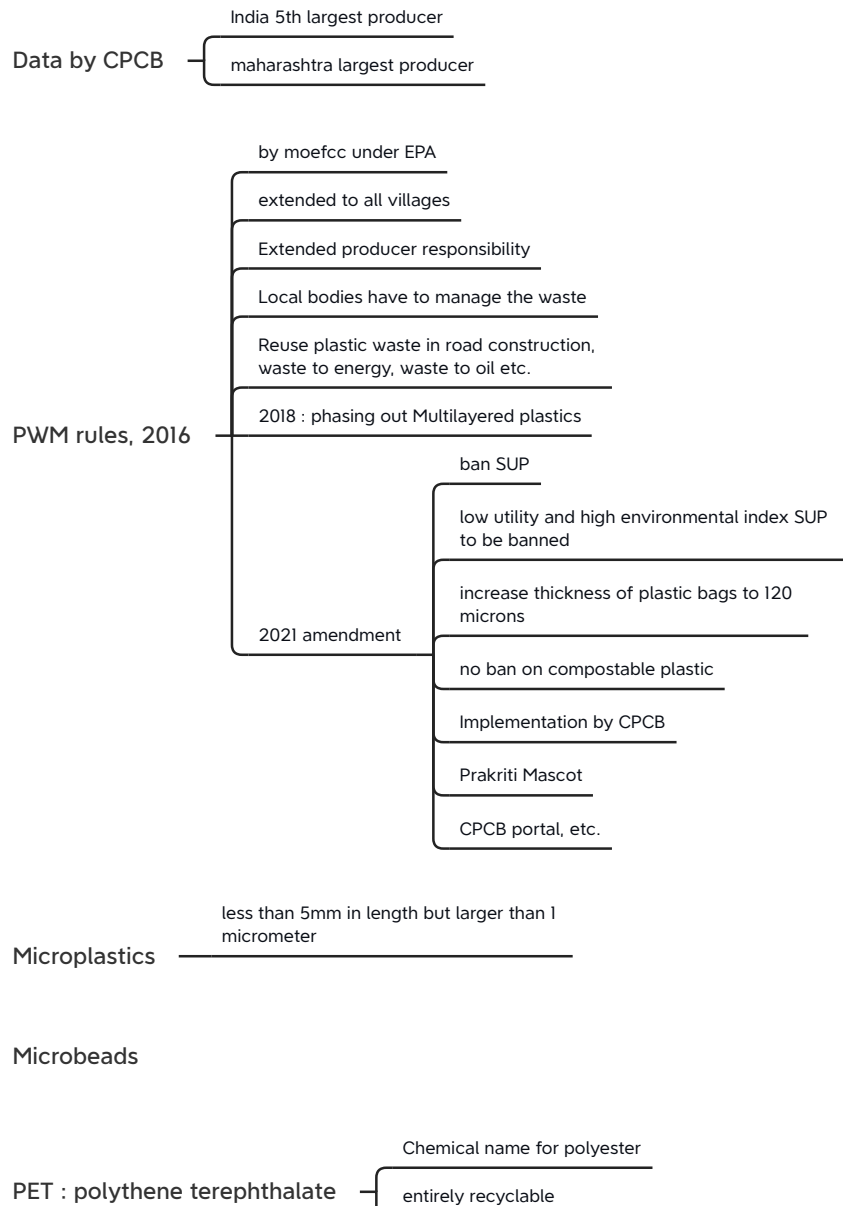


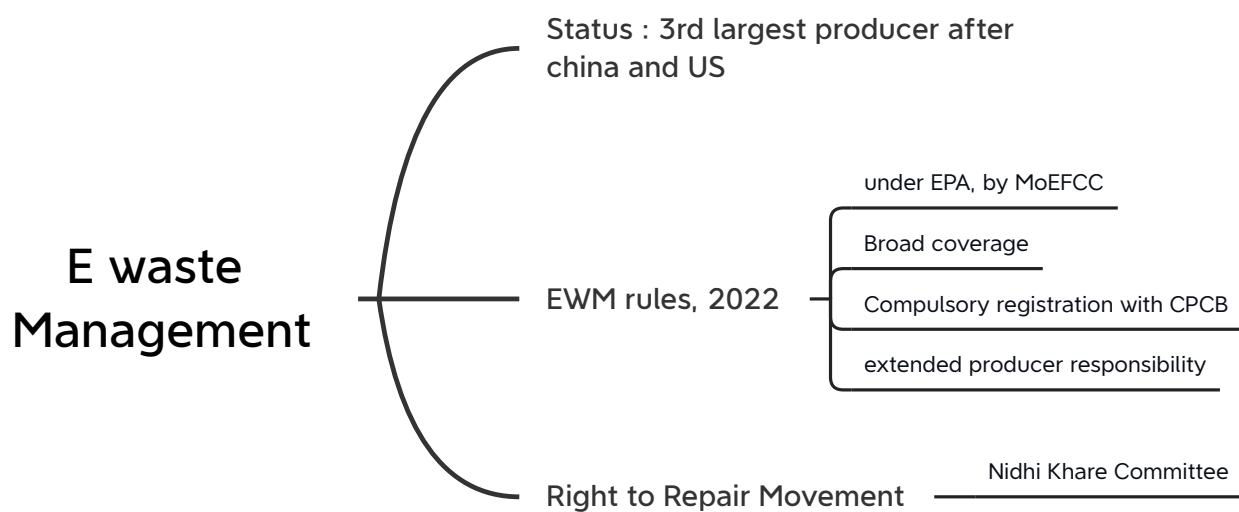


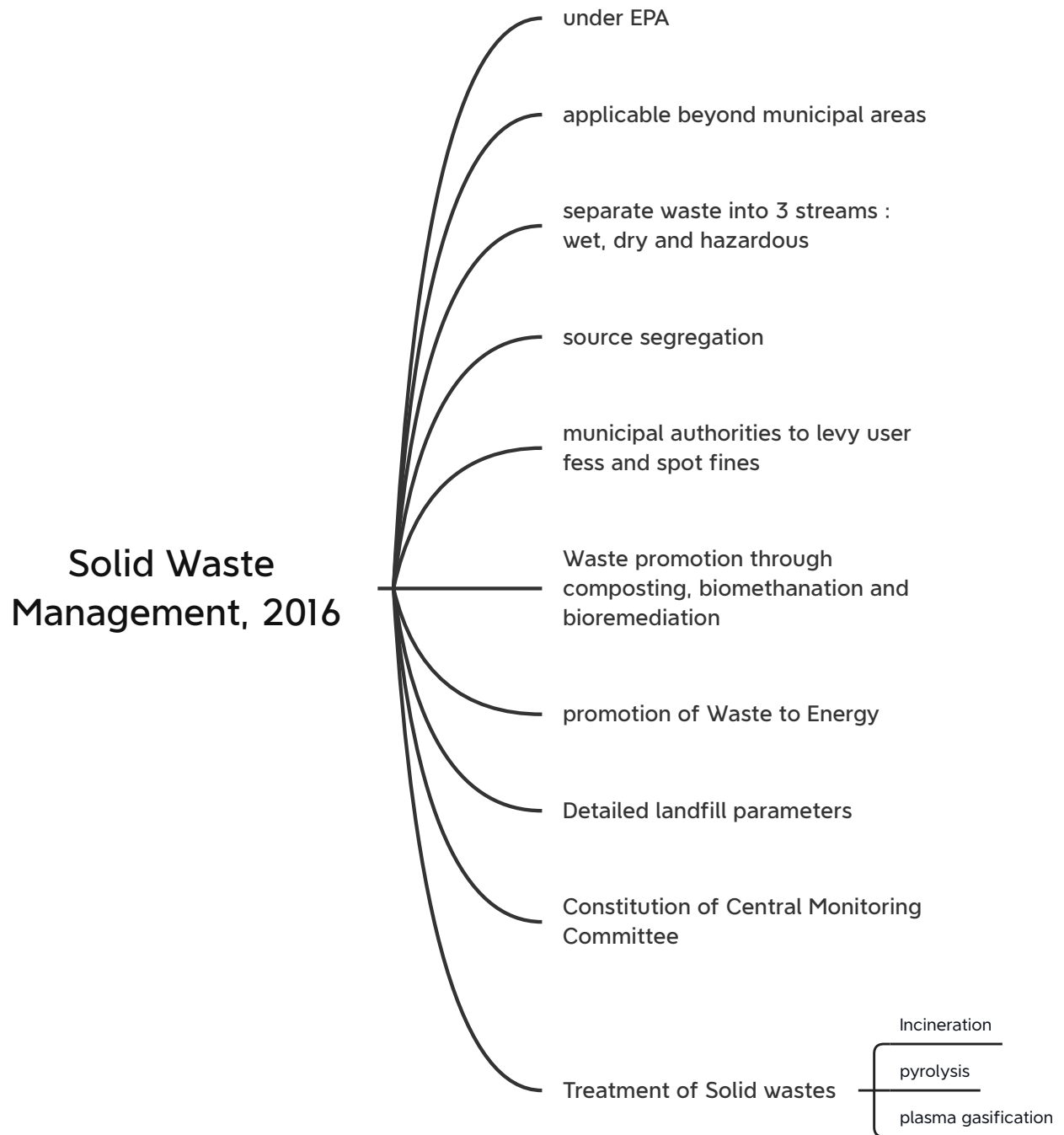




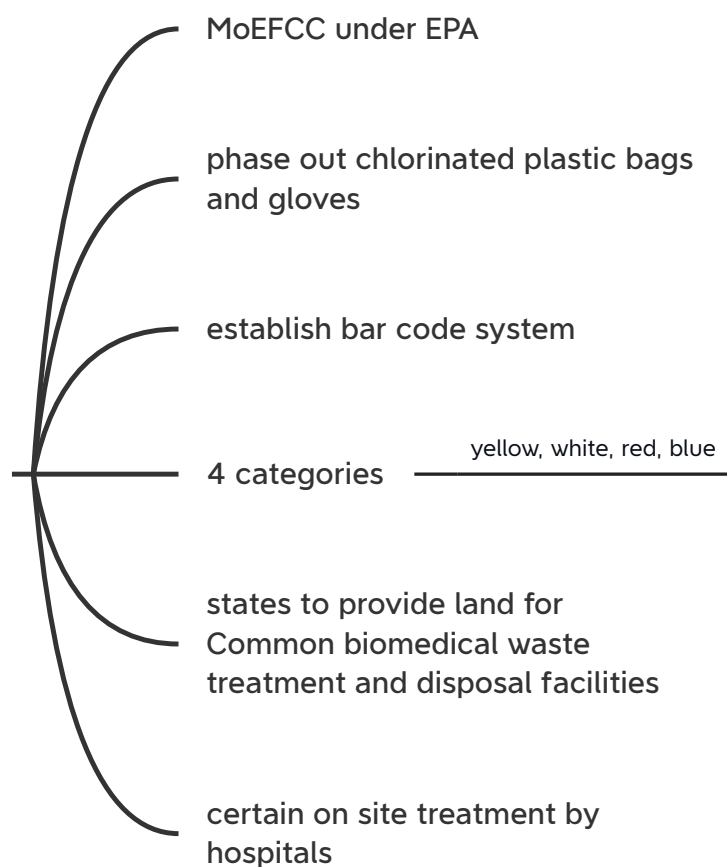
# Plastic Waste Management







## Biomedical waste management rules 2016



## Hazardous Waste Management rules, 2016

MOEFCC under EPA

ambit has been expanded by including other wastes

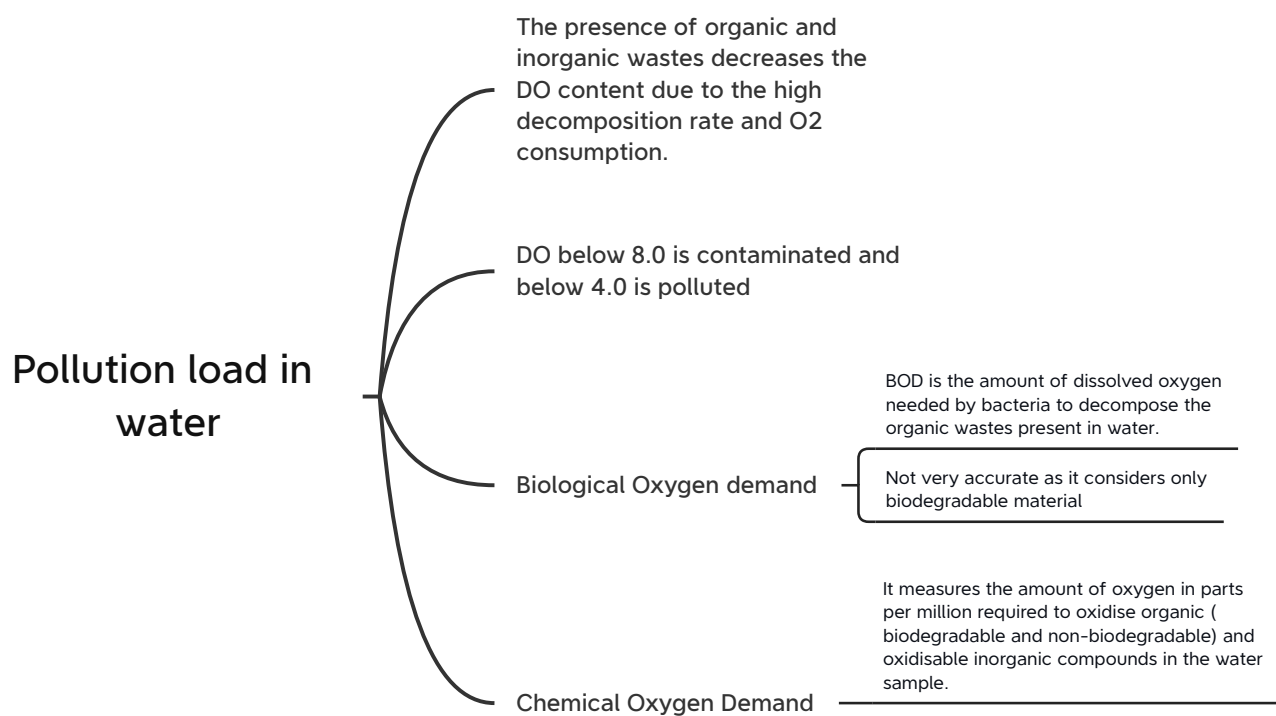
waste management hierarchy

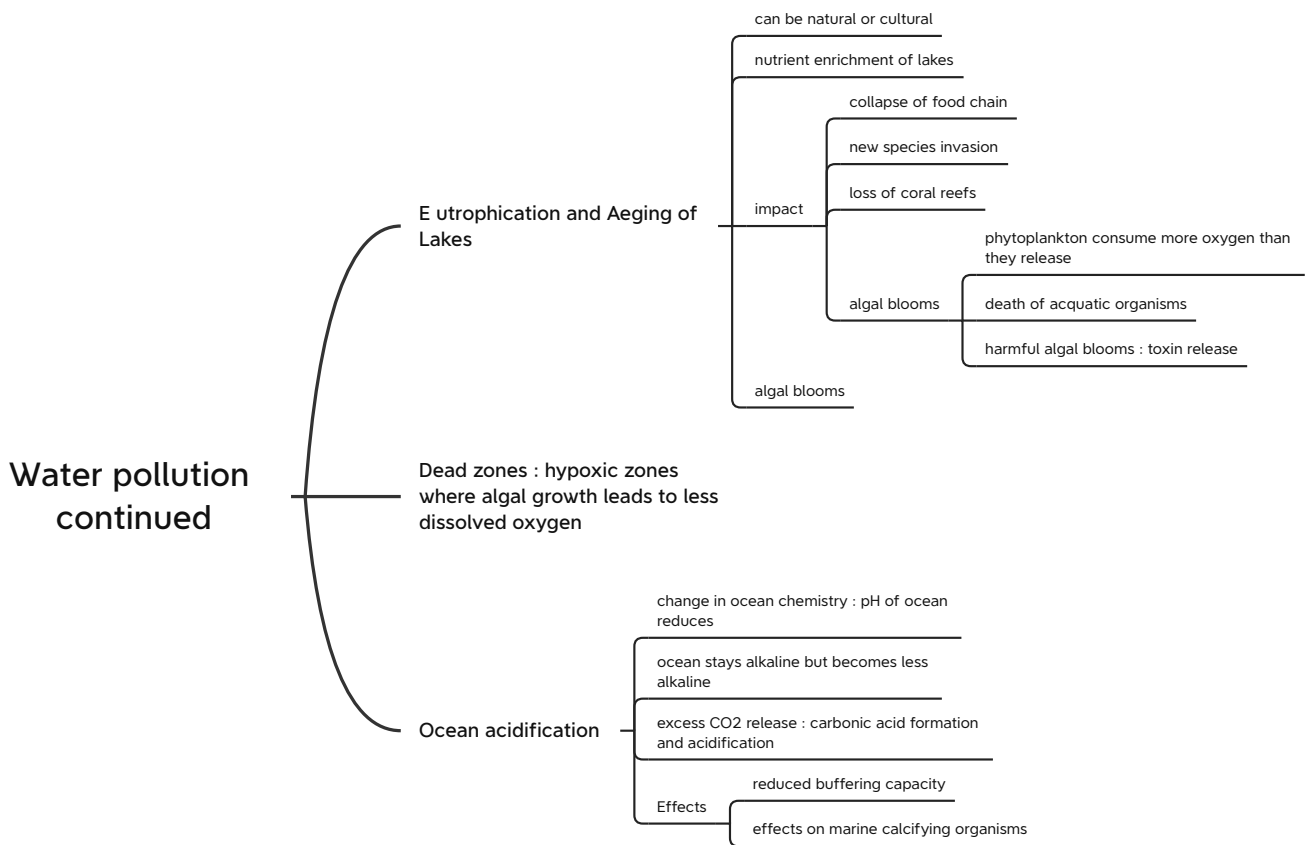
no country can export hazardous waste to india for final disposal

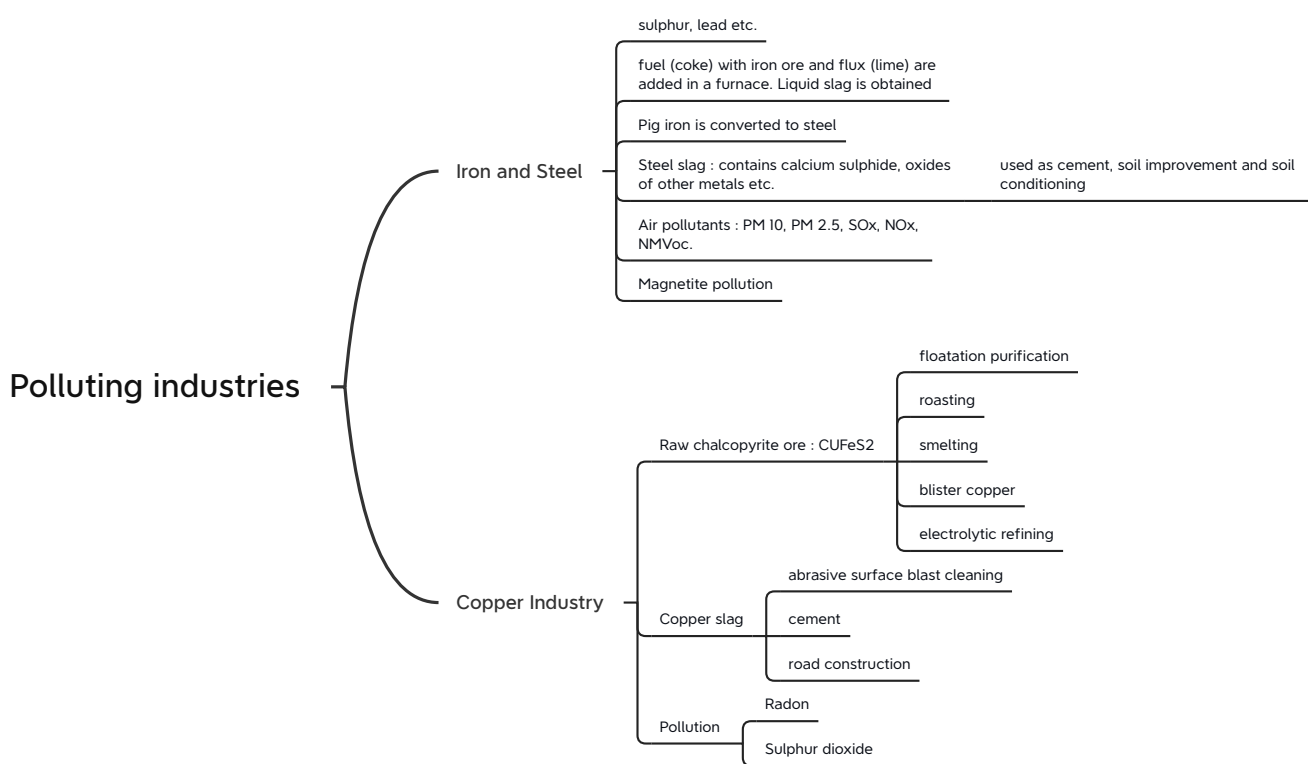
permission from SPCB is required for the layout in this regard, CPCB gives the guidelines

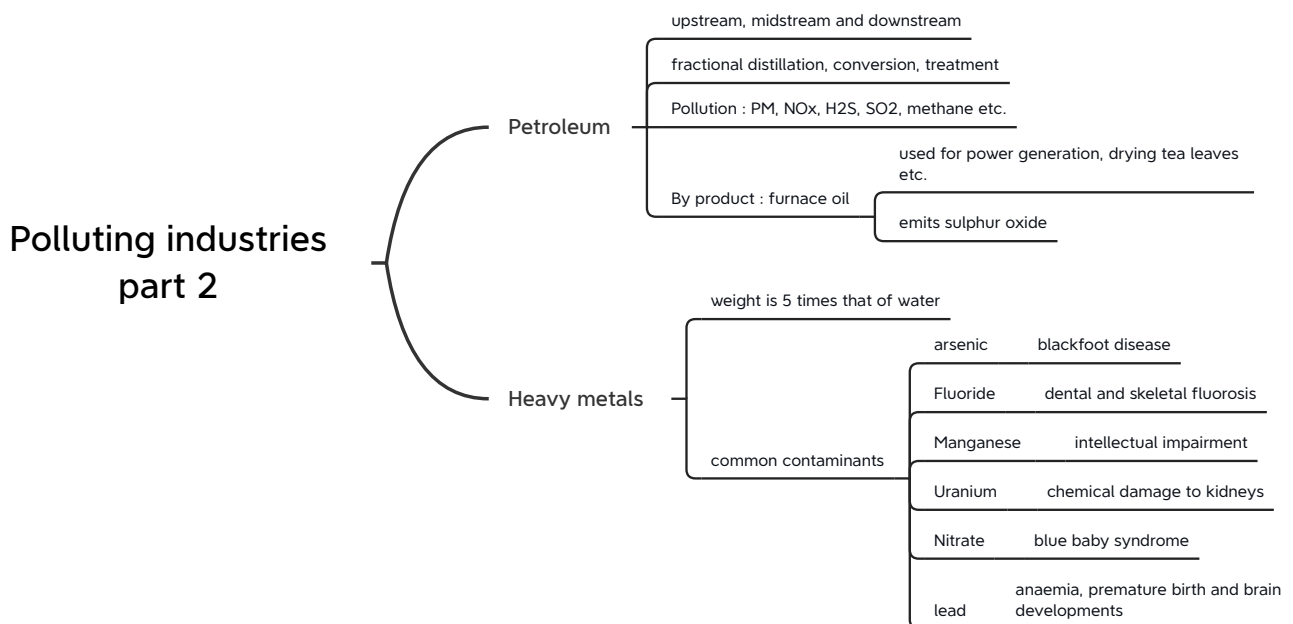
2019 amendment : exemptions to Silk waste, electrical defective waste can be imported back, exemption for certain industries that do not require consent under water and air acts.

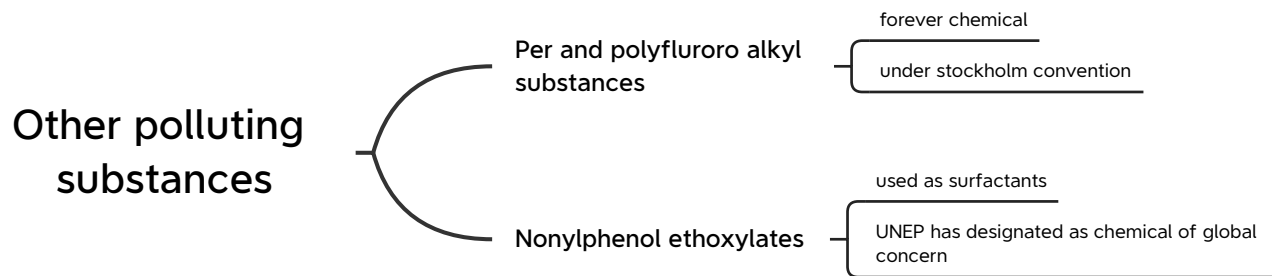


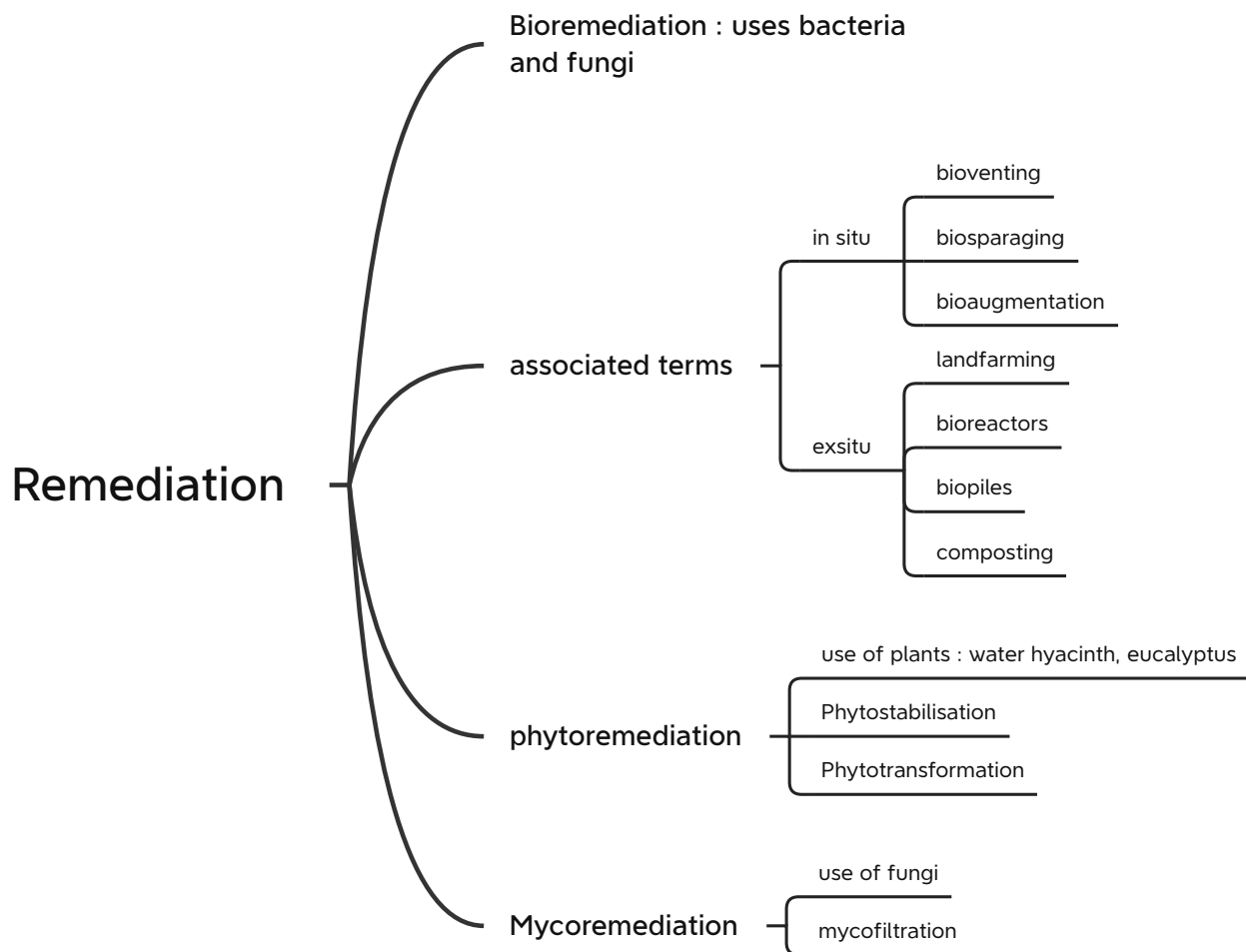


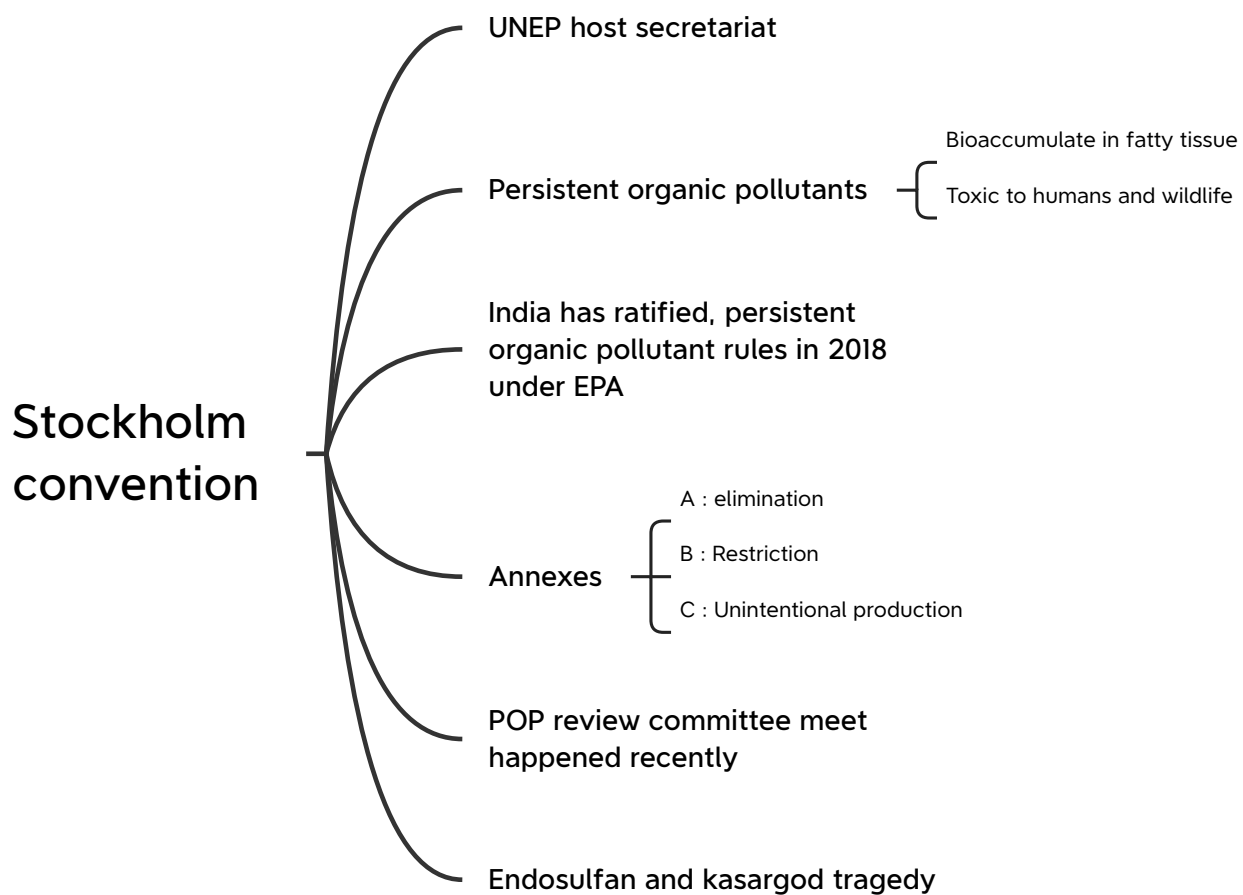


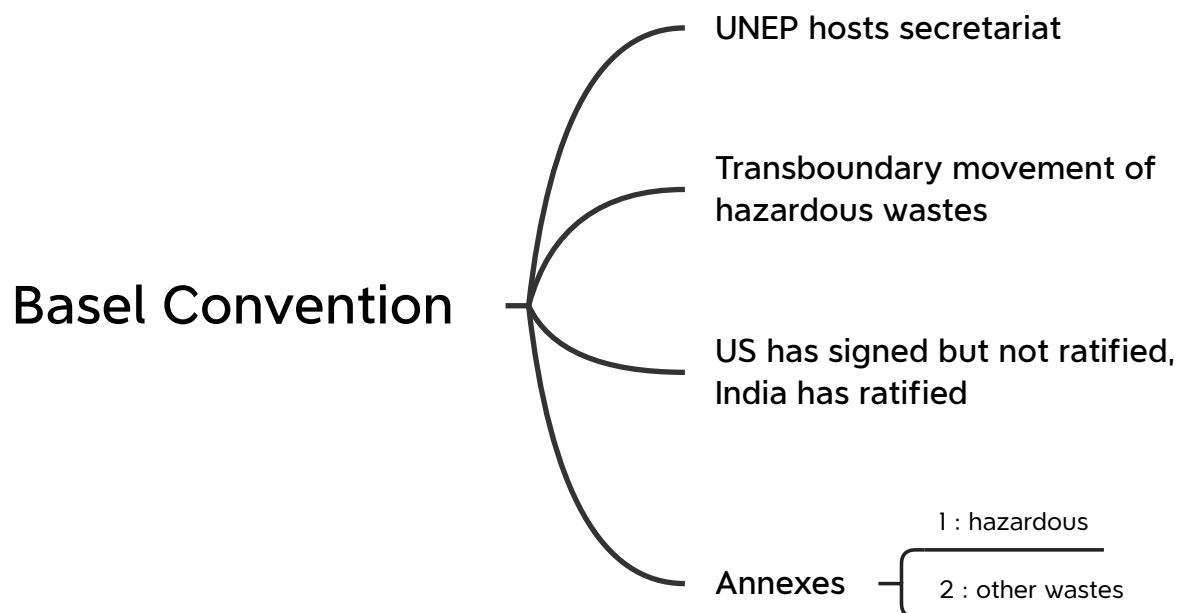












## Rotterdam Convention

UNEP hosts secretariat

Legally binding PIC procedure for pesticides and industrial chemicals

shared responsibility among parties in international trade of hazardous chemicals

Prior informed consent and information exchange

Chemical review committee : subsidiary body iprodione and terbufos recommended

Annex 3 chemicals : subjected to PIC procedure

India Ratified  
One of the largest producers of terbufos, safe as per anupam verma committee report

Related News Pesticide action network report says overuse of pesticides in India



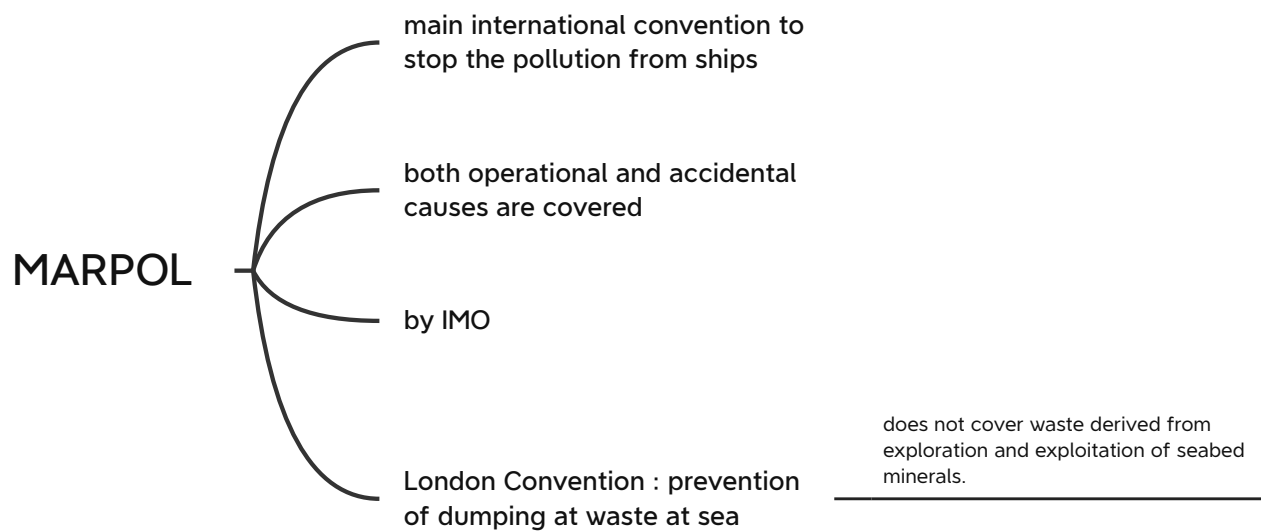
## Honk Kong Convention

recycling of ships

International Labour Organisation  
and Parties to the Basel  
Convention

India is global leader in ship  
recycling and has passed the ship  
recycling act, 2019 to comply with  
guidelines





## Polar Code

for ships operating in polar waters

mandatory and recommendatory measures for safety and pollution

polar ships certificate with categories A, B and C.



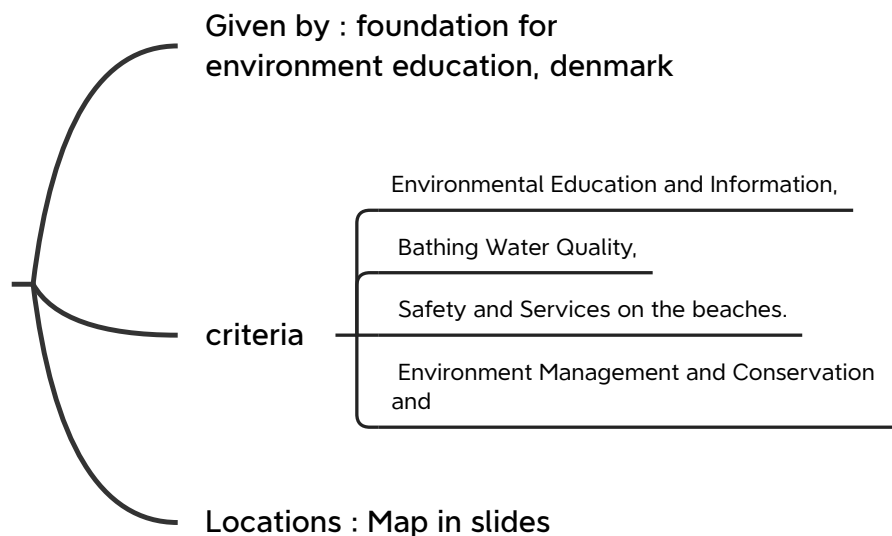
## Under 2 coalition

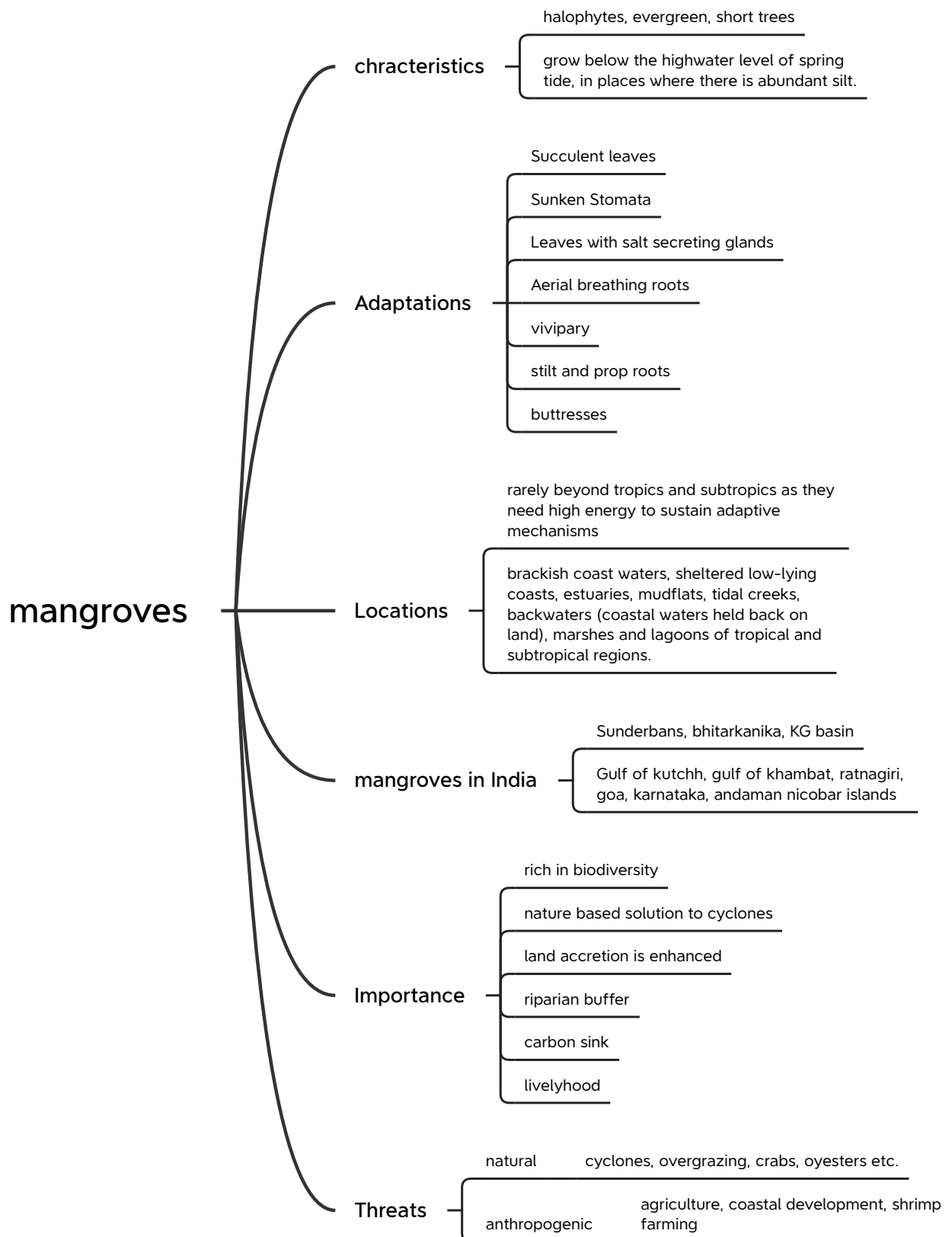
paradiplomacy by subnational governments to keep global GHG emissions low and global temperature rise below 2°C

secretariat : climate group



## Blue Flag Certification





## Initiatives for mangroves

Global Mangrove Alliance : State of mangroves report, 2022

Mangrove Alliance for climate

Launched by UAE and Indonesia at COP 27

India is a member

Building with nature in Indonesia

World Restoration Flagship

MISHTI scheme

this year's budget

mangrove plantation along coastline with MNREGS and CAMPA funds

Sustainable aquaculture in Mangrove ecosystem

Sustainable shrimp cultivation

community based pilot project in west bengal

